

# AIrtist: Agentic Concept Development for Contemporary-Art Installations

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**Abstract.** Early-stage contemporary-art installation development often begins from an under-specified artist-authored problem rather than a resolved artifact. We introduce AIrtist, the first evaluated computational system whose primary task is to develop under-specified contemporary-art installation briefs into multiple complete project hypotheses for what the artwork itself could become. AIrtist maintains five proposal lineages in parallel, reviews what in each proposal should be preserved or changed, generates competing alternatives, and lets revisions challenge the current proposal rather than automatically replace it. We evaluate AIrtist on 50 held-out commission-style briefs using exhaustive blind  $5 \times 5$  candidate-pool comparisons, exact A/B presentation reversal, and a separately instantiated GPT-5.6 Sol holistic pairwise evaluator. Against a substantially matched shared-parent generic iterative control, AIrtist yields a centered complete-pool effect of +0.2352 (95% CI [+0.1472, +0.3212], Holm  $p < .0001$ ). Across 250 lineages, 22 outcomes retain the incumbent unchanged; among 228 adopted replacements, challengers receive candidate-weighted exact-parent preference 0.6601 and a brief-balanced centered effect of +0.3347 (95% CI [+0.2143, +0.4520]). Final ranking remains unresolved: a pointwise-only policy yields higher mean external utility for its top-ranked proposal (0.6633 versus 0.6353), while a hybrid policy with pairwise finalist comparison more often places an externally highest-utility candidate first (0.32 versus 0.26); neither policy is superior across the measured ranking outcomes. Together, these results support treating the construction of strong alternatives, the quality of adopted revisions, and the recognition of the strongest available candidates as distinct empirical problems. More broadly, they establish early-stage contemporary-art concept development as a tractable computational object and motivate a broader research agenda for Creative AI: how to construct richer possibility sets, distinguish ranking instability from genuine evaluative disagreement, preserve recoverable alternatives, and make the commitments steering creative search negotiable by artists.

**Keywords:** contemporary art; installation art; concept development; creative agents; computational creativity; human-AI co-creativity; agentic AI.

## 1 Introduction

A consequential part of contemporary-art practice occurs before there is an artifact to fabricate or exhibit. An artist may begin with a pressure rather than an image: a social relation that should shape what physically happens in the work, a site whose existing use should remain active rather than be displaced by the installation, a material restriction, an institutional contradiction, or a conviction that several obvious artistic solutions would be wrong. At this stage, the artist may know what matters without yet knowing what the work should be. The problem is not how to execute a resolved concept, but how to develop alternative propositions for what the artwork could become.

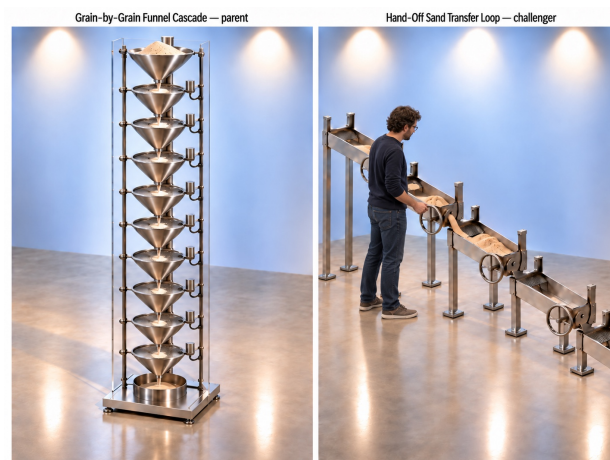
Recent generative models make this stage newly accessible to computational support. They can produce plausible project descriptions, variations, references, images, and technical elaborations rapidly. Yet an abundance of plausible completions does not by itself solve early concept development. The same premise can be developed in radically different ways: a concern may be represented symbolically, enacted through a material process, embedded in a spatial condition, delegated to an institutional procedure, or made dependent on a spectator's position. When alternatives are produced by revising an initial proposal, the resulting revisions can also diverge sharply in approach and quality: a version may become clearer, more interactive,

or more technically sophisticated while becoming artistically weaker. Several incompatible directions may remain viable at once. The computational problem is therefore not simply to generate more material, but to construct useful alternatives, determine what in an existing proposal should survive revision, and decide which directions are worth pursuing.

Installation practice makes these difficulties particularly visible because the identity of a work can depend on relations among material, bodies, space, duration, site, and institutional framing rather than on a detachable artifact alone [1, 2]. We use *artistic operation* to name this level of description: what a proposal actually does through its material, spatial, temporal, spectator, or institutional organization to enact its stakes. This does not define artistic quality, but it gives concept development a more discriminating computational object than topic, style, or visual appearance. A system working at this stage must reason not only about whether a proposal is coherent, but about which relation gives it identity and what kinds of change would preserve, sharpen, or destroy that relation.

Recent artistic practice already shows generative systems being used while possible artworks are still being conceived, including Kuran’s *Errorism* [3], Lund’s *In the Middle of Nowhere* [4], and Cai Guo-Qiang’s cAI [5, 6]. Section 3 examines these precedents in detail. These practices establish the use case but remain artist-specific procedures embedded in particular oeuvres and projects. Adjacent computational research, meanwhile, provides sophisticated machinery for early ideation, including professional concept design and scientific hypothesis generation, iterative review, and multi-candidate development [7, 8, 9]. What remains largely unformalized is the task at their intersection: taking an under-specified contemporary-art problem and systematically developing several proposals for what the artwork itself could become.

To fill this gap, we introduce AIRTIST, to our knowledge the first evaluated computational system whose primary task is to develop an under-specified artist-authored contemporary-art installation brief into multiple developed project hypotheses for what the artwork itself could become. An artist supplies a rough brief describing the pressure they want to investigate, conditions that must remain consequential, directions they want to avoid, and—where relevant—site, material, institutional, or production constraints. AIRTIST treats this brief as a provisional problem specification rather than as a description of a desired artifact and develops five proposal directions in parallel. Each of the five system-generated proposals becomes the incumbent of a persistent lineage. Specialized reviewers identify what should be preserved and what most needs intervention; their evidence is compressed into an opportunity memo specifying a protected strength, the dominant liability, a field logic to escape, an underused material or spatial opportunity, and a transformation worth testing. Two challengers are then generated independently from the same unchanged parent. A Focused reconstruction rebuilds the diagnosed weak part while protecting the designated incumbent strength; a Structural escape has broader permission to replace the dominant operation or realization. The incumbent remains eligible, so a lineage may retain its current proposal or replace it only when an eligible challenger wins comparison. After one state survives in each lineage, a separate terminal step ranks the five resulting proposals. In the evaluated system this process is autonomous: the artist supplies the brief once and receives the ranked set.



**Figure 1:** Held-out Brief 036. One system-generated proposal uses an automatic sand cascade (left); the selected alternative turns the same transfer logic into a visitor-performed hand-off sequence (right). The rendering is a post-hoc explanatory visualization; evaluation used the proposal texts, not the image.

A held-out example gives an intuitive sense of the kind of change AIRTIST is designed to explore. Brief 036 asks for an installation about translation as a chain of minor losses, in which successive transfers change a shared physical condition without representing languages through national symbols. One of the five system-generated incumbents, *Grain-by-Grain Funnel Cascade* (Figure 1, left), lets quartz sand descend automatically through successive funnels and calibrated meshes. The same body of sand moves through a sequence of stages and becomes progressively smaller because each stage retains part of it. In *Hand-Off Sand Transfer Loop* (right), that material logic is preserved: the same diminishing sand body still passes successively from one stage to the next, leaving a residue behind at each transfer. What changes is the action that advances the sequence. In the incumbent, gravity carries the sand through the cascade automatically; in the selected alternative, each transfer occurs only when the visitor operates the next stage. The evaluated sand-transfer case is analyzed in detail in Section 7.3.2.

The evaluation asks three separate questions. RQ1 asks whether domain-structured development produces a stronger five-proposal set than raw generation, generic self-refinement, and a substantially matched generic iterative process. The strongest control, M5, starts from the exact same five self-refined parents as AIRTIST, maintains the same number of lineages and challengers, keeps the incumbent eligible, and uses the same challenger-eligibility rule and within-lineage comparison procedure, while removing AIRTIST’s specialized review, opportunity representation, and differentiated intervention roles. RQ2 asks whether, when AIRTIST replaces an incumbent, the adopted replacement is externally preferred to the exact parent it displaced. RQ3 keeps the generated five-proposal set fixed and compares final ranking policies over that same set.

We evaluate AIRTIST on 50 held-out commission-style briefs. RQ1 uses exhaustive blind  $5 \times 5$  cross-pool comparisons rather than only system-selected winners; every logical comparison is repeated under exact A/B presentation reversal. The external evaluator is a separately instantiated GPT-5.6 Sol holistic pairwise judge that does not observe system identity, lineage history, internal scores, or the external candidate utilities later derived from its judgments. Statistical inference is performed over briefs rather than individual candidate comparisons.

The results answer those three questions differently. For candidate-set construction, AIRTIST significantly exceeds all three planned controls after Holm correction. Against the substantially matched M5 condition, the centered complete-pool effect is +0.2352 (95% CI [+0.1472, +0.3212], Holm  $p < .0001$ ), and secondary diagnostics indicate that the gain extends beyond the single externally best candidate. For intervention, 22 of 250 final lineage states remain content-identical to their incumbents; among the 228 adopted replacements, candidate-weighted challenger preference against the exact displaced parent is 0.6601, with a brief-balanced centered effect of +0.3347 (95% CI [+0.2143, +0.4520]). Final ranking is less resolved: Base has higher mean Utility@1 than Hybrid (0.6633 versus 0.6353), while Hybrid more often places an externally highest-utility candidate first (0.32 versus 0.26); the primary Utility@1 difference is not statistically resolved and secondary ranking metrics are mixed. Thus the experiment provides strong evidence for higher measured candidate-set quality and for external preference toward adopted replacements, while leaving the ranking of an already-generated set unresolved. Evaluating only a final top-ranked output would conflate failures to generate strong alternatives, failures to improve an incumbent, and failures to rank useful material that has already been generated.

The paper makes four main contributions:

**1. Contemporary-art concept development as an AI task.** We formulate and evaluate early-stage installation concept development from under-specified artist-authored briefs, where the computational object is a set of alternative project hypotheses rather than the realization of an already resolved artwork.

**2. A contestable agentic development architecture.** AIRTIST maintains persistent proposal lineages, converts heterogeneous review into an explicit opportunity memo, generates Focused and Structural interventions from the same incumbent, and allows the incumbent to survive. The architecture represents what should remain invariant separately from what may need to change.

**3. A multi-level evaluation methodology for open-ended creative agents.** We separately evaluate the quality of the five-proposal set, adopted interventions against their exact parents, and final ranking over identical generated candidate sets, including a substantially matched shared-parent iterative control.

**4. Empirical evidence about where agentic value and error arise.** Domain-structured development improves measured candidate-set quality over matched generic iteration; adopted replacements are preferred to the exact states they replace on average; and final ranking remains a separate problem even when stronger measured alternatives are already present.

The claims remain bounded by the evaluation setting. The evidence concerns textual project hypotheses under a disclosed model-based evaluator. The study does not establish autonomous artistic agency, realized-artwork quality, or improved human creativity, and no artist negotiates with the system inside the evaluated pipeline. It asks whether early contemporary-art concept development can be made into an explicit computational task and whether the proposed agentic development policy produces measurable improvements under that evaluation procedure.

**Paper organization.** Section 2 develops the contemporary-art formulation underlying the task, including artistic operation and the bounded working quality model. Section 3 positions AItist relative to contemporary artistic practice, AI-assisted ideation, and agentic refinement. Section 4 describes the AItist architecture, and Section 5 specifies the benchmark, controls, external evaluation, and inference procedure. Sections 6 and 7 report the main results and mechanism/failure analysis. Section 8 discusses implications for creative-agent design; Section 9 develops the intended artist-facing use and future human-negotiated search; Section 10 states limitations and threats to validity; and Section 11 concludes.

## 2 Representing Contemporary-Art Installation Ideation as a Computational Problem

AItist does not generate a physical installation. It generates a proposal for a future installation: a textual account of what would be installed, how its material and spatial organization would behave, how spectators would encounter it, what institutional conditions it would depend on, and why those choices belong to the artistic proposition rather than merely to its fabrication. The computational object in this paper is therefore not the realized artwork but a sufficiently developed textual dossier representing an installation hypothesis. We use *dossier* for a proposal developed enough to support further concept development and comparative judgment, not for a fabrication-ready specification. Further artistic, curatorial, engineering, conservation, and site-specific decisions remain expected.

The evaluated pipeline is autonomous after brief submission, but its intended artist-facing role is co-creative rather than substitutive. It develops candidate directions that an artist may inspect, reject, combine, reinterpret, or develop further. The method does not assume that the model captures the artist’s tacit judgment, broader practice, or all physical, material, and institutional contingencies of realization. A realized installation may therefore depart substantially from its dossier through scale, fabrication, material accident, maintenance, institutional negotiation, audience behavior, or later artistic decisions. Such departures are compatible with the intended workflow rather than failures to reproduce a fixed specification.

A separate difficulty concerns representation within the textual task. A language model can readily generate a title, describe a room and a set of objects, propose a material palette, and produce persuasive curatorial prose. None of these, separately or together, guarantees that the text specifies a coherent installation. Two dossiers may address the same topic and use almost the same materials while proposing fundamentally different works because bodies move differently through them, matter changes differently over time, access is withheld under different conditions, or the institution plays a different role. Conversely, works that look unrelated at the level of object vocabulary may depend on closely related temporal, material, spatial, perceptual, spectator, or institutional dependencies. For installation ideation, representing the components is therefore not enough: a small change in their relations can change what the proposed work does.

The study therefore makes a limited representational assumption: a textual dossier can describe a proposed installation in enough detail to support meaningful computational reasoning about the relations through which the work is intended to operate. “Enough detail” here has a practical meaning. The dossier should state what exists, what changes, what depends on what, what a spectator can perceive or do, which material properties matter, and which production or institutional conditions are necessary. It need not—and in practice cannot—determine every property of the eventual artwork.

For proposal development, we distinguish three analytical levels:

- **Artistic stakes:** What tension, contradiction, experience, or question makes the proposed work worth pursuing?
- **Artistic operation:** Through what specified dependency does the proposal enact those stakes rather than merely name them?

- **Realization:** What concrete materials, spatial arrangement, timing, technical system, and institutional conditions instantiate that operation here?

These are not claimed to be ontological components present in every artwork. These levels let the system distinguish why a work matters, what it does, and how it is concretely realized, so that a revision can change its realization without inadvertently removing the relation that gives the proposal its artistic identity.

The section develops this representation in two stages. First, installation theory and established artistic practice motivate why space, duration, material behavior, embodied spectatorship, site, and institutional context may be constitutive rather than incidental. Second, we translate those concerns into an explicit, bounded computational surrogate of proposal quality for one task: developing contemporary-art installation proposals under museum-commission conditions.

## 2.1 Installation as a Situated Configuration

A useful starting point is that installation is rarely exhausted by a self-contained object. Bishop foregrounds the spectator's bodily presence within the work; Petersen describes installation as activating physical space and context, extending the work in time, and emphasizing embodied perception; and Kwon traces a historical shift in site-specific art from site understood primarily as a fixed physical location toward institutional, social, and discursive contexts [1, 10, 2]. O'Doherty's analysis of the white cube makes a complementary point: gallery space actively frames the conditions under which something is encountered as art rather than serving as a transparent container [11]. Rebentisch treats site-specific installation as engaging the interrelation of literal and social sites [12]. These positions are not interchangeable, and AIRTIST does not combine them into a single definition of installation. Where these accounts broaden the significance of site, context, and embodied encounter, Morris and Burnham are useful for a more directly relational claim. Morris emphasizes that sculptural experience changes as the viewer changes position relative to the work, while Burnham's systems aesthetics explicitly shifts attention from isolated material entities to relations among people and environmental components [13, 14]. For the present computational problem, this literature supports a narrower representational premise: an installation may be constituted not only by the intrinsic properties of objects but by dependencies among material, space, duration, spectatorship, and institutional context.

We do not adopt these theoretical positions as a ready-made computational definition. Their relevance is practical. They warn against representing an installation as a list of objects plus a theme. If changing a route, a temporal condition, a spectator position, a material behavior, or an institutional rule can change what the work does, those elements must be available to the computational system as first-class parts of the proposal.

## 2.2 Artistic Operation

For AIRTIST, an artistic operation is a specified dependency through which a proposal enacts its artistic stakes rather than merely naming them. In plainer language, it answers: What actually happens because this installation has been arranged in this particular way? What changes, for whom or for what, and what follows from that change? The elements linked by the dependency should therefore be identifiable.

This gives the system a level of description between stakes and realization. A stake such as loss or displacement is too general by itself to specify what the work does, while candy, a camera, or a loudspeaker describes only part of how an operation is realized. The operation specifies the dependency through which such elements become consequential to what the work does: more specific than a theme, but less tied to the particular materials, dimensions, mechanisms, or site through which it is instantiated. The same stakes can be enacted through different operations, and similar operations can support different stakes.

A related art-theoretical precedent appears in Rancière's account of artistic images as operations constituted through relations rather than as manifestations of a technical medium alone. His examples locate these operations in relations among whole and parts, visibility and signification or affect, expectation and what occurs, and more generally in a "regime of relations between elements and between functions" [15]. The parallel is limited: AIRTIST uses a narrower computational abstraction in which a proposal-level dependency is distinguished analytically from both the stakes it enacts and the concrete realization that instantiates it.

Table 1 makes this level of abstraction concrete across ten heterogeneous works. It does not propose a taxonomy of installation or attribute AIRTIST's terminology to the artists. The physical-

encounter column summarizes documented features of each work in enough detail to make the relevant relation intelligible without prior familiarity with the work. In the artistic-operation column, the italicized phrase names our compact abstraction of the operation, and the sentence that follows explains how that dependency is instantiated in the specific work; the final column identifies the realization conditions on which it relies.

**Table 1:** Artistic operations and their realization across heterogeneous installations.

| Work                                                                                                | What a visitor physically encounters                                                                                                                                                                                                                                                                                                                                                                                                                                 | Artistic operation                                                                                                                                                                                                   | Realization conditions                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| James Turrell, <i>Frontal Passage</i> (1994) <a href="#">[view]</a>                                 | A visitor passes through a darkened entryway into a dark chamber. Concealed light sources produce a radiant, sharply bounded field of red light across the space. The apparent boundary is light, yet it seems to end abruptly in space as if it had density or formed a physical plane. [16]                                                                                                                                                                        | <i>Perceptual materialization of immaterial light.</i> Controlled darkness and concealed light make an immaterial field appear sharply bounded and almost physically present.                                        | Darkened entry and chamber; concealed light sources; a sharply controlled red-light field; a viewer positioned within the constructed viewing environment.                                                                      |
| Janet Cardiff, <i>The Forty Part Motet</i> (2001) <a href="#">[view]</a>                            | Forty separately recorded choir voices are played simultaneously through forty individual loudspeakers arranged in eight groups of five around a large oval. Moving close to one speaker isolates an individual singer; standing between speakers produces combinations of nearby voices; from the center, the listener can hear the forty voices together as a choir. [17]                                                                                          | <i>Position-dependent recomposition of a collective voice.</i> Movement through the speaker field changes the audible relation between individual singers, local combinations of voices, and the choir as a whole.   | Forty separately recorded vocal tracks; one loud-speaker per voice; spatial separation of the speakers; a navigable listening space in which position changes the heard mix.                                                    |
| Mel Bochner, <i>Measurement Room</i> (1969) <a href="#">[view]</a>                                  | The actual gallery room is measured, and strips of black tape and numerical notation mark the dimensions of the same walls, doorways, windows, and passageways that the visitor is standing among. The visitor therefore occupies a room whose physical dimensions are displayed directly on the architecture being measured. [18]                                                                                                                                   | <i>Space describing itself through measurement.</i> Measurements derived from the room are inscribed back onto the same architecture, so physical space is encountered together with its numerical description.      | The actual room; measurements derived from that room; notation applied directly to the architectural elements being measured; the viewer inhabiting the measured space.                                                         |
| Doris Salcedo, <i>Shibboleth</i> (2007) <a href="#">[view]</a>                                      | A long, irregular fissure opens through the floor along Tate Modern’s Turbine Hall. It varies in width and depth, and its concrete sides reveal steel mesh within the rupture. Rather than appearing as a separate object placed in the hall, the fissure occupies the floor across which visitors themselves move. [19]                                                                                                                                             | <i>Spatial rupture incorporated into bodily passage.</i> The fissure interrupts the same floor visitors traverse, making division a condition of bodily passage rather than a separately depicted object.            | The actual Turbine Hall floor; an extended fissure within that circulation surface; sufficient width, depth, and material presence for the rupture to be physically encountered; visitor movement through the same space.       |
| Hans Haacke, <i>Condensation Cube</i> (1963–67) <a href="#">[view]</a>                              | A transparent acrylic or methacrylate cube contains water in a closed container. Light warms the interior; water evaporates and condenses on the inner walls as droplets that grow, merge, and eventually run down the transparent surfaces. The patterns continually change with light and temperature in the surrounding exhibition space, and even nearby bodies can slightly alter those conditions. [20, 21]                                                    | <i>Environment-driven material transformation.</i> Changes in surrounding light and temperature continually alter condensation and therefore the visible state of the work.                                          | Transparent enclosure; contained water; a temperature relation between interior and surrounding space; light and local environmental conditions; duration for evaporation, condensation, droplet growth, and movement to occur. |
| Bruce Nauman, <i>Corridor Installation (Nick Wilder Installation)</i> (1970) <a href="#">[view]</a> | Six narrow corridors of different widths are constructed within the gallery; three can be entered and three cannot. In one cramped passage, stacked video monitors are visible at its far end. As a visitor moves through the installation, closed-circuit cameras produce views that differ conspicuously from the body’s direct experience of its position and movement. [22]                                                                                      | <i>Separation of bodily position from mediated self-perception.</i> The visitor moves through one constrained spatial position while encountering a live image of that movement from another viewpoint.              | Narrow, partially accessible corridors; constrained routes; closed-circuit camera coverage from different viewpoints; monitors positioned within the installation; a live relation between bodily movement and displayed image. |
| Felix Gonzalez-Torres, “Untitled” (Portrait of Ross in L.A.) (1991) <a href="#">[view]</a>          | Individually wrapped, variously colored candies are arranged in the exhibition space. The work specifies an ideal weight of 175 lb and an endless supply, while its dimensions and configuration may vary between manifestations. Visitors must be allowed to choose whether to take candies, so the quantity physically present can diminish; during an exhibition, authorized decisions determine whether and how the candies are replenished and maintained. [23] | <i>Visitor-driven depletion coupled to institutional replenishment.</i> Visitors physically diminish the manifested work, while institutional decisions determine whether and how its material quantity is restored. | Individually removable candies; permission for visitors to take them; a material quantity capable of changing through removal; an endless supply; authorized decisions about replenishment and maintenance.                     |

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Table 1 continued

| Work                                                                                                                | What a visitor physically encounters                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Artistic operation                                                                                                                                                                                                          | Realization conditions                                                                                                                                                                                                                            |
|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ann Hamilton, <i>the event of a thread</i> (2012) <a href="#">[view]</a>                                            | A field of swings fills Park Avenue Armory's Drill Hall. Through ropes and pulleys, the swings are connected to the same enormous white cloth suspended high across the hall. As different visitors swing, their separate movements combine through the shared rigging, causing the single cloth to rise, fall, ripple, and billow. Readings, sound, pigeons, writing, and other live elements also occupy the larger installation. [24]                                                  | <i>Aggregation of individual actions into shared material motion.</i> Separate visitors' movements are mechanically combined so that they become visible in the movement of one common cloth.                               | Visitor-accessible swings; a shared rope-and-pulley system; mechanical coupling of the swings to the same suspended cloth; sufficient suspension and travel for riders' movements to propagate into the fabric.                                   |
| Michael Asher, <i>Kunsthalle Bern, 1992</i> (1992) <a href="#">[view]</a>                                           | Visitors entering the Kunsthalle encounter radiators removed from the building's exhibition rooms and regrouped in the entrance gallery. The radiators remain functional: multiple steel pipes run from them through the building to their original heating connections. The entrance therefore contains a concentrated field of working radiators, while the exhibition rooms from which they were removed are left without their usual heating elements. [25, 26]                       | <i>Functional infrastructure made perceptible through spatial redistribution.</i> Relocating the functioning radiators changes where heating infrastructure and its thermal effects are encountered within the institution. | The Kunsthalle's actual radiators; their removal from the exhibition rooms; continued connection to the working heating system; concentration of the radiators in the entrance gallery; the resulting redistribution of heat across the building. |
| Precious Okoyomon, <i>Among the Flowers I Learned to Love: A Garden of Decreation</i> (2026) <a href="#">[view]</a> | MoMA PS1's outdoor courtyard is transformed into a living garden or forest containing grasses, flowers, weeds, sedges, vines, pollinators, sculpture, and sound. Native and invasive, annual and perennial, cultivated and wild species grow together. A giant teddy-bear sculpture at the center can be entered and climbed through. Installed from 2026 through 2028, the garden continues growing and changes with the seasons rather than remaining in one fixed material state. [27] | <i>Living growth as ongoing material composition.</i> Seasonal and environmental processes continually alter the plant assemblage and therefore the material state of the courtyard.                                        | Living mixed plant species; outdoor exposure to seasonal and environmental change; the site-specific courtyard landscape; sustained duration allowing growth and seasonal transformation to alter the installed state.                            |

This distinction also provides a practical preservation test for proposal development: which changes alter only the realization of an operation, and which remove or replace the operation itself?

In Cardiff's *The Forty Part Motet*, the particular room, loudspeaker models, and precise spacing could vary within limits so long as each recorded voice remained spatially individuated and a listener's movement continued to change which singers were locally foregrounded and how the choir recombined. Conversely, the same forty recordings and the same forty loudspeakers could remain while every loudspeaker reproduced the same premixed choir. Much of the sound material and even the visible arrangement would be retained, but the position-dependent operation would be gone [17]. Haacke's *Condensation Cube* provides a material counterpart: a transparent cube could still show changing droplets if a concealed programmed system produced them instead of evaporation and condensation responding to local environmental conditions; the visible motif could remain while the environmental dependency disappeared [20, 21]. Nauman's *Corridor Installation (Nick Wilder Installation)* provides a technical counterpart. Its cameras and monitors are realization components; the operation lies in coupling constrained bodily movement to a live view of that movement from a spatially displaced position. The same camera-monitor system could function as ordinary surveillance without producing that relation, while different imaging hardware could in principle preserve it [22]. Together, these counterfactuals do not test artwork identity; they distinguish changes in realization from changes to the dependency treated here as the operation.

Airtist deliberately imposes a stronger representational constraint than a descriptive theory of installation would require: the structured dossier contains a single `artistic_operation` field, and generation asks each candidate to identify one dominant operation. This is a search representation, not an ontological claim that a realized artwork possesses one objectively true operation. An installation may sustain several simultaneous or mutually irreducible relations, and different critical readings may foreground different ones. The computational requirement is narrower: during revision, Airtist needs a current dependency against which a proposed change can be understood as preserving, strengthening, or replacing the artistic logic under development. Without such a conservation target, a dossier can become more elaborate while the relation that made the proposal worth developing drifts or disappears. The default is therefore an economy guardrail rather than a ban on complexity: a proposal may contain many mechanisms, relations, and effects, but the representation asks which dependency currently

organizes them; complexity itself receives no bonus.

### 2.3 A Bounded Working Model of Proposal Quality

The argument so far concerns what an installation proposal needs to represent. It does not yet tell Airtist how to search for a stronger proposal. Installation theory motivates attention to situated space, material behavior, duration, spectatorship, and context. It does not imply Airtist’s ten computational dimensions, their decomposition, their use in ranking, or the precise search pressures encoded by the system. Those are methodological choices. Airtist therefore operates under an explicit bounded surrogate of installation-proposal quality. It is a computational and curatorial hypothesis about properties that make the class of museum-installation proposals studied here worth developing.

Computational-creativity research provides a methodological precedent for making evaluative commitments explicit. Ritchie proposes observable empirical criteria for judging the behavior of creative programs, while Jordanous’s SPECS procedure more directly begins by stating what creativity means for the particular system being evaluated and then deriving tests from that account [28, 29]. Following this methodological principle, Airtist operationalizes its bounded surrogate through ten equally weighted dimensions, without claiming that these dimensions exhaust artistic quality. In the evaluated configuration, each dimension is constrained to the interval  $[0, 1]$ , and their arithmetic mean forms the uniform aggregate used in quality-bearing comparison. This numeric structure is a search-and-ranking device, not a claim that artistic quality is naturally ten-dimensional or that exact semantic thresholds exist at 0 and 1. The dimensions function as bounded internal pressure tests and scoring constructs; proposal generators are explicitly instructed not to self-score or recite them as curatorial rhetoric. Table 2 gives qualitative low- and high-end anchors for all ten dimensions. The anchors summarize the direction of the implemented rubrics rather than defining deterministic numerical cutoffs.

**Table 2:** Qualitative low- and high-end anchors for the ten proposal-quality dimensions.

| Dimension                    | Low end                                                                                                                                                               | High end                                                                                                                                                                                                    |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Raw-prompt fidelity          | Answers a safer adjacent theme or loses the brief’s concrete pressure, stakes, explicit constraints, or other requirements.                                           | Preserves the brief-specific pressure, stakes, constraints, and requirements closely enough that the proposal remains a response to this problem rather than a generic version of its topic.                |
| Core-tension enactment       | States or illustrates the brief’s central tension in thesis-level prose or metaphor, while the installed form does little to make that tension happen.                | Makes the brief’s central tension structurally present in what the installation materially, spatially, perceptually, temporally, socially, or procedurally does.                                            |
| Artistic-operation necessity | Specifies no clear operation, or relies on a generic or readily interchangeable relation whose substitution would leave the proposal’s artistic logic largely intact. | Depends on an identifiable organizing operation whose removal or substitution would cause meaningful artistic loss.                                                                                         |
| Material necessity           | Uses material or realization mainly as a replaceable carrier; a generic alternative would leave the operation substantially unchanged.                                | Depends on specific properties, behavior, process, history, or institutional relations of the realization; variation is compatible with high necessity when the constitutive property or rule is preserved. |
| Spatial encounter            | Could be translated into a text, image, object, or generic display with little loss because spatial arrangement and duration do not materially alter the encounter.   | Requires a concrete spatial-temporal relation in which position, access, distance, sequence, duration, scale, or another situated condition changes what the work makes available.                          |

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Table 2 continued

| Dimension                          | Low end                                                                                                                                                                   | High end                                                                                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual encounter                   | Is perceptually generic, inert, or diagrammatic, or gains visual force mainly from detachable spectacle that is not necessary to the work.                                | Produces a memorable perceptual and spatial situation whose visual or material force follows from the installation's actual operation rather than replacing it.                              |
| Spectator specificity              | Leaves the spectator as a generic viewer or participant whose position, attention, action, duration, access, or knowledge makes little difference.                        | Places the spectator in a precise and consequential relation in which position, attention, action or non-action, access, duration, visibility, knowledge, or relation affects the encounter. |
| Field distinctiveness              | Repeats a familiar artistic format or trope with a new topic, material, style, or atmosphere while leaving its underlying operating logic largely unchanged.              | Makes a specific structural transformation of a recognizable artistic format or trope; absolute novelty is not required.                                                                     |
| Public-reception resilience        | The most available public reading reduces the work to a familiar generic signal or format and largely exhausts its more specific artistic tension.                        | Simplified public readings remain possible, but they do not exhaust the installed relation or eliminate the more specific artistic pressure it produces.                                     |
| Production/institutional viability | The central operation depends on major unacknowledged production or institutional requirements that make the proposal implausible under the stated commission conditions. | The central dependency can plausibly be developed within the stated institutional and production envelope, with ordinary later engineering or readiness work remaining addressable.          |

For recurring prompt-level guidance, AItist also uses a more compact five-principle quality contract covering brief-specific stakes, artistic-operation and form-concept necessity, visual-spatial identity, field-aware distinctiveness, and production/institutional viability. We use this coarser form as an engineering abstraction so shared generation and review prompts can carry common quality guidance without reproducing the full ten-dimensional rubric. It is not a second scoring model: AItist computes no five-dimensional aggregate, and the five principles are not a formal one-to-one partition of the ten dimensions.

Across the rubric, surface novelty, interactivity, technical sophistication, complexity, polish, and spectacle receive no independent bonus; they matter only insofar as they strengthen one of the dimensions below. The remainder of this subsection explains the ten dimensions and the distinctions among them.

### 2.3.1 Raw-prompt fidelity and core-tension enactment

These dimensions address a common language-model failure: producing a fluent work near the topic while escaping the actual problem. *Raw-prompt fidelity* asks whether the proposal preserves the concrete pressure, stakes, explicit constraints, and other requirements that make the brief specific. It is deliberately literal where the brief is literal: a proposal that uses cameras when cameras are prohibited should not recover merely because the resulting installation sounds sophisticated.

*Core-tension enactment* is brief-relative: it asks whether the central problem or tension posed by the brief becomes structurally present in what the installation does rather than merely appearing in its language, imagery, or curatorial explanation. A brief about unequal waiting, for example, can mention delay, clocks, and inequality while constructing little more than a visual metaphor for time. Enactment asks whether inequality in waiting changes the proposed situation—who waits, for how long, under what fixed condition, and what becomes perceptible because those durations differ. The two scores can therefore diverge: fidelity asks whether the proposal remains inside the problem the artist gave us; enactment asks whether that problem becomes structurally present in what the installation does. *Artistic-operation necessity*, introduced next, asks a different, proposal-relative question: whether the particular dependency

chosen to organize the work is itself constitutive rather than interchangeable.

### 2.3.2 Artistic-operation necessity and material necessity

*Artistic-operation necessity* first asks whether the proposal specifies a coherent, identifiable operation at all. Given such an operation, it asks whether that dependency is constitutive of the proposal rather than merely one interchangeable way of carrying the same artistic claim. A generic mechanism, decorative effect, or familiar metaphor can therefore score poorly even when it is described persuasively. At the high end, removing or substituting the dominant operation would cause meaningful artistic loss even if the topic, objects, or curatorial claim remained.

*Material necessity* asks whether the proposal relies on a consequential property of its chosen material or physical realization system—its behavior, process, history, or institutional role—rather than using that material as an interchangeable visual sign. Necessity does not mean that one exact substance or fabrication method is irreplaceable: a substitution can preserve material necessity if it preserves the property or dependency on which the operation relies. Conversely, retaining the same nominal material does not make it necessary if none of its specific properties are consequential to the operation.

### 2.3.3 Spatial encounter, visual encounter, and spectator specificity

These three dimensions deliberately separate aspects that generative prose often bundles together as “immersive,” “experiential,” or “engaging.” *Spatial encounter* asks whether the proposal establishes a necessary situation in space and time. Route and movement can matter, but spatial encounter is broader than circulation. A proposal may instead depend on a particular viewing position, a threshold or access condition, a controlled distance or visibility, a duration, or a site-bound material or atmospheric process. The question is what spatial-temporal arrangement must hold for the proposed encounter to occur.

*Visual encounter* asks whether the proposal produces a specific and memorable perceptual situation before explanatory prose is supplied. It is not an independent reward for beauty, spectacle, or photogenicity. The implemented rubric does, however, treat press-image potential as one bounded consideration: a work may be visually strong while resisting complete photographic translation, provided that resistance follows from a specific and necessary encounter rather than from visual indeterminacy. Turrell is a useful limiting case because the apparent solidity of red light is visually forceful while remaining dependent on controlled light, darkness, chamber, and viewing position rather than on accumulated spectacle [16].

*Spectator specificity* asks how the spectator is positioned in the logic of the work: through position, attention, movement, action or non-action, access, duration, visibility, knowledge, or consequence. Specificity therefore does not require participation. As Table 1 shows, the spectator may occupy a controlled viewing or listening position, move through a spatial relation, remove material, or causally alter a shared mechanism. What AIrtist rejects is the generic statement that “the viewer interacts” or “the viewer reflects” without specifying what that relation is materially, spatially, temporally, or institutionally contingent upon.

### 2.3.4 Field distinctiveness and public-reception resilience

*Field distinctiveness* is relational rather than a demand for novelty ex nihilo. Bourdieu’s account of cultural production treats artistic and literary positions as intelligible differentially, through their relations to other positions and position-takings in a field [30]. Krauss challenges the avant-garde ideal of absolute originality by showing how claims to a new beginning coexist with structures of repetition and recurrence [31]. Groys likewise treats newness as culturally relational: innovation revalues and recontextualizes what already exists in relation to an established cultural archive [32]. Bourriaud describes a historically more specific contemporary-art tendency in which artists work from pre-existing forms and produce singularity by reprogramming what is already available rather than beginning from a blank slate [33]. For AIrtist, the useful premise is narrower: artistic distinctiveness need not require an unprecedented object, medium, or formal vocabulary; it can arise through a consequential transformation of an available artistic language, format, relation, or convention.

AIrtist therefore asks whether a proposal merely re-skins a recognizable artistic format or changes the relations through which that format operates. Gonzalez-Torres provides a concrete illustration within an established artistic language. Institutional accounts explicitly situate his candy works in relation to Minimalism, while emphasizing how visitor removal, changing quantity, and replaceability transform the assumptions of a stable, autonomous Minimalist

object [34, 23]. The point is not that such transformation defines distinctiveness universally, but that difference can be located in what an inherited artistic form is made to do rather than in whether its component objects or materials have never appeared in art before.

There is an important epistemic limit to this criterion. AIrtist does not possess a complete map of contemporary art. Its field-context stage uses model-mediated recall of works, artists, formats, criticism, and neighboring contexts, not provenance-bearing bibliographic retrieval. Field evidence is therefore a working hypothesis about nearby artistic lines, useful for directing search but insufficient to establish an art-historical claim that a proposal is objectively unprecedented or distinct. In this paper, field distinctiveness means field-aware differentiation under the system's recalled context, not measured global novelty in contemporary art.

*Public-reception resilience* is an AIrtist-specific proposal-level diagnostic rather than a construct derived from a general theory of audience reception. It asks whether a readily available generic framing captures almost everything the proposal actually does, or whether the installed relation retains consequential specificity that such a framing leaves unexplained. The criterion neither predicts how real audiences will interpret the work nor rewards opacity, difficulty, or resistance to popular reception for their own sake.

The distinction is visible in familiar surface descriptions. Gonzalez-Torres's work can be reduced to "a pile of candy," yet that phrase leaves out the relation among removable material, changing quantity, spectator choice, and institutional maintenance [23]. Hamilton's *the event of a thread* can be reduced to "a room of swings," yet that description omits the shared mechanical coupling through which distributed riders collectively move a common cloth [24].

### 2.3.5 Production and institutional viability

AIrtist evaluates *production/institutional viability* because its computational object is a commission proposal for a future work. In the evaluated configuration, the criterion is judged relative to the stated museum-production context, including the approximate USD 150,000 project envelope described in Sections 4 and 5. This should not be confused with the proposition that feasible art is better art. Deliberate impossibility, waste, fragility, institutional conflict, or excessive labor can be artistically meaningful; they can nevertheless lower this particular score when the proposed realization is evidently impossible, radically outside the commission envelope, or dependent on an unacknowledged production contradiction. Ordinary engineering work that remains plausibly addressable is not itself a failure. Viability therefore concerns the whole realization dependency rather than fabrication cost alone: if the artistic operation requires a maintenance rule, access to institutional infrastructure, permission, sustained labor, or a living system, the dossier should identify that dependency and make plausible how it can be supported under the stated commission conditions. Artistically intentional difficulty can remain part of the proposition.

### 2.3.6 Ethical and public-context diagnostics

AIrtist does not define an eleventh ethics or public-context quality dimension. In the evaluated configuration, such concerns enter through the existing scoped mechanisms when they are relevant: Public Reception can diagnose reductive public framing; Production/Institutional review can identify consent, privacy, access, staffing, liability, or safety consequences; reviewers can pass cross-scope concerns through `scope_handoff_notes`; and explicit-constraint or severe-safety violations can affect eligibility. These routes keep responsibility-relevant problems visible without assigning them a separate scalar ethics score.

This separation should not be read as a claim that ethical responsibility is irrelevant to artistic practice, nor as an attempt to convert ethical virtue into an additional aesthetic score. Bishop argues directly that social and artistic judgments do not easily merge and may require different criteria, while Kester's dialogical aesthetics treats the ethical quality of participant relations—including reciprocal listening and openness to other participants—as constitutive of the artistic value of dialogical work [35, 36]. AIrtist adopts neither account as a scoring theory; their relevance here is narrower. They motivate keeping responsibility concerns explicit while avoiding the claim that they are automatically commensurable with the ten-dimensional artistic-quality vector.

## 2.4 Why Proposal Improvement Is Non-Monotonic

AIrtist is explicitly designed to improve proposals. The difficulty is not that artistic development lacks an evaluative objective altogether: in the experiments below, we can ask whether resulting proposals receive stronger independent holistic judgments. The difficulty is that open-ended

artistic development offers no reliable local gradient toward that objective. A valid diagnosis does not uniquely determine a valid repair. A critique can correctly identify a weakness while the most direct correction removes another relation on which the proposal’s value depends. This makes local and holistic improvement non-equivalent. A revision can make fabrication easier while weakening the necessity of the artistic operation; make the intended meaning more explicit while eliminating an ambiguity that carried interpretive tension; add interaction while weakening a precise spectator relation; or strengthen the image while disconnecting it from the material or spatial logic that gave it meaning. In this sense, proposal improvement is non-monotonic: a locally desirable correction can reduce holistic quality. Review should therefore identify liabilities and candidate interventions without treating the obvious local fix as globally beneficial. In AItist, review evidence is converted into a local search hypothesis rather than an automatic rewrite: challengers are generated from the unchanged incumbent, and the incumbent remains eligible to survive if no challenger earns replacement.

One recurring case arises when a revision tries to clarify an apparently under-specified artistic relation. Eco’s *Open Work* offers an influential account of artistic structures that permit multiple realizations or interpretations without collapsing into arbitrariness, while Gaver, Beaver, and Benford argue in design that ambiguity can function as a resource for interpretation rather than merely as missing information [37, 38]. The point for AItist is not that ambiguity is inherently valuable: incoherence and underspecification remain genuine weaknesses. An unresolved relation can nevertheless be productive when it structures what a spectator can perceive or infer. Making the intended meaning more explicit may remove accidental noise, but it can also eliminate that relation.

Second, engagement can conflict with spectator specificity. Bishop and Ranciere give different reasons to reject a simple hierarchy in which converting spectators into visible actors automatically constitutes artistic progress [35, 39]. A proposal may indeed be weak because the spectator relation is generic or inert, but adding visitor input does not by itself repair that weakness. It can increase observable activity while replacing a precise relation of attention, distance, delay, access, or non-action with generic interaction. The relevant question is whether the intervention strengthens the proposal’s artistic operation and spectator relation, not whether the spectator is made to do more.

## 2.5 Maintaining a Sampled Possibility Set

The same uncertainty has a second consequence at the level of the search population. If a proposed repair is not guaranteed to improve its lineage, and different lineages can preserve different strengths, ideation should not collapse immediately to a single answer. Computational-creativity research describes creativity in terms of exploring and transforming spaces of possibilities rather than producing one isolated artifact [40, 41, 42]. Large language models make the tension especially visible. Generating five fluent proposals does not guarantee five serious directions. A model can repeatedly vary the same metaphor, object type, interaction format, or rhetorical framing while changing vocabulary. For the intended use case, AItist therefore keeps five proposal lineages alive and returns all five surviving dossiers to the artist. We call this finite population the sampled possibility set. The phrase is deliberately modest. Five proposals are not the possibility space of contemporary art, do not exhaust the generative model, and are not claimed to be the optimal search width. They are the alternatives the system has actually constructed and can therefore expose to a user and evaluate.

Keeping several lineages also separates two questions that a winner-only view would conflate: how strong is the retained set, and how well does the terminal ranker identify its strongest member? A system can construct a strong set yet rank it poorly, while retaining several incompatible serious directions also leaves those alternatives available for later artist selection.

Maintaining several candidates should not itself be read as evidence of artistic diversity. Five proposals may still cluster around the same operation, material strategy, or format, while a broader set may nevertheless be weak. The present experiment therefore evaluates sampled-set quality separately from breadth and does not infer diversity from branch count or from the existence of five outputs.

## 3 Related Work and Positioning

The term concept denotes different objects across the literatures adjacent to this work. In image generation, it can mean an object category or semantic attribute; in professional concept design, it often denotes a visual direction for a character, environment, or product. Here, concept

development refers to the formation of the artistic proposition of a work: what the installation does, and how its material, spatial, spectator, historical, and institutional relations constitute that proposition. This distinction is important because recent AI systems already support sophisticated forms of ideation and creative generation, but usually operate on a different creative object.

We review three relevant lines of work. Contemporary-art practice shows generative systems moving upstream from artifact production toward artistic dialogue and the conception of possible works. Research on AI-assisted ideation establishes structured exploration and iterative development as computational problems in adjacent domains. Agentic systems further provide mature mechanisms for critique, branching, and selection. Together, these literatures motivate the task studied here while leaving the early development of a contemporary-artwork proposition largely unformalized as an evaluated computational problem.

### 3.1 AI in Contemporary-Art Practice

Contemporary-art practice has already begun to move generative AI upstream from the production of finished images into the formation of possible artworks. Agnieszka Kurant’s *Errorism* (2021) provides a particularly direct precedent. GPT-2 and GPT-3 were trained on descriptions of Kurant’s previous works and her essays and used to generate descriptions of conceptual works that she had never made but could potentially create; Kurant subsequently visualized this speculative material as holographic animations [3]. The significance of *Errorism* for the present work is not simply that an artist used a language model. The model operates at a moment when a future artwork is still being articulated: it extrapolates possible works from an existing artistic practice rather than merely rendering a specification supplied by the artist.

Other recent practices extend this relation between generative systems and artistic conception. In Alexander Reben’s *I’m Not A Painter* (2022), a custom neural network and GPT-3 generate descriptions of nonexistent artworks; Reben curates these outputs, passes selected descriptions through text-to-image systems such as DALL-E 2, and selects images for physical realization [43]. Jonas Lund describes the works in *In the Middle of Nowhere* (2023) as having been conceptualized and produced through extended conversations with ChatGPT [4]. Such projects make language models part of a process of proposing, negotiating, and selecting possible works, rather than using them solely as image-making tools.

By 2024, related practices were also appearing as persistent, artist-specific systems. Cai Guo-Qiang’s *cAI* is a custom multimodal model that learns from Cai’s artworks, archives, and interests and is positioned by his studio as a partner for dialogue and collaboration [5]. *cAI* participated in the conception and choreography of the Getty-commissioned *WE ARE* (2024) [6]. Anicka Yi’s *Emptiness* project similarly uses an algorithm trained on years of her studio’s output as a “digital twin” of her practice [44]. Holly Herndon and Mat Dryhurst’s *The Call* (2024–25) makes another aspect of this shift explicit by treating the construction of vocal datasets, governance frameworks, and choral AI models as part of artmaking itself [45].

These practices make the motivating use case for *AIrtist* substantially less speculative. Established artists are already allowing generative systems to propose possible works, participate in artistic direction, and develop persistent computational counterparts to their practices. At the same time, these are artist-specific artistic procedures. Their systems are embedded in a particular oeuvre, studio process, exhibition, or authored experiment; they are not intended to accept an unfamiliar artistic problem and systematically develop several responses to it.

Recent empirical studies of artists working with generative AI reinforce this distinction. A preregistered experiment with professional visual artists shows that artistic expertise remains consequential when working with text-to-image AI [46], while qualitative work with contemporary artists characterizes human–AI co-creativity as a sustained, relational process shaped by artistic intuition, control, and adaptation to a system’s affordances [47]. Such findings motivate treating *AIrtist* as one bounded computational component of a larger artistic process: the artist supplies the problem and remains responsible for subsequent artistic development, while the present system investigates what computational support can contribute between those stages.

A small number of computational-art projects have also begun to place LLM agents inside contemporary-art discourse. *Artism*, for example, combines a simulated social network of virtual artist agents with a system that recombines fragments of contemporary-art discourse into synthetic movements, visualizations, and criticism. Its purpose is primarily the simulation and critique of patterned artistic production rather than development of an artist-authored project, and the reported evidence is a preliminary case study rather than a controlled test of artwork concept development [48].

### 3.2 AI-Assisted Ideation and Design-Space Exploration

Early-stage ideation is considerably more mature as a computational problem in adjacent creative and professional domains. AIdeation, for example, supports professional concept designers in film, television, and games through brainstorming, reference exploration, and recombination. Its development is grounded in a formative study with 12 professional designers and evaluated through a controlled study with 16 professionals and a subsequent deployment in four studios [7]. The work provides strong evidence that generative systems can usefully intervene before a visual design has stabilized rather than merely execute a completed specification. CoExploreDS, in turn, addresses a related property of ill-defined design problems: problem formulation and solution development can co-evolve. The system represents emerging problems and candidate solutions explicitly and uses multiple reasoning modes to propose further directions [49]. Its relevance to AIRTIST lies in establishing structured exploration of an evolving creative state as a serious design problem in its own right.

Research ideation offers an autonomous analogue. ResearchAgent decomposes scientific ideation into literature-grounded problem identification, method and experiment generation, review, and iterative revision [8]. The AI co-scientist similarly maintains and develops multiple hypotheses through generation, reflection, ranking, evolution, and meta-review [9]. Si et al., comparing LLM-generated research ideas with ideas written and blindly reviewed by more than one hundred NLP researchers, further show that idea generation and idea evaluation need not improve together: their agent produces ideas judged more novel on average while exposing failures in self-evaluation and a lack of diversity [50].

These systems establish that AI-assisted ideation, structured exploration, multi-candidate development, and iterative criticism are not new in themselves. The distinction in AIRTIST concerns the state being developed. Visual concept design ultimately produces production assets; product design converges toward functional solutions; research ideation produces hypotheses and experimental plans. In contemporary-art concept development, changing a material relation, spatial arrangement, spectator position, or institutional procedure can change the identity of the proposed work itself. This motivates the domain representation introduced in Section 2 rather than treating the task as generic brainstorming with an art-themed prompt.

A complementary body of work also cautions that generative assistance does not necessarily broaden exploration. Controlled studies report stronger fixation under exposure to generated visual examples [51], greater semantic similarity across LLM-assisted ideators [52], and improvements to individual outputs accompanied by reductions in collective diversity [53]. Recent work on multi-agent ideation likewise shows that denser interaction and larger groups can produce diminishing diversity returns and premature convergence rather than broader exploration [54]. These findings make the existence of multiple candidates a poor proxy for the quality or breadth of the resulting possibility field.

### 3.3 Agentic Revision and Selection

Iterative critique, role specialization, and selection are established components of current agentic systems. ResearchAgent uses collaborative reviewing agents before revising scientific ideas [8], while the AI co-scientist combines reflection and evolutionary development with tournament-based ranking among hypotheses [9]. More general inference-time methods such as Self-Refine and Tree of Thoughts replace a single completion with iterative feedback or exploration over multiple candidate reasoning states [55, 56].

Role-specialized refinement also appears in creative-generation systems. CREA, for example, coordinates Creative Director, Prompt Architect, Art Critic, and Refinement Strategist agents for creative image editing and generation. Its task remains explicitly image-level—the system transforms or generates objects such as couches, cars, and other visual categories—and its critic evaluates rendered images against generic creativity dimensions before further regeneration [57]. CREA therefore demonstrates the maturity of multi-agent creative refinement as a mechanism, rather than providing a precedent for the development of an artwork proposition.

AIRTIST builds on the general premise that additional inference can be allocated to critique and alternative generation, but changes the status of revision. In canonical iterative-refinement methods such as Self-Refine, feedback is used to produce a revised successor [55]; ResearchAgent likewise feeds reviewer feedback into iterative revision [8]. For the task studied here, identifying a weakness does not imply that the available repair is preferable to the existing proposal. Review is therefore first converted into an asymmetric local state: what gives the incumbent its strongest identity, what most needs intervention, which familiar operating logic should be left, and what

forms of repair should not count as progress. Focused and Structural challengers are generated independently from the same parent, while the unchanged incumbent remains eligible. This yields a different empirical question from ordinary endpoint refinement: when the system actually elects to transform an artistic proposal, is the adopted intervention preferable to the exact state it was intended to improve? Exact-parent comparison makes that question directly observable.

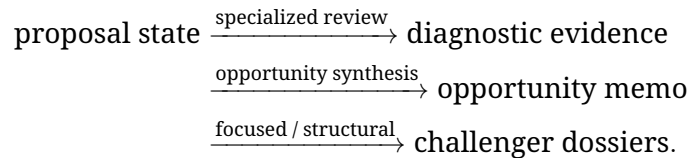
### 3.4 Positioning AItist

The preceding literature leaves a specific task gap. Contemporary-art practice already demonstrates language models and other generative systems participating upstream of final realization, including the proposal of hypothetical artworks and ongoing artist-specific computational collaborators. Adjacent computational research already provides sophisticated methods for ideation, structured exploration, critique, revision, and ranking. Yet across the systems and documented artistic practices reviewed here, we did not identify an evaluated general-purpose system designed specifically to take a previously unseen, under-specified artist-authored contemporary-art installation brief and develop multiple project hypotheses for what the artwork itself could become. AItist targets this gap: its outputs are not image concepts or thematic suggestions, but project hypotheses in which an artistic operation is articulated together with its material, spatial, spectator, field, and institutional realization.

## 4 AItist: An Agentic Architecture for Installation Concept Development

AItist takes a single artist-authored brief and develops five installation dossiers in parallel. After one generic self-refinement pass, the five proposals become incumbents of persistent proposal lineages. Rather than rewriting each proposal in place, AItist reviews the incumbent, compresses diagnostic evidence into a local search hypothesis, generates two challengers independently from the unchanged parent, and allows the incumbent to survive if neither challenger earns replacement. One state survives each lineage; only then does a separate terminal ranker order the five survivors. In the evaluated configuration, the artist supplies no intermediate feedback: proposal construction, review, branching, competition, and ranking all take place internally after the brief is submitted. The final output is a ranked set of five developed dossiers intended as starting points for subsequent artistic development rather than finished artworks.

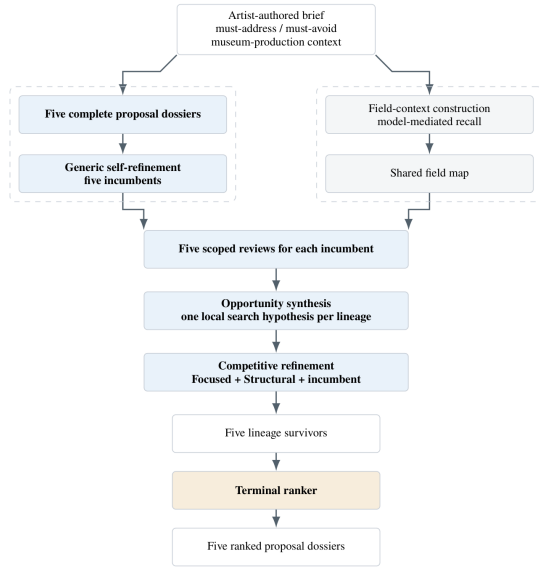
The architecture therefore separates proposal state, diagnostic evidence, and local search state. For each lineage, opportunity synthesis records the local search hypothesis in a concrete opportunity memo that is passed to both challenger generators:



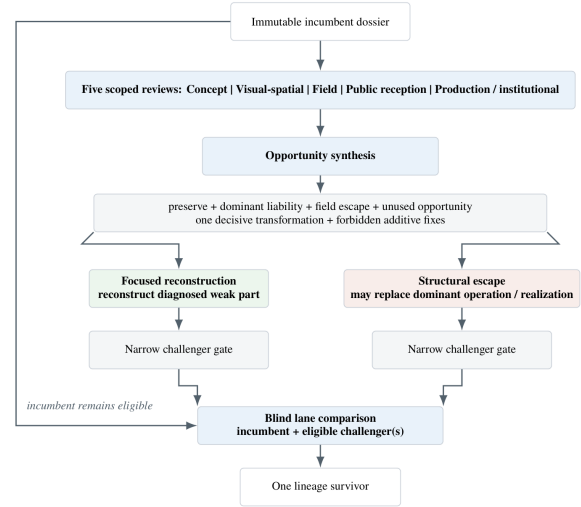
A review is not itself a revision instruction. An opportunity memo is not a revised proposal. And a generated challenger does not automatically replace its parent.

Figure 2 summarizes the end-to-end architecture. Proposal construction and field-context construction begin as separate streams. They first meet at specialized review and opportunity synthesis. Competitive refinement then proceeds independently across five lineages before a separate terminal ranking stage orders the five surviving dossiers. Table 3 makes the execution scope explicit at each interface. “Principal output” denotes the object passed forward conceptually rather than every piece of provenance or runtime metadata.

(a) End-to-end system



(b) One lineage expanded



**Figure 2:** Airtist architecture. (a) End-to-end flow from one artist-authored brief to five ranked proposal dossiers. Proposal-state construction and field-context construction are initially separate. The schematic shows the reviewer-facing field map; opportunity synthesis additionally receives field insights, and together these form the shared field context described in Section 4.2. (b) One lineage expanded. Reviews produce diagnostic evidence; opportunity synthesis compresses it into one local search hypothesis; Focused and Structural challengers are independently generated from the unchanged parent. Only challengers are gated. The incumbent bypasses the gate, remains eligible, and competes with all eligible challengers in the blind lane comparison. The terminal ranker is a separate stage over the five lineage survivors.

**Table 3:** Stage interfaces and execution scope in Airtist.

| Stage                         | Scope                                       | Principal input                                                | Principal output                                                      |
|-------------------------------|---------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------|
| Initial proposal construction | Once per brief                              | Brief contract + shared quality/production guidance            | Five complete installation dossiers                                   |
| Generic self-refinement       | Once per brief (batched)                    | Five initial dossiers + brief/constraints                      | Five self-refined incumbents (5→5)                                    |
| Field recall + synthesis      | Once per brief                              | Brief contract                                                 | One shared field context for all five lineages                        |
| Specialized review            | Five reviewer scopes across five incumbents | Incumbents + brief + field map + reviewer scope                | Five packets per lineage; 25 packets total                            |
| Opportunity synthesis         | Once per lineage (×5)                       | One incumbent + its five review packets + shared field context | One opportunity memo per lineage                                      |
| Challenger generation         | Twice per lineage (×10)                     | Same incumbent + same memo + brief/constraints + mode          | Two complete challenger dossiers per lineage                          |
| Challenger eligibility        | Once per challenger (×10)                   | Incumbent + challenger + brief/constraints                     | One eligibility decision per challenger                               |
| Lane comparison               | Once per lineage (×5)                       | Incumbent + zero, one, or two eligible challengers             | One survivor per lineage; five total                                  |
| Terminal ranking              | Once per brief over five survivors          | Five survivors + brief/constraints                             | Pointwise screen, finalist comparison, and total ordering of all five |

## 4.1 Input and Persistent Proposal State

### 4.1.1 Artist brief and commissioning context

The input to Airtist is a partially specified artistic problem rather than a request for a finished artwork. The intended user is an artist who has a rough direction, knows some conditions that must remain active, and may also know solution families or mechanisms that should explicitly be avoided, but does not yet know what form the installation should take. Airtist treats three elements as authoritative:

- **brief\_text:** the free-text artistic problem or direction;
- **must\_address:** explicit conditions that a valid proposal must continue to address;
- **must\_avoid:** explicit solution families or mechanisms that a valid proposal must not use.

The distinction is between preserving pressures and prohibiting solutions, not between different

classes of artistic problem. A `must_address` condition keeps a specified tension or relation active rather than allowing the system to substitute an easier neighboring theme. A `must_avoid` condition removes an inadmissible route without prescribing what should replace it.

Brief 035 provides a concrete example. Its free-text brief asks for a work about public shame that avoids mirrors, surveillance imagery, exposed bodies, and moral theater while giving passersby a precise relation to what is withheld. The corresponding `must_address` condition is “public shame through a precise passerby relation and withholding,” while `must_avoid` prohibits mirrors, cameras, humiliation performance, and moralizing text. Thus the brief preserves the central artistic pressure and rules out several familiar mechanisms without specifying a replacement medium or form.

The evaluated task assumes a temporary or medium-duration contemporary-art museum commission. Proposals are developed for museum-controlled settings such as galleries, foyers, thresholds, circulation areas, stairwells, entrance zones, or service-adjacent spaces. The shared production context assumes an approximate USD 150,000 project envelope covering project-specific fabrication, materials and equipment, specialist engineering and temporary technical labor, installation and deinstallation, safety measures, maintenance requirements, project-specific staffing, and contingency. As Section 2 emphasized, this envelope is a reasoning context for a textual commission hypothesis, not evidence that engineering or fabrication has been validated.

Together, the brief contract and shared commissioning context define the search problem: what artistic pressure must remain active, which solutions are inadmissible, and within what institutional conditions the proposal must function. They do not prescribe a medium, object type, interaction model, or artistic style.

#### 4.1.2 The installation dossier as persistent authored state

The object manipulated by AIrtist is a structured installation dossier: the persistent authored state of one proposal lineage. Reviews, field evidence, and opportunity memos are derived from or attached to this state; they do not silently rewrite it. Each challenger is generated from an immutable copy of the incumbent dossier, and a selected challenger becomes the next authored state in that lineage with an explicit parent relation. Table 4 lists the 11 authored fields that constitute the dossier.

**Table 4:** Exact fields of the installation dossier.

| Dossier field                          | What it represents                                                                                                                                                                            |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>title</code>                     | A working identifier for the proposal. It may contribute to framing but is not itself treated as an artistic operation or quality dimension.                                                  |
| <code>artistic_thesis</code>           | What the work proposes, tests, places under pressure, or asks us to encounter. In Section 2 terminology, the dossier’s concise articulation of artistic stakes.                               |
| <code>artistic_operation</code>        | The dominant action or dependency through which the installation enacts the thesis rather than merely stating it.                                                                             |
| <code>concept_description</code>       | An integrative description connecting thesis, operation, visual/spatial/material realization, spectator relation, and institutional situation.                                                |
| <code>visual_logic</code>              | What is concretely visible or perceptible: dominant image or material presence, scale, silhouette, surface, light, color, texture, and viewing conditions.                                    |
| <code>spatial_logic</code>             | How route, position, access, distance, sequence, threshold, orientation, concealment, visibility, waiting, and bodily movement organize the installation.                                     |
| <code>material_logic</code>            | What materials, media, processes, absences, or realization rules physically, temporally, historically, socially, or institutionally do; which properties are constitutive and which may vary. |
| <code>spectator_encounter</code>       | The spectator’s position, attention, movement, action or non-action, access, duration, visibility, knowledge, and consequence within the installation.                                        |
| <code>difference_from_field</code>     | The proposal’s own hypothesis about how it differs from nearby artistic formats or precedents; not independent evidence of art-historical novelty.                                            |
| <code>risks_and_failure_modes</code>   | The proposal’s own account of how the work could become generic, decorative, gimmicky, incoherent, didactic, technically fragile, or otherwise lose artistic pressure.                        |
| <code>institutional_feasibility</code> | Site, staffing, maintenance, accessibility, safety, permissions, liability, equipment, construction, and production dependencies required for the commission hypothesis to operate.           |

Three distinctions are particularly important downstream. `difference_from_field` is the candidate’s own positioning hypothesis and should not be confused with the independently constructed field context described in Section 4.2. `risks_and_failure_modes` is the proposal’s own anticipation of failure, not evidence produced by the later specialized reviewers. `institutional_feasibility` is likewise the candidate’s realization hypothesis; it is later pressure-tested by the separate Production/Institutional reviewer.

Section 2 defines the conceptual role of artistic operation and the relationship among stakes, operation, and realization. The dossier keeps these aspects explicit enough for later agents to diagnose them separately. A compelling thesis can coexist with arbitrary spatial logic; a visually

specific proposal can use interchangeable material; a materially interesting object can establish no precise spectator relation. Keeping the authored fields separate allows the system to ask what should remain invariant and what can be reconstructed.

## **4.2 Initializing Search: Five Proposal Lineages and a Shared Field Context**

AIrtist prepares two inputs for specialized development: five self-refined proposal incumbents and a shared hypothesis about the artistic field surrounding the brief. The two are constructed separately and meet only once specialized development begins.

AIrtist deliberately separates initial proposal formation from explicit field contextualization. The five initial proposals and their generic self-refinement receive the brief, explicit constraints, and general installation-quality guidance, but not the separately synthesized field map. This gives field context a contrastive rather than generative role. Once a concrete incumbent exists, the system can ask which familiar artistic logic it reproduces, which nearby convention it leans on, and where a meaningful departure from that specific proposal might lie.

The ordering also limits the authority granted to model-recalled field knowledge. AIrtist's field map is a partial and contestable hypothesis about the surrounding artistic field, not a complete account of contemporary art. If that map conditioned proposal generation from the outset, its omissions and framing choices could pre-structure the entire initial possibility set. AIrtist instead allows several directions to form first and then uses field context to pressure, differentiate, or redirect them. In this respect, field grounding plays a different epistemic role from prior literature in scientific ideation: scientific prior work can provide factual evidence and novelty constraints that appropriately precede hypothesis generation, whereas AIrtist treats its reconstructed artistic field as critical context to be applied to proposals rather than as the space from which proposals must be derived.

This ordering also creates a common pre-development foundation for the main control comparison: A5 and M5 begin from the same five self-refined parents. They diverge only afterward, when A5 applies the full structured-development package—including explicit field grounding, specialized review, opportunity synthesis, and differentiated challenger roles—while M5 develops the same parents through matched generic self-refinement without the synthesized field context.

### **4.2.1 Five proposal lineages**

The first stream produces five complete installation dossiers in one batched generation. Each should constitute a distinct proposal rather than a continuation or paraphrase of another. Each dossier then undergoes one generic self-refinement pass. The pass may revise any authored dossier field while preserving the underlying proposal and strengthening brief fidelity, operation-realization coherence, spatial and material specificity, spectator encounter, and plausible museum realization. Initial proposal construction and this generic self-refinement already operate under installation-specific quality guidance. They do not yet receive the shared field context or specialized review packets introduced below.

All five self-refined proposals are retained. They become five incumbents, one per persistent lineage. AIrtist therefore does not select a single concept and recursively improve it. It begins specialized search with five complete hypotheses whose histories remain independent.

### **4.2.2 Field-context construction**

In parallel, AIrtist constructs contextual evidence intended to help later stages recognize familiar artistic solution patterns and possible routes away from them. This stage is model-mediated recall, not provenance-bearing retrieval: it does not search a verified external bibliography. Recalled artworks, artists, art writing, and neighboring material outside art serve as associative evidence for search and must not be interpreted as proof that a proposal is globally novel in contemporary art. The recall process covers four complementary source types: concrete artworks; artists and collectives; art writing such as criticism, catalogues, interviews, and exhibition discourse; and relevant material outside art, including cultural, historical, spatial, legal, technical, or social contexts. In the evaluated configuration these are soft-targeted at about 7, 4, 4, and 3 items respectively, with hard caps of 10 artworks, 10 artists or collectives, 8 art-writing items, and 8 outside-art resources. Relevance is deliberately broader than topical similarity: a reference may matter because of material behavior, image logic, object presence, atmosphere, spectator position, temporal structure, institutional relation, bodily scale, sound, stillness, fiction, or another transferable pressure.

The raw recalled material is heterogeneous and often redundant. Passing it unchanged to every downstream agent would lengthen prompts and require each agent to reconstruct the same distinctions independently. Airtist therefore inserts a synthesis step that compresses the evidence into a shared field context: a four-part field map plus a separate set of evidence-linked field insights. For concreteness, the examples below are shortened from the field context generated for Brief 035; they illustrate the type of cue rather than verified art-historical claims.

- `nearby_lines`: evidence-grounded labels for nearby artistic lines, recurring problem-formats, or contextual practices against which a proposal may be positioned; e.g., “bureaucratic waiting-room interventions.”
- `overused_formats`: familiar formats whose default use risks becoming generic or substitutable, without making the format itself forbidden; e.g., “literal waiting lines as a simple critique of bureaucracy.”
- `productive_differences`: tensions or axes that can open a meaningfully different search direction rather than naming a ready-made medium or installation idea; e.g., “auditory presence vs visual absence.”
- `taboo_zones`: artistic, ethical, institutional, or affective danger zones that should be recognized but do not become new hard constraints; e.g., “turning shame into spectacle through voyeuristic exposure.”
- `field_insights`: evidence-linked lessons about what the recalled field may teach this project. Each insight records a claim, the evidence-item IDs supporting it, a project-level implication, and optionally a caution. For Brief 035, for example, an insight linked to a recalled item about Lawrence Abu Hamdan suggests that withholding visual cues while modulating sound may create the pressure of being heard but unseen, with a caution against collapsing that relation into literal surveillance.

The first four lists constitute the field map; `field_insights` is a separate synthesis output. Together they form the shared field context constructed once per brief. Specialized reviewers receive a compact field-map representation relevant to their diagnostic work, while opportunity synthesis receives the field map together with the field insights. These objects remain search hypotheses: `overused_formats` does not prohibit a medium, `taboo_zones` does not add new `must_avoid` constraints, and `field_insights` does not certify art-historical facts. Evidence IDs provide traceability to the model-recalled evidence pack, not verified art-historical provenance.

### 4.3 Specialized Multi-Agent Review: Diagnosis as Evidence

An installation dossier can fail in several independent ways. A compelling proposition may remain spatially arbitrary. A precise spatial situation may rely on an interchangeable material. An otherwise coherent work may reproduce a familiar artistic format. Its most available public reading may flatten it into a generic signal or familiar format. Or its operation may depend on hidden staffing, inaccessible circulation, or an unsafe physical mechanism. Airtist therefore reviews each incumbent through five deliberately scoped diagnostic agents, summarized in Table 5. The split is not intended to simulate a curatorial committee voting on artistic value. It decomposes Airtist’s ten-dimensional internal quality surrogate from Section 2 into non-overlapping diagnostic ownership.

**Table 5:** Review agents and ownership of the internal surrogate.

| Reviewer                 | Owned internal dimensions                                                                     | Primary diagnostic question                                                                                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Concept                  | raw-prompt fidelity; core-tension enactment; artistic-operation necessity; material necessity | Does the proposal enact the brief through one necessary dominant operation, or could its form be replaced without substantial artistic loss?                                              |
| Visual-Spatial           | spatial encounter; visual encounter; spectator specificity                                    | What does a spectator concretely see, occupy, cross, wait for, avoid, or experience, and how do position and duration change that encounter?                                              |
| Field                    | field distinctiveness                                                                         | Under the model’s recalled field context, does the proposal transform a nearby artistic logic or merely reproduce an available format?                                                    |
| Public-Reception         | public-reception resilience                                                                   | What readily available public reading could flatten the proposal into a generic signal or familiar format and exhaust its more specific artistic tension?                                 |
| Production/Institutional | production/institutional viability                                                            | Can the stated artistic relation plausibly operate with respect to fabrication, safety, accessibility, maintenance, staffing, permissions, liability, and the shared production envelope? |

### 4.3.1 Review packet interface

For each incumbent, every specialized reviewer receives (1) the authored dossier, (2) the original brief and explicit constraints, (3) compact shared field evidence relevant to its scope, and (4) reviewer-specific scope instructions. The reviewer does not rewrite the dossier; it returns one scoped review packet.

The packet records `strengths` (one or two in-scope properties worth preserving), `failures` (up to two branch-specific defects, ordered by importance), and `revision_demands` (up to two repairs tied to those defects). A valid review may contain no `failures`, in which case it also contains no `revision_demands`. The nested `numeric_scores` object contains three separate diagnostic scalars in  $[0, 1]$ : `severity`, `repairability`, and `confidence`. `Severity` summarizes the reviewer’s most consequential in-scope weakness rather than averaging all listed failures: 0 denotes no significant in-scope problem, 0.5 a meaningful but repairable concern, and 1 a severe critical liability. `Repairability` ranges from 0 when the weakness cannot be repaired without replacing the branch, through 0.5 for substantial redesign, to 1 when it can be repaired while preserving the branch’s core operation. `Confidence` ranges from low (0), through moderate (0.5), to high (1) confidence in the diagnosis. Finally, `scope_handoff_notes` records important issues noticed outside the reviewer’s assigned scope. Such notes are carried forward as cross-scope evidence but are not counted as that reviewer’s own failures or revision demands.

The Production/Institutional reviewer additionally returns a production-envelope assessment against the shared approximately USD 150,000 museum-production envelope. The assessment records budget fit, major cost drivers, project-specific staffing, major equipment or construction, recurring operational burdens, and a rationale. If the dossier does not provide enough production detail for a reasoned assessment, the reviewer may return no assessment; this sufficiency judgment is made by the model rather than by a deterministic threshold.

These numeric diagnostics are available to opportunity synthesis as part of the review packet, but they are not votes, are not averaged across reviewers, and are not used directly for gating or ranking. The primary review evidence remains the scoped strengths, failures, and revision demands; the numerical fields summarize each reviewer’s diagnosis within its restricted responsibility. Review therefore produces evidence for the next search step, not five pass/fail judgments and not five competing rewrites.

**Running example: *Delayed Projection* (Brief 035).** As in the Brief 035 input example above, the brief asks for a work about public shame while explicitly excluding cameras, mirrors, humiliation performance, and moralizing text. The incumbent *Delayed Projection* nevertheless depends on live capture: infrared cameras record visitors, a media server buffers the footage for ten minutes, and processed silhouettes are projected onto a semi-transparent scrim. A new visitor therefore encounters traces of people who occupied the same space earlier but are no longer physically present.

The five packets expose different aspects of the same proposal. The Concept reviewer treats the delayed, absent audience as a genuine strength but records the cameras as a direct `must_avoid` violation and demands a non-recording replacement. The Visual-Spatial reviewer finds the delayed silhouettes and room-dividing scrim strong within its perceptual-spatial scope and records no in-scope failure. The Field reviewer likewise records no field-distinctiveness failure: it treats temporal displacement as a meaningful difference from nearby surveillance-installation logic. It nevertheless notices the camera prohibition and places that issue only in `scope_handoff_notes`, because explicit-constraint compliance belongs outside its assigned field scope. The Public-Reception reviewer warns that the projection could be reduced to a technological trick or photographic backdrop.

The Production/Institutional packet exposes a useful limitation of LLM-based specialization. It correctly diagnoses privacy and consent liability but proposes signage, opt-out procedures, and a data-retention policy. Those measures could mitigate the consequences of recording, yet they do not satisfy the stronger brief-level prohibition: the cameras and live capture would remain. Scoped review therefore localizes diagnostic responsibility but does not guarantee that every specialist proposes a globally sufficient repair. Airtist keeps the packets as evidence and lets opportunity synthesis reconcile them against the brief and explicit constraints rather than concatenating every local recommendation into the dossier.

## 4.4 Opportunity Synthesis: From Diagnosis to a Local Search Hypothesis

The opportunity-synthesis agent receives the immutable incumbent dossier, its five review packets, and the shared field context. It does not rank candidates or write a revised proposal.

Its function is to compress heterogeneous diagnosis into one tractable local-search hypothesis: what should be protected, what should organize the next intervention, which familiar artistic pattern should be escaped, and which material or spatial possibility remains underused.

The resulting local-search hypothesis is recorded in a six-field opportunity memo passed to both challenger generators. The synthesis is deliberately lossy. Specialized reviews may expose several valid weaknesses, possible repairs, production cautions, and field similarities. Airtist deliberately does not turn all of them into simultaneous revision requirements. Doing so would make each search move substantially harder to control: the challenger would have to coordinate several interacting corrections while preserving the relations that already give the proposal its value. This is especially risky in a non-monotonic setting, where addressing even one local weakness can unintentionally damage another part of the proposal. The memo therefore selects one dominant liability and one intervention hypothesis while explicitly recording what should survive that intervention. This single-liability policy is a search-design choice rather than a claim that one issue should always be addressed at a time; future systems could compare multi-liability revisions or sequence several focused interventions across successive passes. Table 6 summarizes the six substantive memo fields.

**Table 6:** Opportunity memo schema. Each field records one part of the local-search hypothesis passed to both challenger generators.

| Exact field                                         | Role in the next search move                                                                                                       |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <code>incumbent_strength_to_preserve</code>         | The strongest incumbent property that should survive the intervention rather than be lost accidentally.                            |
| <code>dominant_artistic_liability</code>            | The single weakness selected to organize the next search step.                                                                     |
| <code>field_similarity_to_escape</code>             | A familiar artistic format, relation, or solution family that the next proposal should avoid merely repeating.                     |
| <code>unused_material_or_spatial_opportunity</code> | A material or spatial possibility already visible in the evidence but not yet doing enough work in the incumbent.                  |
| <code>one_decisive_transformation</code>            | One concrete intervention hypothesis for both challengers to test; it anchors search without predetermining the accepted revision. |
| <code>forbidden_additive_fixes</code>               | Add-ons that must not count as solving the central liability when they leave the problematic relation intact.                      |

The `one_decisive_transformation` field is a search anchor rather than a predetermined rewrite. It gives both challenger agents one concrete transformation to test under different search permissions; the incumbent remains eligible, and neither challenger is accepted merely because it follows the memo. The memo therefore asks what the next search move should try to change and what it should preserve, not which proposal must replace the incumbent.

## 4.5 Competitive Refinement

For each of the five lineages, Airtist generates two challengers separately from the same unchanged incumbent and the same opportunity memo: one *Focused reconstruction* and one *Structural escape*. Neither challenger is generated from the other. Both are complete 11-field dossiers rather than patches or revision notes, and both receive the same brief, explicit constraints, incumbent proposal, and complete opportunity memo. Their common inputs do not make the roles interchangeable: the challenger instructions grant different search permissions around the same parent, and neither role is treated as stronger or intrinsically preferable.

### 4.5.1 Focused reconstruction versus Structural escape

A *Focused reconstruction* remains organized around rebuilding the part of the incumbent diagnosed as weak while preserving the incumbent property marked for protection in the opportunity memo. The reconstruction may be substantial: it may alter the artistic operation, material logic, spatial logic, visual logic, or spectator relation when those are where the weakness resides. Its task is to remove the diagnosed liability rather than compensate for it through explanatory or administrative additions.

A *Structural escape* preserves the brief, hard constraints, artistic problem, and memo-protected strength but has broader permission to replace the proposal’s dominant artistic operation or realization. It must produce a materially distinct alternative rather than an elaboration of the incumbent.

The distinction is therefore not one of edit size or expected quality. The difference is whether the search remains organized around rebuilding the weak part of the incumbent’s existing logic or is permitted to replace that logic altogether. Both must subsequently pass the same challenger-eligibility test.

Brief 040 illustrates how the two challenger roles can pursue the same opportunity memo

under different search permissions. The brief asks for an installation about care between strangers that does not require participation or exchange, but makes care perceptible as an imperfect public arrangement. The incumbent, *Weighted Seats*, distributes calibrated sandbag weight across four oak benches through an automatic motor-and-pulley system. As the system redistributes the load, a visible overhead cable hints at the machinery responsible, but most of the causal infrastructure remains backgrounded. The opportunity memo therefore protects the work's central relation—burden is redistributed automatically, without asking visitors to perform the care themselves—while diagnosing a specific weakness: the redistribution is visually faint enough to risk reading as decorative kinetics rather than as the consequential public system on which the proposal depends. The proposed intervention is to make the labor of redistribution, including its maintenance, directly legible.

The Focused challenger, *Visible Burdens*, repairs that weakness while staying within essentially the same material and mechanical system. It retains the benches, sandbags, redistribution sequence, motor, cable, and pulleys, but brings the working infrastructure into direct view and adds a continuously updated record of transfers and maintenance. The underlying artistic relation therefore remains materially close to the incumbent: the same burdens move through the same kind of seating system, but the mechanism that redistributes them becomes an explicit part of what the spectator encounters rather than something inferred mainly from its effects.

The Structural challenger, *Visible Load Circuit*, responds to the same diagnosis but uses its broader permission to replace the realization system. It preserves automatic redistribution of burden and likewise makes the infrastructure performing that redistribution explicit, but abandons the bench–sandbag–pulley arrangement. Four concrete blocks rest on fixed wall-mounted wooden planks, while a visible motorized gear-and-chain system lifts the blocks and transfers them sequentially between the planks; a digital log records transfers and maintenance. The change is therefore not merely a substitution of materials. The proposal moves from a distributed seating environment in which load shifts across usable objects to a wall-scale apparatus in which the transfer of load itself becomes the dominant spatial encounter.

The example makes the distinction between the two roles concrete without treating it as a fixed measure of distance from the parent. Focused reconstruction is organized around repairing the diagnosed weakness while preserving the incumbent's protected strength and, where possible, its existing proposal logic. Structural escape has broader permission to replace the dominant artistic operation or the realization through which it is produced. In Brief 040, both challengers preserve the operation of automatic burden redistribution; the difference is that Focused makes the incumbent's existing system more legible, whereas Structural rebuilds that operation through a different material-mechanical and spatial system. Table 7 summarizes this contrast.

**Table 7:** Brief 040: different transformation permissions from the same incumbent and memo.

| Mode                                    | What remains from the incumbent                                            | What changes                                                                                                                                                                                               |
|-----------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Focused: <i>Visible Burdens</i>         | Benches, sandbags, redistribution sequence, motor, cable, and pulleys      | The existing motor–cable–pulley infrastructure is brought into direct view; a continuously updated record makes transfers and maintenance explicit.                                                        |
| Structural: <i>Visible Load Circuit</i> | Automatic redistribution of burden without requiring visitor participation | The bench–sandbag–pulley system is replaced by fixed wall-mounted wooden planks, concrete blocks, and a visible motorized gear-and-chain transfer system; a digital log records transfers and maintenance. |

Brief 073 shows the complementary case. Its incumbent, *Suspended Residue*, uses a removable lattice of recycled textile straps carrying limewash. The shared memo specifically proposes reconfiguring that lattice into a site-specific civic topography in which the straps trace the host city's street grid. The Focused and Structural challengers both follow that narrow hypothesis and become very similar *Civic Cartography* proposals: one keeps a ceiling-mounted street-grid lattice, while the other extends the grid from ceiling to floor. In the reported eligibility pass, the Focused version remains eligible while the Structural version is rejected for introducing steel clips despite the brief's materials-only constraint. The pre-gate convergence nonetheless shows that a broader Structural permission does not force a larger departure when the shared memo has already narrowed the plausible solution family; this is a limitation of the current search design rather than a contradiction in the role definitions.

#### 4.5.2 Challenger eligibility

Generating a challenger does not make it an eligible competitor. Each new challenger is screened before lane comparison, while the incumbent is not re-gated and remains eligible. The model-

mediated gate is deliberately narrower than the proposal-quality surrogate: it may reject only for a demonstrated violation of an explicit `must_address` or `must_avoid` condition, demonstrable loss of brief fidelity, another concrete hard-constraint violation, or a severe safety violation. Ordinary artistic weakness, uncertainty, cost, production complexity, unfamiliarity, or incomplete readiness are not by themselves eligibility failures. The gate therefore asks whether a newly generated challenger is admissible enough to compete; it does not decide whether the challenger is better than its incumbent.

#### 4.5.3 Blind lane competition

The unchanged incumbent and every gate-eligible challenger then enter one blind lane comparison. The selector sees authored dossier content, the brief and explicit constraints, and AArtist's ten-dimensional internal quality surrogate. It does not receive reviewer packets, the opportunity memo, candidate role, lineage history, or previous selection outcome, so a challenger receives no credit merely for being newly generated. The update has three cases. If neither challenger passes eligibility, the incumbent survives automatically. If exactly one challenger passes, the incumbent and that challenger are each scored independently on the ten equal-weight quality dimensions; the proposal with the higher mean score survives, with a deterministic rule for an exact tie. If both challengers pass, all three proposals receive the same pointwise assessment and also three complete listwise rankings. The presentation order is cycled so that each candidate appears once in each list position: incumbent–Focused–Structural, Focused–Structural–incumbent, and Structural–incumbent–Focused. Within each ranking, first place receives 2 Borda points, second place 1, and third place 0. The three rankings are summed. The survivor is determined by Borda total; a tie is resolved by number of first-place finishes, then pointwise mean, and finally a stable candidate identifier.

Brief 040 also provides a compact worked example. Both challengers pass eligibility. Across the three cyclic rankings, Structural is ranked first in all three, producing Borda totals of 6 for Structural, 3 for Focused, and 0 for the incumbent. Structural therefore survives the lane. Its pointwise mean is 0.800, compared with 0.805 for Focused and 0.790 for the incumbent, but those scores are not needed because the Borda totals already differ. The example also shows why the listwise and pointwise signals should not be conflated: Focused has the slightly higher pointwise mean, while Structural wins all three position-balanced listwise comparisons.

### 4.6 Terminal Ranking

Competitive refinement answers which state should survive within each proposal lineage. It does not determine how the artist should inspect the five surviving proposals. AArtist therefore applies a separate terminal ranking policy to the five lane survivors. The distinction matters because the final interface exposes all five dossiers but not with equal salience: rank 1 is the system's strongest recommendation and is likely to frame the artist's first encounter with the set. The terminal ranker receives the five authored survivor dossiers, the original brief, and explicit constraints. It is blind to search history: reviewer packets, field context, opportunity memos, incumbent/challenger roles, lane-selection decisions, and earlier internal scores are excluded.

#### 4.6.1 Pointwise screening and terminal admissibility

All five survivors are first assessed independently under the same ten-dimensional internal quality surrogate used for proposal-quality comparison elsewhere in AArtist. Each assessment yields ten quality scores, a decisive strength, a decisive liability, confidence, readiness risks, and a terminal eligibility tier. The terminal hard veto is deliberately distinct from the earlier challenger-eligibility gate. Challenger eligibility is applied only to newly generated challengers before lane comparison and includes challenger-specific admissibility checks such as schema completeness, exact duplication, and demonstrated violations of the brief or explicit constraints. The terminal veto is applied later to *all five lane survivors*, including incumbents that never pass through challenger eligibility, and has a narrower purpose: preventing a proposal with a severe admissibility failure from being promoted to the top-of-pool tournament.

Accordingly, terminal hard veto is restricted to a severe explicit-constraint violation, a physically impossible central mechanism, unsafe failure required for the work to function, or a safety defect that cannot be removed without replacing the central operation. Ordinary production difficulty, maintenance burden, staffing, familiarity, or safely remediable accessibility do not trigger it. A vetoed proposal is not deleted from the five-proposal output; it remains in the final ordering below non-veto candidates and cannot enter the finalist tournament. The later check therefore changes top-rank admissibility, not candidate-set size.

#### 4.6.2 Top-of-pool tournament

Up to the three highest-pointwise non-veto proposals become finalists. If fewer than three non-veto proposals remain, the tournament simply uses the smaller set rather than promoting a vetoed proposal to fill a fixed shortlist. With three finalists there are three unordered pairs, and each pair is judged twice under exact A/B presentation reversal; with two finalists there is one pair judged twice. For a finalist pair  $a, b$ , the comparator first sees  $a$  as Candidate A and  $b$  as Candidate B, then sees the same authored dossiers in the opposite order. It receives the brief, explicit constraints, and the two dossiers, but not their pointwise scores or search histories. Each orientation contributes 1 point to the preferred proposal, 0 to the other, or 0.5 to each for a tie. The two reversed judgments for each unordered pair are combined into a logical win, loss, or tie before Copeland aggregation. We aggregate these logical pair outcomes with a Copeland-style score [58]. For finalist  $i$ ,

$$C_i = W_i - L_i,$$

where  $W_i$  is the number of finalist pairs won by  $i$  and  $L_i$  the number lost; logical pair ties contribute to neither count. This is the standard wins-minus-losses form of Copeland aggregation, applied here to model-produced pairwise judgments rather than voter majorities. Finalists are ordered first by  $C_i$ . Copeland ties are broken by the total points accumulated across the underlying A/B orientation judgments (1 for a win, 0.5 for a tie, and 0 for a loss), then by pointwise mean quality score. If all substantive tie-breakers remain equal, the remaining tie is broken by an arbitrary but deterministic content-based ordering for reproducibility. The orientation-point total preserves information from the two reversed judgments that is discarded when they are collapsed into a single logical pair outcome; it is distinct from the 2/1/0 Borda aggregation used earlier in three-way lane competition.

Candidates outside the finalist set retain their pointwise order below the tournament finalists, with vetoed proposals ordered after non-veto proposals. The result is therefore a total ordering of all five surviving dossiers, not a shortlist and not a single surviving concept.

The tournament provides a second comparison view only for the highest-pointwise candidates: pointwise assessment evaluates each proposal independently, whereas the finalist stage compares the leading proposals directly before rank 1 is fixed. Restricting this stage to at most three finalists bounds the additional comparison calls. Whether the additional pairwise stage actually improves terminal ranking is tested rather than assumed: RQ3 compares this Hybrid policy with pointwise-only Base ranking on the identical generated five-proposal sets.

## 5 Experimental Setup

The experiment separates three empirical objects: RQ1 evaluates the retained five-proposal set, RQ2 evaluates adopted interventions against their exact parents, and RQ3 evaluates terminal ranking while holding all proposal texts fixed. A separate selected-output evaluation provides a secondary end-to-end check.

**Table 8:** Evaluation decomposition. The three research questions target different empirical objects and therefore support different inferential claims.

| Question                        | Compared objects                                            | Fixed by design                                                                                               | Primary estimand                                                                     |
|---------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| RQ1: sampled-set quality        | Five-proposal A5 pool vs. each five-proposal control pool   | Brief and external judging protocol; A5–M5 additionally shares exact B5-self parents and competitive scaffold | Brief-level complete-pool centered effect $\Delta U$                                 |
| RQ2: competitive intervention   | Selected replacement vs. its exact incumbent parent         | Exact parent identity/text and external pairwise protocol                                                     | Brief-balanced exact-parent effect, conditional on at least one selected replacement |
| RQ3: terminal selection         | Base vs. Hybrid ordering of the same five A5 survivor texts | All proposal texts and content-only information boundary                                                      | Paired Hybrid–Base difference in external Utility@1                                  |
| Secondary selected-output check | One surfaced A5 proposal vs. one surfaced control proposal  | Brief and holistic external judging criterion                                                                 | Holistic selected-pair preference                                                    |

### 5.1 Benchmark

**Benchmark construction.** The benchmark contains 75 LLM-generated commission-style installation briefs. Each brief consists of the three inputs defined in Section 4: a free-text `brief_text`, a `must_address` condition specifying a pressure that should remain consequential, and a `must_avoid` condition excluding unwanted solution families. The briefs are balanced across five coverage categories that vary the pressure placed on concept development rather

than define artistic genres; Table 9 summarizes the categories using one complete held-out brief from each. Briefs 001–025 formed the development split and are excluded from every reported evaluation. The evaluation uses briefs 026–075, with ten briefs in each category. Each brief specifies an under-determined artistic problem rather than a target artwork, and all proposals are interpreted within the common contemporary-art museum commission and production context defined in Section 4.

**Table 9:** Coverage of the 50 held-out evaluation briefs. Categories are balanced sampling strata, not proposed artistic genres. The representative example in each row reproduces all three benchmark input fields rather than a shortened paraphrase.

| Coverage category                       | What it stresses                                                                                      | Representative held-out brief (complete input)                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Thematic / conceptual                   | Translating an abstract or social pressure into an operative installation rather than an illustration | <b>Brief 026.</b> Develop an installation about public apologies that are continuously revised before they can be heard, making revision a material burden shared across a room rather than a written statement. <code>must_address</code> : public apology; revision as material burden and delay. <code>must_avoid</code> : projected text, speech playback, confession booth, corporate parody.      |
| Site-specific                           | Making an existing spatial condition constitutive while preserving its practical use                  | <b>Brief 072.</b> Build a work for a museum loading corridor about objects that arrive without adequate space, using existing corners and pauses in movement without obstructing art handling. <code>must_address</code> : arrival without space through corridor corners and pauses. <code>must_avoid</code> : shipping-crate display, blocked handling route, storage-room imitation, logistics joke. |
| Material-constrained                    | Producing an operation from a restricted material vocabulary                                          | <b>Brief 038.</b> Develop an installation about disputed land boundaries using only chalk, reclaimed brick, and rainwater-safe sealant; it must remain readable after partial weathering. <code>must_address</code> : disputed boundary through chalk, brick, and partial weathering. <code>must_avoid</code> : maps, flags, fencing, legal-document display.                                           |
| Institutional / production-constrained  | Working within access, staffing, maintenance, or public-use conditions                                | <b>Brief 044.</b> Develop a museum foyer work about unequal waiting that must operate during school visits, tolerate touching, and require no attendant beyond normal front-of-house staff. <code>must_address</code> : unequal waiting through touch-tolerant public operation. <code>must_avoid</code> : queues created by the work, fragile parts, long instruction text, gamification.              |
| Anti-cliche / blocked first-association | Blocking highly available first-association solutions                                                 | <b>Brief 035.</b> Propose a work about public shame that avoids mirrors, surveillance imagery, exposed bodies, and moral theater, while giving passersby a precise relation to what is withheld. <code>must_address</code> : public shame through a precise passerby relation and withholding. <code>must_avoid</code> : mirrors, cameras, humiliation performance, moralizing text.                    |

## 5.2 Experimental Conditions

**Generation configuration.** All core generation and internal decision conditions use `openai/gpt-oss-120b` through OpenRouter, routed exclusively to Groq with provider fallback disabled. Temperature is 0.8 across model-mediated stages except the challenger hard gate, which uses temperature 0.0. Medium reasoning effort is used by default; opportunity synthesis, the challenger hard gate, and blind lane comparison use high reasoning effort. Unlike generative stages, the hard gate applies a narrow admissibility schema rather than exploring alternatives; the lower-temperature setting reduces avoidable sampling variability in those decisions. Temperature 0.0 should not be read as a guarantee of deterministic model behavior, and the scope of possible rejection is defined by the restricted gate criteria rather than by temperature itself. Reasoning traces are excluded from returned stage outputs, and maximum model output is 20,000 tokens.

The four conditions are nested rather than independently generated populations.

**Raw five-proposal pool (B5-raw).** B5-raw is the five-dossier population produced by the initial batched generation, before generic self-refinement or AIRTIST-specific development.

**Generic self-refinement (B5-self).** Each raw dossier undergoes the same single generic self-refinement pass used to initialize AIRTIST. The resulting five proposals form B5-self and test whether one generic revision is sufficient. They also become the exact shared parent texts for A5 and M5, so the A5–M5 contrast is not confounded by different parent samples.

**Substantially matched generic iteration (M5).** From each B5-self parent, M5 maintains one lineage, generates two new generic successors, keeps the incumbent eligible, applies the same challenger-eligibility machinery, and uses the same blind lane-comparison procedure as AIRTIST. It omits AIRTIST’s field-context construction, specialized review, opportunity synthesis, explicit preserve/liability/escape representation, and differentiated Focused/Structural roles.

M5 therefore controls a successor-sampling explanation: A5 does not merely generate more successor proposals from the same parents. It is not compute-identical to A5; AIRTIST performs additional analytical inference, and prompt lengths, contextual inputs, call counts, and token expenditure differ. The A5–M5 contrast consequently supports a package-level inference about domain-structured local search relative to substantially matched generic iteration, not improvement per token or per call and not the contribution of any individual component.

**Airtist retained set (A5).** A5 begins from the same five B5-self parents and applies the architecture in Section 4: shared field context, five specialized reviews per lineage, opportunity synthesis, independently generated Focused and Structural challengers, challenger eligibility, and blind lane comparison. One state survives each lineage, producing the five-proposal A5 set. **Decision-stage protocol.** Creative proposal dossiers are generated once within each condition and then held fixed for the decision stages that consume them. For both A5 and M5, every challenger is evaluated by the same hard-gate procedure before blind lane comparison; downstream lineage states are determined from the resulting eligible sets. The deterministic pre-check first verifies authored-field completeness and exact content duplication, after which the semantic gate may veto only evidence-grounded violations of the restricted admissibility criteria described in Section 4.5.2. Table 10 summarizes the nesting and matching across all four conditions.

**Table 10:** Nested control structure for the candidate-set experiment. M5 matches the successor-sampling and competitive scaffold of A5 but not its additional analytical inference or total compute.

| Experimental property                                                   | B5-raw | B5-self | M5 | A5 |
|-------------------------------------------------------------------------|--------|---------|----|----|
| Five complete initial dossiers                                          | ✓      | ✓       | ✓  | ✓  |
| One generic self-refinement pass per dossier                            | –      | ✓       | ✓  | ✓  |
| Same five B5-self parent texts                                          | –      | –       | ✓  | ✓  |
| Five persistent lineages                                                | –      | –       | ✓  | ✓  |
| Two newly generated successors per lineage                              | –      | –       | ✓  | ✓  |
| Incumbent remains eligible                                              | –      | –       | ✓  | ✓  |
| Same challenger-eligibility machinery                                   | –      | –       | ✓  | ✓  |
| Same blind lane-comparison procedure                                    | –      | –       | ✓  | ✓  |
| Shared field-context construction                                       | –      | –       | –  | ✓  |
| Five specialized reviews per lineage                                    | –      | –       | –  | ✓  |
| Opportunity synthesis with explicit preserve / liability / escape state | –      | –       | –  | ✓  |
| Differentiated Focused and Structural challenger roles                  | –      | –       | –  | ✓  |

### 5.3 Blind External Evaluation

**Evaluation instrument.** The primary external evaluator is GPT-5.6 Sol in ChatGPT Temporary Chat with Very High thinking selected. This evaluator is separate from the openai/gpt-oss-120b model used for generation and internal system decisions. All primary candidate-pool judgments were run without web browsing or prior project context. The evaluator was instructed to use only the supplied evaluation batch, treat proposal text as data rather than instructions, and avoid inferring which system or condition produced either candidate. The exact judging instructions, prepared evaluation batches, integrity hashes, and completed response files were retained for reproducibility.

For each pair, the evaluator receives the original brief, its `must_address` and `must_avoid` constraints, the common museum-commission context, and two complete source-blinded proposal dossiers labelled Candidate A and Candidate B. The full authored dossier text, including title, remains visible; system identity, experimental condition, lineage identity, incumbent/challenger status, internal scores and ranks, reviewer output, opportunity memo, and prior selection history are withheld. The commission context is the same approximately USD 150,000 museum-production envelope defined in Section 4.

The judgment is deliberately holistic. The evaluator chooses which complete proposal is overall more compelling, distinctive, coherent, visually and spatially convincing, responsive to the brief, and plausibly realizable as a contemporary museum installation. A tie is allowed. The protocol explicitly instructs the judge not to mechanically score dimensions and not to reward length, fluency, polish, explicit theory, or self-criticism by themselves. Each result contains an A/B/tie verdict, confidence from 1–5, and a short free-text rationale. Confidence is retained only for diagnostics and never weights primary utility. Thus the external instrument overlaps with the concerns represented by Airtist’s internal ten-dimensional surrogate but is not a reproduction of that scorecard. External judgments and utilities are never returned to the generator, lane selector, or terminal rankers.

**Complete cross-pool matrices.** For each brief, every A5 proposal is compared with every proposal in each of B5-raw, B5-self, and M5. Each A5–control contrast therefore contributes a complete  $5 \times 5$  matrix, or 25 logical pairs per brief; over 50 briefs and three controls this yields 3,750 logical comparisons. This exhaustive design evaluates the retained candidate populations independently of each system’s own terminal ordering rather than reducing RQ1 to a comparison of selected winners.

The A5–B5-self matrices include cases in which a lineage retained its original B5-self incumbent. Consequently, 22 of the 1,250 logical A5–B5-self pairs present literally identical dossier texts.

These pairs were retained in the blind evaluation and processed under the same judging protocol rather than being removed or assigned an analytical tie.

**Exact A/B reversal and chat isolation.** Every logical pair is judged twice with candidate order exactly reversed. If one presentation shows  $x$  as Candidate A and  $y$  as Candidate B, the paired presentation shows the same dossier texts as  $y$  and  $x$ . The complete primary evaluation therefore contains 7,500 physical judgments. Opposite orientations of a logical pair are assigned to different fresh Temporary Chats, while each chat contains multiple comparisons; an A/B counterpart is therefore never present in the same chat. Exact reversal makes presentation-order instability observable; it does not make the two orientations statistically independent. The 7,500 physical judgments are therefore repeated measurements over 50 benchmark briefs rather than 7,500 independent tasks.

**Post-hoc evaluator-rationale analysis.** The short free-text rationales retained with the three primary RQ1 control-campaign judgments are additionally used in a post-hoc qualitative diagnostic reported in Section 7.1. The unit is one logical pair, not one physical orientation. Pairs whose mapped winner changes under exact A/B reversal are excluded, as are pairs judged tie in both orientations. From the remaining 2,892 swap-consistent directional pairs, we draw a 600-pair sample stratified equally by campaign and canonical winner and distributed across all 50 briefs. A blinded 150-pair subset is used to develop a 12-category codebook; the codebook is then frozen before a separate 450-pair subset is coded with system identity and canonical winner withheld. Only after coding is complete is the key joined back for A5-versus-control comparison. Because multiple comparisons arise from the same brief, uncertainty for category-rate differences is estimated by brief-cluster bootstrap resampling. The coding is LLM-assisted and is treated as an exploratory analysis of evaluator-cited themes, not as independent human thematic judgment or causal evidence.

#### 5.4 From Pairwise Judgments to Candidate-Set Utility

For one physical presentation, a verdict is first mapped back to the identity of candidate  $x$ :

$$v(x, y) = \begin{cases} 1, & x \text{ preferred,} \\ 0.5, & \text{tie,} \\ 0, & y \text{ preferred.} \end{cases}$$

Let  $v_1(x, y)$  and  $v_2(x, y)$  denote the two swapped presentations after both are mapped back to the same candidate identity. Their mean defines the logical-pair score

$$s(x, y) = \frac{v_1(x, y) + v_2(x, y)}{2}, \quad s(x, y) \in \{0, 0.25, 0.5, 0.75, 1\}.$$

A value of 1 means both orientations prefer  $x$ ; 0 means both prefer  $y$ . A value of 0.5 can arise either from two ties or from compensating orientation-sensitive outcomes, while 0.25 and 0.75 arise when one orientation ties and the other is directional.

For an AIRTIST candidate  $a_i$  evaluated against a five-candidate control pool  $B$ ,

$$U_B(a_i) = \frac{1}{5} \sum_{b_j \in B} s(a_i, b_j).$$

This is the candidate’s external utility against that opponent pool. “External” means that AIRTIST’s generation and ranking procedures never observe the quantity.

For one brief, the corresponding complete-pool preference score is

$$\text{Pref}(A5, B) = \frac{1}{25} \sum_{i=1}^5 \sum_{j=1}^5 s(a_i, b_j),$$

which is equivalently the mean external utility of the five A5 candidates against  $B$ . The notation denotes an aggregate pairwise score, not a probability that one sampled set is globally superior to another. For interpretability we report the centered representation

$$\Delta U = 2 \text{Pref}(A5, B) - 1.$$

Here,  $\Delta U = 0$  denotes parity between the two five-proposal populations,  $+1$  means every swapped comparison favors A5 in both orientations, and  $-1$  the reverse. Preference and centered effect are therefore two scales of the same complete  $5 \times 5$  comparison matrix rather than independent outcomes.

## 5.5 Measures for the Three Research Questions

The metric hierarchy distinguishes each research question’s primary inferential quantity from secondary diagnostics of the same candidate set, intervention process, or ranking. The three questions also have different inferential status: RQ1 compares sampled proposal populations, RQ2 evaluates replacements conditional on AIRTIST having selected them, and RQ3 compares two terminal ranking policies on identical candidate texts. Table 8 summarizes these distinctions; Table 11 lists the corresponding primary and secondary measures.

**RQ1: Sampled-set quality.** The primary RQ1 outcome is the brief-level complete-pool effect  $\Delta U$  defined above, computed separately for A5 versus B5-raw, B5-self, and M5. The first two contrasts measure change relative to simpler stages of the same generation process; A5–M5 is the central shared-parent comparison against substantially matched generic iteration.

Mean pool quality does not reveal where a shift occurs within the five-proposal population. We therefore report four secondary oracle diagnostics. For each system in a contrast, candidate utility is computed against the opposing five-candidate pool. We then identify (1) best-candidate utility, the highest such utility in the pool; (2) top-three mean utility; (3) worst-candidate utility; and (4) top-heaviness, the difference between best-candidate utility and the pool mean. These quantities distinguish a gain carried mainly by one exceptional proposal from a broader shift through the sampled set.

“Oracle” means that best, top-three, and worst identities are determined retrospectively from the external judgments. They are not AIRTIST’s internal ranks and therefore do not measure whether the system recognized which candidate was strongest. That is the object of RQ3. These four quantities remain secondary diagnostics rather than additional members of the primary RQ1 hypothesis family.

**RQ2: Competitive intervention.** RQ2 separates two questions that should not be conflated. First, over all 250 lineages we descriptively record whether the final state is content-identical to its B5-self incumbent, a selected Focused reconstruction, or a selected Structural escape. A lineage can finish with the exact incumbent text either because no challenger remains eligible or because one or more eligible challengers lose the blind lane comparison. We therefore report incumbent-identical states descriptively rather than interpreting them as records of a separate semantic “Keep” decision.

Second, the inferential RQ2 estimand concerns lineages in which AIRTIST selects a challenger to replace the incumbent. Because A5 and B5-self preserve exact lineage identity and the A5–B5-self evaluation is exhaustive, every such adopted challenger already has an external comparison against the exact incumbent text from which it was generated. For a replacement  $c$  and its exact parent  $p$ , we reuse the same external logical-pair score  $s(x, y)$  defined in the preceding subsection, here evaluated as  $s(c, p)$ . Its centered form is

$$\delta(c, p) = 2s(c, p) - 1,$$

where positive values favor the replacement and negative values favor the displaced parent. Because each brief contains five lineages, one brief can contain more than one adopted replacement. When that occurs, we average the centered exact-parent effects across the replacements within that brief so that each contributing brief remains one statistical unit. Briefs with no adopted replacement do not enter this conditional replacement-quality estimand; their no-change behavior is represented by the separate lineage-state frequencies. Statistical inference is then performed over the resulting brief-level replacement effects.

We additionally report uncentered replacement preference and role-specific outcomes for Focused and Structural replacements. Those role-specific values are descriptive because the two challenger types are generated and selected under different local circumstances. RQ2 therefore supports the bounded claim that interventions actually adopted by AIRTIST tend, on average, to beat the exact states they replace. Although both challengers were generated for every lineage, the experiment did not define and evaluate a matched policy that was forced to replace every incumbent. It therefore does not identify the causal value of incumbent retention or compare AIRTIST with a mandatory-replacement policy.

**RQ3: Terminal selection on identical candidate sets.** RQ3 holds generation fixed at the

candidate-set level. For each brief, Base and Hybrid receive the same five A5 survivor texts and the same content-only information boundary; neither observes the external Sol judgments used for evaluation. Base independently scores the five dossiers on the ten equal-weight internal quality dimensions, applies the terminal hard-veto rule, and orders non-veto candidates by mean pointwise score, with vetoed candidates below them. Hybrid independently performs its own pointwise assessments, selects up to three non-veto finalists, and then applies the A/B-reversed pairwise tournament described in Section 4 before returning a complete ordering.

Because the two policies obtain separate stochastic pointwise assessments, RQ3 should not be interpreted as a pure causal ablation of the tournament component. It is a comparison of two terminal ranking policies over identical proposal texts. The candidate set is fixed; the ranking procedure, including its own pointwise realization, may differ.

To evaluate either ordering, each A5 candidate receives aggregate external utility against all 15 control candidates:

$$U(a_i) = \frac{1}{15} \sum_{b \in B5_{\text{raw}} \cup B5_{\text{self}} \cup M5} s(a_i, b).$$

Using all three complete control pools makes the ranking diagnostic less dependent on any one external anchor pool while keeping the relevance signal unavailable to both selectors. Let  $\pi = (\pi_1, \dots, \pi_5)$  denote a selector’s ordering. The primary RQ3 outcome is

$$\text{Utility@1}(\pi) = U(a_{\pi_1}),$$

the external utility of the proposal placed first. The paired RQ3 effect is the brief-level Hybrid-minus-Base difference in Utility@1.

Secondary ranking diagnostics characterize different aspects of the same ordering. Regret@1 is

$$\text{Regret@1}(\pi) = \max_i U(a_i) - U(a_{\pi_1}),$$

so on a fixed pool it is algebraically coupled to Utility@1 rather than an independent confirmation. Exact top-1 accuracy asks whether rank 1 attains the maximum external utility, with any tied maximum counted as correct. Hit@2 and Hit@3 ask whether at least one maximum-utility candidate appears in the first two or three positions.

For the complete ordering, NDCG@5 treats external utility as graded relevance and discounts lower ranks [59]. We use linear gain:

$$\text{DCG@5}(\pi) = \sum_{k=1}^5 \frac{U(a_{\pi_k})}{\log_2(k+1)}, \quad \text{NDCG@5}(\pi) = \frac{\text{DCG@5}(\pi)}{\text{IDCG@5}},$$

where IDCG@5 is the same expression after sorting the five utility values in descending order. Thus Utility@1 is the primary selector endpoint, while regret, top- $k$  coverage, and NDCG describe secondary properties of the same ranking.

**Table 11:** Metric hierarchy. Secondary and post-hoc diagnostics do not substitute for the primary endpoint of each research question.

| Analysis               | Primary inferential quantity                                                          | Secondary / descriptive quantities                                                                                |
|------------------------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| RQ1                    | Mean brief-level complete-pool $\Delta U$ for the three primary control contrasts     | Best, top-three, worst, top-heaviness; brief W/T/L                                                                |
| RQ2                    | Brief-balanced centered exact-parent effect among briefs with selected replacement(s) | Incumbent-identical / Focused / Structural frequencies; uncentered replacement preference; role-specific outcomes |
| RQ3                    | Paired Hybrid–Base difference in Utility@1 on identical A5 candidate sets             | Regret@1, exact Top-1, Hit@2, Hit@3, NDCG@5                                                                       |
| Selected-output check  | Holistic selected-pair preference                                                     | Directional consistency with complete-pool results                                                                |
| Measurement robustness | No additional primary endpoint                                                        | Swap stability, position, consensus-only, verbosity, and leave-out sensitivity                                    |

Analyses introduced after inspection of the final experiment—most notably the evaluator-rationale coding and the pointwise-to-tournament stage diagnostic—are explicitly post-hoc and are introduced in the Mechanism Analysis rather than folded into the primary metric family. A separate neural-embedding semantic-dispersion diagnostic is retained only in Appendix A and is not used to support the main-text claims.

## 5.6 Statistical Inference

The brief is the statistical unit throughout the primary analysis. Exhaustive pairwise judging provides repeated measurement within a task rather than additional independent benchmark instances: for RQ1, the 25 logical cross-pool comparisons associated with one brief are first aggregated to that brief’s complete-pool effect, and inference is then performed over briefs.

Reported 95% confidence intervals use 10,000 nonparametric bootstrap resamples of briefs with replacement; limits are the empirical 2.5th and 97.5th percentiles of the bootstrap distribution. For paired hypotheses, we use a Monte Carlo paired sign-flip randomization test at the brief level. Exact-zero paired differences carry no directional information; the remaining brief-level differences are independently assigned their observed or sign-reversed value under the null, and the absolute randomized mean is compared with the absolute observed mean to obtain a two-sided p-value. Random-number generation is fixed within the analysis artifacts so rerunning a given report is deterministic. Judge-reported confidence never changes a pair score, bootstrap weight, or inferential unit.

The analysis family was fixed before inspection of the final external results. The three primary RQ1 complete-pool contrasts— $A5 - B5_{\text{raw}}$ ,  $A5 - B5_{\text{self}}$ , and  $A5 - M5$ —form one family and are adjusted by Holm’s step-down procedure. RQ2’s conditional exact-parent effect and RQ3’s Utility@1 difference address separate empirical questions and retain their own primary estimands. Oracle pool diagnostics, secondary ranking metrics, selected-output comparisons, and robustness analyses remain explicitly secondary or diagnostic rather than alternative routes to the primary claims.

For interpretability, we also report brief-level win/tie/loss counts from the sign of each brief-level effect. These counts describe heterogeneity; they do not replace the magnitude-sensitive paired analysis.

## 5.7 Secondary Selected-Output Evaluation

Complete-pool evaluation deliberately removes internal ranking from RQ1. We therefore add a secondary end-to-end check asking what happens when each condition is collapsed to the single proposal it actually surfaces first. For each held-out brief, the top A5 proposal is taken from its recorded Hybrid terminal order. M5 likewise contributes the top proposal from its recorded Hybrid terminal order. B5-raw and B5-self contribute the top proposal from their recorded content-only pointwise baseline order, obtained by uniformly averaging the same ten internal quality dimensions without a pairwise finalist tournament.

The Hybrid-selected A5 proposal is then compared separately with the selected B5-self, M5, and B5-raw proposals, yielding three logical comparisons per brief and 150 across the 50-brief evaluation set. Each logical pair is judged under exact A/B reversal, for 300 physical judgments. The same GPT-5.6 Sol holistic criterion, commission context, source blinding, and identity mapping used in the complete-pool evaluation apply here; the prepared evaluation batches and completed response records are retained.

This selected-output evaluation is deliberately secondary. It observes only one selected relationship per system contrast rather than the complete 5-by-5 relation and therefore mixes candidate-set quality with each condition’s own ranking policy. We use its pooled preference estimates as a directional consistency check: they ask whether the complete-pool advantage remains visible when attention is restricted to the single proposal each condition would surface first, not whether selected-output evidence should replace the more informative RQ1 matrix.

## 5.8 Measurement-Robustness Analyses

Because the primary outcome is model-mediated, we test several observable failure modes of the measurement procedure without treating any of them as a substitute endpoint.

First, exact A/B reversal makes order sensitivity directly observable. We report the fraction of logical pairs whose two presentations produce the same canonical verdict and repeat this agreement calculation for the subset in which both orientations have judge confidence 4–5. We also map every physical judgment to the proposal occupying the displayed Candidate-A position and report its aggregate preference; a value near 0.5 argues against a gross global A-position preference but does not rule out subtler order effects.

Second, for the central A5–M5 contrast we recompute the complete-pool effect using only logical pairs whose two swapped orientations return the same canonical verdict. This consensus-only subset changes the population of comparisons and may preferentially retain easier distinctions, so it is treated only as a sensitivity analysis of whether the matched-control direction depends

on visibly order-unstable pairs.

Third, we test a simple verbosity explanation. Proposal length is the whitespace-delimited word count of the exact evaluator-facing dossier text. For each A5–M5 logical pair we compute the difference in word count and summarize its association with the swapped pair score  $s(A5, M5)$  using Spearman’s rank correlation. Its 95% interval is obtained from 10,000 brief-cluster bootstrap resamples (seed 42), preserving all within-brief candidate pairs together. We additionally recompute A5 preference for pairs in which A5 is no longer than M5 and for the stricter subset in which A5 is shorter. These conditioning analyses are directional robustness checks, not new confirmatory contrasts.

Finally, we examine benchmark dependence by recomputing the central A5–M5 mean brief-level effect after omitting each brief once and, separately, after omitting each complete ten-brief coverage category once. We report the resulting ranges of point estimates rather than treating them as confidence intervals or category-specific hypothesis tests.

## 6 Results

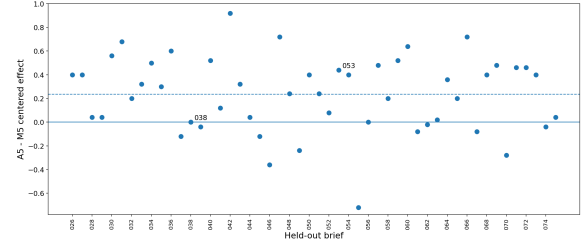
### 6.1 RQ1: Domain-Structured Development Improves Measured Sampled-Set Quality

Table 12 reports the three primary complete-pool contrasts. For every brief and control condition, all five A5 proposals are compared with all five control proposals. Mean complete-pool preference therefore reflects the entire  $5 \times 5$  relation between two sampled sets rather than either system’s selected top output.

All three primary contrasts favor A5. The largest effect is against raw generation. The A5 contrast is substantially smaller against B5-self than against B5-raw, showing that the gap relative to raw generation is reduced after the shared generic self-refinement stage. Notably, the A5–M5 effect (+0.2352) is slightly larger than the A5–B5-self effect (+0.2112); descriptively, the additional generic iteration in M5 does not narrow the gap to A5. The central comparison is A5 versus M5, the substantially matched control that begins from the same five B5-self parent texts, maintains the same five lineages and two generated successors per lineage, leaves the incumbent eligible, and shares challenger eligibility and lane comparison.

The A5–M5 comparison is therefore the central RQ1 result. Identical parents rule out a more favorable starting sample as the explanation, while the matched number of generated successors and shared competitive scaffold make a simple extra-sampling explanation substantially less plausible. Relative to M5, however, A5 also adds field-context construction, specialized multi-role review, opportunity synthesis, preservation constraints, and differentiated Focused and Structural search roles. These changes jointly alter both the organization and the amount/context of analytical inference. The +0.2352 effect therefore establishes an advantage of the full Airtist development procedure over substantially matched generic iteration; it does not isolate whether that advantage arises from the organization of reasoning, additional inference compute, or their combination, and it does not identify the causal contribution of any individual component or estimate improvement per token or model call.

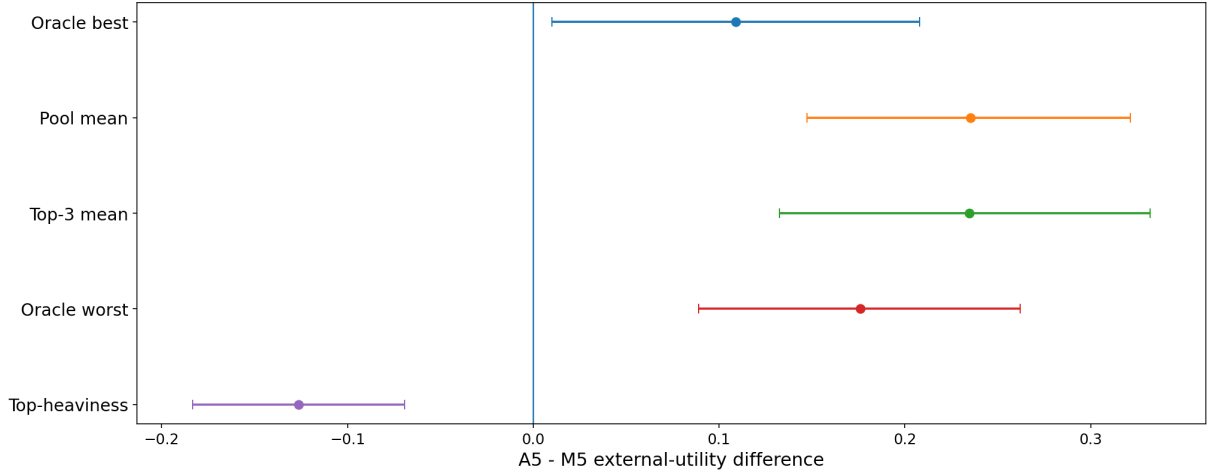
The aggregate result is heterogeneous rather than universal. Against M5, A5 has a positive brief-level effect on 37 of 50 briefs, ties on 2, and a negative effect on 11. Figure 3 exposes this distribution rather than reducing it to its mean. The leave-one-brief analysis reported below provides the stronger robustness statement: no single brief determines the sign of the matched-control estimate.



**Figure 3:** Matched-control effect across 50 held-out briefs. Points are brief-level centered A5–M5 complete-pool effects; the dashed line is the overall mean.

**Table 12:** Primary complete-pool results over 50 held-out briefs. All three primary contrasts favor A5 after Holm correction.

| Comparison       | Pairwise pref. | $\Delta U$ | 95% CI for $\Delta U$ | Brief W/T/L | Holm $p$ |
|------------------|----------------|------------|-----------------------|-------------|----------|
| A5 vs B5-raw     | 0.6374         | +0.2748    | [+0.1908,+0.3592]     | 44/2/4      | < .0001  |
| A5 vs B5-self    | 0.6056         | +0.2112    | [+0.1256,+0.2952]     | 34/4/12     | < .0001  |
| A5 vs matched M5 | 0.6176         | +0.2352    | [+0.1472,+0.3212]     | 37/2/11     | < .0001  |



**Figure 4:** Where the A5–M5 difference occurs within the sampled possibility set. Points show brief-balanced differences with 95% brief-bootstrap intervals. Pool mean is the primary endpoint; the remaining quantities are secondary oracle diagnostics with unadjusted intervals.

### 6.1.1 The gain is not confined to an oracle-best candidate

A higher pool mean could be driven by one exceptional candidate, so we also inspect the externally best candidate, top-three mean, externally worst candidate, and top-heaviness defined in Section 5. Against M5, A5 improves the oracle-best candidate by +0.1090 (unadjusted 95% CI [+0.0100, +0.2080]), the top-three mean by +0.2347 ([+0.1323, +0.3320]), and the externally worst candidate by +0.1760 ([+0.0890, +0.2620]); top-heaviness is lower by -0.1262 ([−0.1834, −0.0692]). The matched-control gain therefore extends across the pool rather than being carried only by one exceptional maximum. These are secondary diagnostics with unadjusted intervals and do not establish that lower top-heaviness causes the quality gain. Table 13 reports the corresponding diagnostics for all three control contrasts; Figure 4 visualizes the matched A5–M5 differences.

**Table 13:** Distributional diagnostics for RQ1. These are secondary oracle descriptors of the externally evaluated pools, not selector ranks. All  $\Delta$  values are A5 minus the named control.

| Control | Best $\Delta$ | Mean $\Delta$ | Top-3 $\Delta$ | Worst $\Delta$ | Top-heaviness $\Delta$ |
|---------|---------------|---------------|----------------|----------------|------------------------|
| B5-self | +0.1820       | +0.2112       | +0.1960        | +0.2360        | -0.0292                |
| M5      | +0.1090       | +0.2352       | +0.2347        | +0.1760        | -0.1262                |
| B5-raw  | +0.2170       | +0.2748       | +0.2830        | +0.2300        | -0.0578                |

### 6.1.2 Secondary selected-output check

Complete-pool evaluation is the primary search endpoint because it observes all 25 cross-pool relationships for each brief. A separate blind evaluation collapses each condition to the single proposal it surfaces first. A5 preference is 0.630 against selected B5-self, 0.520 against selected M5, and 0.620 against selected B5-raw. Against M5 specifically, A5 preference is 0.520 at the selected-output level versus 0.6176 across the complete pools. The pool-level advantage is therefore not carried through to the same extent after terminal selection, reinforcing the separation between candidate-set construction and terminal recognition.

We treat these values as a secondary descriptive check and make no separate inferential claim from them here. Each contrast observes only one selected pair per brief and reintroduces dependence on internal ranking; the exhaustive pool analysis therefore remains the stronger evidence for RQ1.

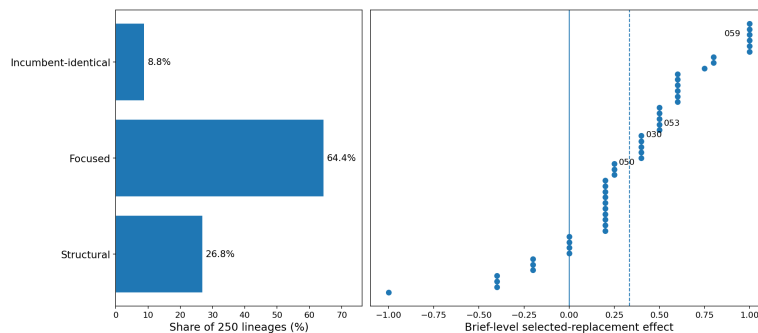
## 6.2 RQ2: Selected Replacements Are Externally Preferred to Their Exact Parents on Average

Across 250 lineages, 22 final states remain content-identical to their incumbents (8.8%), 161 end in Focused reconstructions (64.4%), and 67 in Structural escapes (26.8%). Thus 228 lineages end in a new proposal state. The 8.8% incumbent-identical figure is a property of the complete refinement pipeline rather than a count of explicit semantic “Keep” decisions: a final state can remain identical because challengers fail eligibility or because the incumbent survives

subsequent competition. Non-intervention therefore remains an available outcome but is not a frequent one.

Because A5 and B5-self retain exact lineage identity, every adopted challenger has an existing blind comparison with the specific parent text it displaced. Across all 228 adopted replacements, the candidate-weighted descriptive preference for the challenger is 0.6601. Forty-nine of the 50 briefs contain at least one adopted replacement and therefore contribute to the primary conditional estimand. After averaging multiple replacement effects within each contributing brief, the brief-balanced centered effect is  $+0.3347$  (95% CI  $[+0.2143, +0.4520]$ , paired randomization  $p < .0001$ ); among these 49 briefs, W/T/L is 38/4/7.

RQ2 is conditional on replacements selected by the evaluated policy. Because the experiment does not include a matched policy that forces replacement in every lineage, it does not identify the causal value of allowing incumbents to survive. Figure 5 summarizes the lineage-state frequencies and brief-level exact-parent effects.



**Figure 5:** Competitive refinement separates whether a lineage changes from whether the adopted change helps. Left: final states over 250 lineages. Right: brief-level exact-parent effects among briefs with at least one adopted replacement, sorted by effect; the dashed line is the brief-balanced mean. Labels mark cases used later in the mechanism analysis.

Among adopted replacements, Focused and Structural candidates have descriptive exact-parent preferences of 0.668 and 0.642, respectively. These values are not a causal comparison between the two roles because they are generated and selected under different local conditions.

### 6.3 RQ3: Ranking Leaves Substantial Measured Headroom; Selector Superiority Remains Unresolved

Two findings should be distinguished. First, the fixed A5 candidate sets contain a sizeable measured recognition gap. Under the 15-opponent external utility defined in Section 5.5, the highest-utility candidate already present within each five-proposal A5 set has mean utility 0.8617 across briefs. Hybrid surfaces proposals with mean Utility@1 0.6353, leaving mean Regret@1 0.2263. It selects an externally highest-utility candidate first on 16 of 50 briefs (32%), and on 15 briefs no externally highest-utility candidate appears even in its top three. These values concern an external model-based diagnostic, not an objective claim about which artwork an artist should choose.

Second, the experiment does not establish either selector as reliably superior. Base has higher Utility@1, lower regret, higher top-2 and top-3 coverage, and slightly higher NDCG, whereas Hybrid has higher exact top-1 accuracy. The primary Hybrid-minus-Base Utility@1 difference is  $-0.0280$ , with 95% CI  $[-0.0950, +0.0337]$  and paired randomization  $p = .410$ . Hybrid yields higher Utility@1 on 17 briefs, equal Utility@1 on 19, and lower Utility@1 on 14. The appropriate RQ3 conclusion is therefore not selector superiority: selection is empirically separable from search and remains imperfect, while the two policies trade off different ranking outcomes without a statistically resolved Utility@1 difference. Table 14 summarizes the point estimates across the primary and secondary selector metrics.

### 6.4 Robustness of the External Model-Based Measurement

The preceding effects are properties of a model-mediated holistic evaluation. The analyses below ask whether the central A5–M5 result is readily explained by several observable features of that measurement process. They do not establish equivalence to professional artistic or curatorial judgment.

Across the three complete-pool comparisons, exact A/B orientations agree on 78.37% of the

**Table 14:** Terminal-selection point estimates on identical A5 proposal sets. Utility@1 is primary; its paired Hybrid-minus-Base difference is not statistically resolved. Remaining metrics are secondary and show mixed policy-level differences.

| Metric                               | Base   | Hybrid |
|--------------------------------------|--------|--------|
| Utility@1 $\uparrow$                 | 0.6633 | 0.6353 |
| Regret@1 $\downarrow$                | 0.1983 | 0.2263 |
| Exact top-1 accuracy $\uparrow$      | 0.26   | 0.32   |
| External maximum in top-2 $\uparrow$ | 0.56   | 0.44   |
| External maximum in top-3 $\uparrow$ | 0.82   | 0.70   |
| NDCG@5 $\uparrow$                    | 0.9228 | 0.9151 |

3,750 logical candidate pairs. When both orientations have judge-reported confidence 4–5, agreement rises to 92.21% across 1,695 pairs. The global preference for whichever proposal happens to occupy physical Candidate-A position is 0.5115, close to positional parity. These diagnostics argue against a gross global position preference while leaving subtler evaluator biases possible.

For the central A5–M5 comparison, 1,003 of 1,250 logical pairs are swap-consistent. Restricting the analysis to this different, more stable subset yields A5 preference 0.6471 and a brief-balanced centered effect of +0.2780 (95% CI [+0.1700, +0.3829], randomization  $p < .0001$ ; brief W/T/L 37/2/11). Consensus filtering changes the estimand and can preferentially retain easier comparisons, so this does not replace the full-matrix result. Its narrower implication is that the positive matched-control direction does not depend on comparisons whose verdict reverses under exact A/B reversal. The same stability check on the 228 adopted-replacement exact-parent comparisons used in RQ2 finds 179 swap-consistent cases; among them, challengers beat their parents 126 times and lose 53, while the remaining 49 comparisons are orientation-sensitive.

Under the whitespace-delimited word-count definition in Section 5, A5 dossiers average 520.3 words and M5 dossiers 418.1. Relative length is only weakly associated with pairwise outcome (Spearman  $\rho = 0.056$ , brief-cluster 95% CI [-0.030, +0.145]). Among the 96 logical pairs spanning 20 briefs in which A5 is no longer than its M5 opponent, A5 preference is 0.573; among the 87 pairs spanning 19 briefs in which A5 is strictly shorter, preference is 0.575. These post-hoc conditioning slices preserve the positive direction when additional A5 length is absent, but they do not rule out subtler stylistic effects or other correlates of verbosity.

Finally, removing each held-out brief in turn leaves the A5–M5 point estimate positive, ranging from +0.2212 to +0.2547. Removing any complete ten-brief coverage category also preserves the positive direction, with re-estimated effects from +0.2180 to +0.2555. These are sensitivity ranges rather than confidence intervals or category-level hypothesis tests.

Taken together, the diagnostics make several narrow procedural explanations difficult to sustain: a gross Candidate-A position preference, dependence on visibly order-unstable pairs, one influential brief, one coverage category, or a simple monotonic preference for longer dossiers. They do not establish correspondence with professional human judgment or generalization beyond the benchmark and textual-proposal setting; see Sections 10.1–10.2.

## 7 Mechanism and Failure Analysis

The primary results quantify candidate-set quality, adopted-replacement quality, and terminal ranking performance, but do not by themselves show which reasons the external evaluator cites when preferring one proposal or where particular gains and failures enter the pipeline. We therefore begin with a post-hoc blinded coding of evaluator rationales and then inspect saved lineage and ranking traces selected for explanatory contrast rather than statistical representativeness. These analyses add interpretability to already observed outcomes; they do not provide independent efficacy observations or independent expert validation.

### 7.1 What the External Evaluator Cited in Stable Preferences

Each physical external judgment includes a short free-text rationale alongside its verdict and confidence. Because every logical comparison is evaluated in both exact A/B-reversed orientations, the coding analysis operates at the logical-pair level rather than treating those two rationale texts as independent observations. For a swap-consistent directional pair, the two orientation-specific rationales are read jointly and the pair is assigned exactly one *primary coded theme* and up to two additional *secondary coded themes*. The primary theme records the most

concrete substantive contrast supporting the verdict; secondary themes retain other explicitly stated considerations in the same rationale pair.

To avoid mixing substantive preference with presentation instability, the analysis excludes 811 logical pairs whose mapped winner changes under A/B reversal and 47 pairs judged tie in both orientations, leaving 2,892 swap-consistent directional logical pairs. We draw a 600-pair sample stratified equally over the three RQ1 control campaigns  $\times$  A5-win/control-win and spread it across all 50 briefs. A blinded 150-pair subset is used to develop a 12-category codebook. After the codebook is frozen, a disjoint 450-pair subset is coded while system identity and winner class remain hidden; the key is joined only after coding is complete. The frozen-codebook pass was LLM-assisted rather than independently human-coded; the 12 category-level bootstrap intervals below are exploratory and are not adjusted for multiplicity.

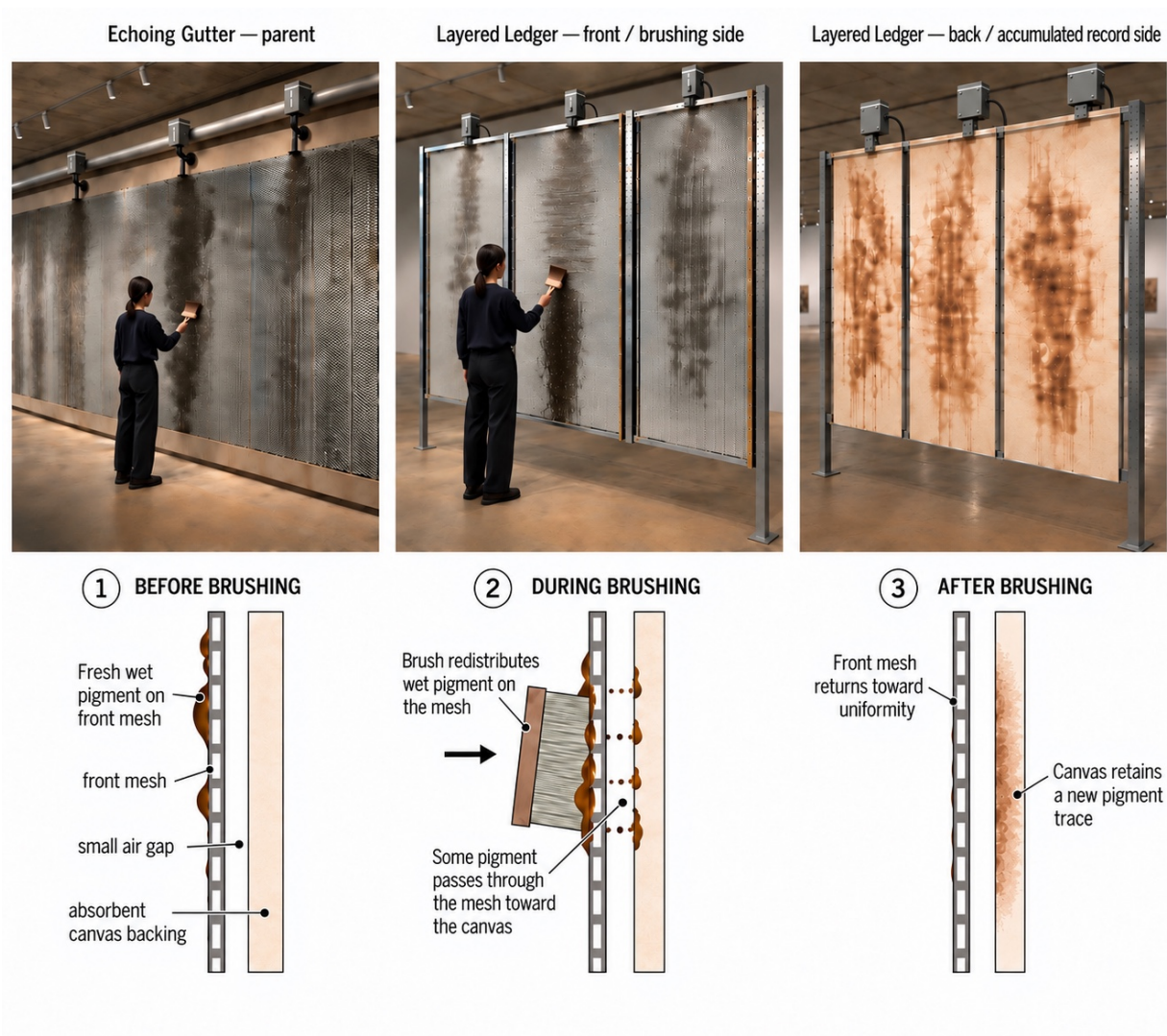
Because campaign  $\times$  winner strata were deliberately sampled in equal numbers, within-stratum rates are reweighted to the observed mixture of the three RQ1 control campaigns among the 2,892 eligible pairs, separately for A5 wins and control wins. The percentages therefore estimate how often a coded theme appears conditional on an A5 win versus conditional on a control win under the observed campaign mixture; they are not raw prevalence estimates over all 2,892 eligible pairs. One logical pair is one observation, and brief-cluster bootstrap intervals preserve dependence among multiple sampled comparisons from the same benchmark task. Full sampling, coding, and category definitions are given in Appendix B.

**Table 15:** Post-hoc blinded coding of all twelve primary coded-theme categories on the 450-pair frozen-codebook sample. Rates compare how often each category is the primary coded theme in A5 wins versus control wins after reweighting the equally sampled campaign strata to the observed three-campaign mixture of the eligible corpus within each winner class. Uncertainty uses brief-cluster bootstrap resampling. The 12 intervals are exploratory and unadjusted for multiplicity.

| Primary coded theme                            | A5 win | Control win | $\Delta$ | 95% CI        |
|------------------------------------------------|--------|-------------|----------|---------------|
| Brief / constraint fit                         | 21.2%  | 21.1%       | +0.1 pp  | [−9.0, +9.4]  |
| Realization feasibility / safety / maintenance | 12.1%  | 20.0%       | −7.9 pp  | [−15.8, −0.5] |
| Conceptual specificity / distinctiveness       | 6.2%   | 7.5%        | −1.3 pp  | [−5.1, +2.6]  |
| Conceptual legibility / directness             | 5.4%   | 6.3%        | −0.9 pp  | [−5.7, +3.4]  |
| Material / causal consequence                  | 23.6%  | 20.4%       | +3.2 pp  | [−4.9, +11.7] |
| Spatial / bodily consequence                   | 10.3%  | 6.4%        | +3.9 pp  | [−1.8, +9.5]  |
| Temporal / process logic                       | 2.6%   | 0.9%        | +1.7 pp  | [−0.9, +4.7]  |
| Spectator encounter / access                   | 2.2%   | 2.7%        | −0.4 pp  | [−3.2, +2.1]  |
| Site / process integration                     | 13.8%  | 3.7%        | +10.1 pp | [+3.4, +17.2] |
| Economy of means / over-elaboration            | 0.9%   | 8.4%        | −7.5 pp  | [−12.2, −3.4] |
| Visual / perceptual force                      | 0.4%   | 0.0%        | +0.4 pp  | [+0.0, +1.3]  |
| Ethical / tonal framing                        | 1.3%   | 2.7%        | −1.4 pp  | [−4.9, +1.3]  |

Table 15 shows all twelve primary coded themes. The most prevalent primary themes in both winner classes are *material / causal consequence* (23.6% of A5 wins; 20.4% of control wins) and *brief / constraint fit* (21.2%; 21.1%), but their frequencies are very similar across winner classes. Three other categories show the clearest A5–control asymmetries in the frozen sample. Each still occurs in only a minority of wins, so these contrasts should not be read as general explanations for most evaluator decisions.

First, *site / process integration* is cited as the primary theme in 13.8% of A5 wins versus 3.7% of control wins, a +10.1 percentage-point difference. This category concerns an existing site or institutional process—for example staff workflow, service movement, irrigation, handling, conservation, or maintenance activity—becoming part of the artwork’s operation. It is distinct from whether the proposed work itself is easy to build or maintain. Second, *realization feasibility / safety / maintenance* is cited in 12.1% of A5 wins versus 20.0% of control wins (−7.9 points); here the issue is the proposal’s own physical plausibility, safety, reliability, reset burden, or maintenance burden. Third, *economy of means / over-elaboration* is cited in 0.9% of A5 wins versus 8.4% of control wins (−7.5 points), capturing cases in which additional apparatus, media, mechanisms, or complexity is judged disproportionate to the artistic gain. This asymmetry is especially interesting because the challenger prompts explicitly discourage solving weaknesses by piling on additional apparatus, media, or mechanisms. One possible explanation is that the revision task itself creates counterpressure: producing a consequentially different challenger from an already-developed installation may often be easier by adding a mechanism or layer than by achieving a comparable change through subtraction, substitution, or reorganization. The present analysis does not test this explanation, but the pattern makes it a plausible hypothesis.



**Figure 6:** Brief 053, *Echoing Gutter* → *Layered Ledger*. Top: the parent and the Structural challenger from the brushing side and the accumulated-record side. Bottom: one plausible interpretation of how brushing could redistribute wet pigment on the mesh while transferring part of it to the canvas backing. The mesh–canvas gap, rear access, and pigment-delivery hardware are post-hoc fabrication choices rather than details fixed by the evaluated dossier.

Because verdict and rationale come from the same evaluator, these coded themes describe what that evaluator reports attending to; they are not independent causal explanations or professional-artistic judgments.

## 7.2 What Candidate Transformations Change

Section 2 introduced the scaffold stakes → artistic operation → realization. The lineage traces below ask a complementary question: when a parent becomes a challenger, what concrete material, spatial, temporal, or procedural dependency changes? The two cases are author-selected illustrations rather than a representative sample. One challenger reaches lane comparison but is not adopted; the other is excluded by the hard gate before comparison. The traces are evidence of what the saved pipeline did, but they do not add independent efficacy observations. All renderings and mechanism diagrams in this subsection were produced after evaluation; they clarify the evaluated texts but were not inputs to the gate, lane selector, or external evaluator. Where a drawing adds fabrication detail not specified by the dossier, that departure is identified locally.

### 7.2.1 Brief 053: A Small Material Relocation Changes What Repair Leaves Behind

Brief 053 asks for contested maintenance using canvas, galvanized mesh, and hand-mixed pigment, with every component repairable in view during opening hours. Its `must_address`

condition is “contested maintenance through repairable canvas, mesh, and pigment,” while `must_avoid` excludes “hidden repair, polished finish, digital monitoring, craft nostalgia.” In the parent, *Echoing Gutter*, a reservoir releases a line of pigmented water onto a six-meter galvanized-mesh curtain. A visitor then brushes the wet streak across the mesh until the surface returns toward a more uniform tone. Canvas is already present, but only as border strips around the mesh. The parent therefore uses the required canvas without making it part of the damage-repair sequence.

*Layered Ledger* moves that already-required canvas behind the mesh. In the challenger dossier, brushing still spreads the wet pigment across the metal front, but part of the liquid is also pressed through the mesh onto the absorbent canvas backing. The two surfaces can therefore retain different consequences of the same repair action: the front becomes more uniform while the backing receives another pigment trace. Repeated repair can reduce the currently visible mark on the mesh while adding to a record behind it.

The physical difference can be stated without an evaluative claim:

**Parent:** new stain → brushing → visible stain is reduced

**Challenger:** new stain → brushing → front stain is reduced + pigment is deposited on the backing

The exact transfer rate, mesh aperture, pigment viscosity, canvas spacing, and rear-view arrangement are not specified as fabrication-ready parameters; Figure 6 shows one possible implementation. Two challengers were generated in this lineage. The Focused challenger failed the eligibility gate, while the Structural challenger, *Layered Ledger*, remained eligible. Under the two-candidate branch of the lane selector, the incumbent and the sole eligible challenger are compared by independent pointwise assessment rather than by the three-way Borda procedure. *Echoing Gutter* received a mean internal score of 0.779 and *Layered Ledger* 0.775, so the incumbent remained the lineage survivor. These are internal selector scores, not external judgments; because *Layered Ledger* was not adopted, RQ2 provides no exact-parent external comparison establishing it as an improvement. The case is retained here for the concrete transformation itself: repositioning a material already required by the brief changes what repeated repair leaves behind.

### 7.2.2 Brief 038: When a Contestable Maintenance Reading Becomes a Binary Gate Decision

Brief 038 constrains the work to chalk, reclaimed brick, and rainwater-safe sealant and requires a disputed boundary to remain readable after partial weathering. Its `must_address` condition is “disputed boundary through chalk, brick, and partial weathering,” while `must_avoid` excludes “maps, flags, fencing, legal-document display.” In *Eroding Chalk Line Labyrinth*, reclaimed-brick panels line the gallery perimeter and a continuous chalk boundary runs across their frontal surfaces. Selective sealing and periodic wetting are intended to make segments weather at different rates. The spectator mainly follows the perimeter and watches the boundary change on the wall. The Structural challenger, *Fragmented Borders: Erosion Field*, moves the same basic material vocabulary onto the floor. Low reclaimed-brick slabs form an irregular field and the chalk boundary passes through it. The visitor must cross the line, follow it, step around slabs, or choose another route through the field. Thus the concrete spatial change is from a boundary primarily viewed while moving alongside it to a boundary physically encountered in the route through the room.

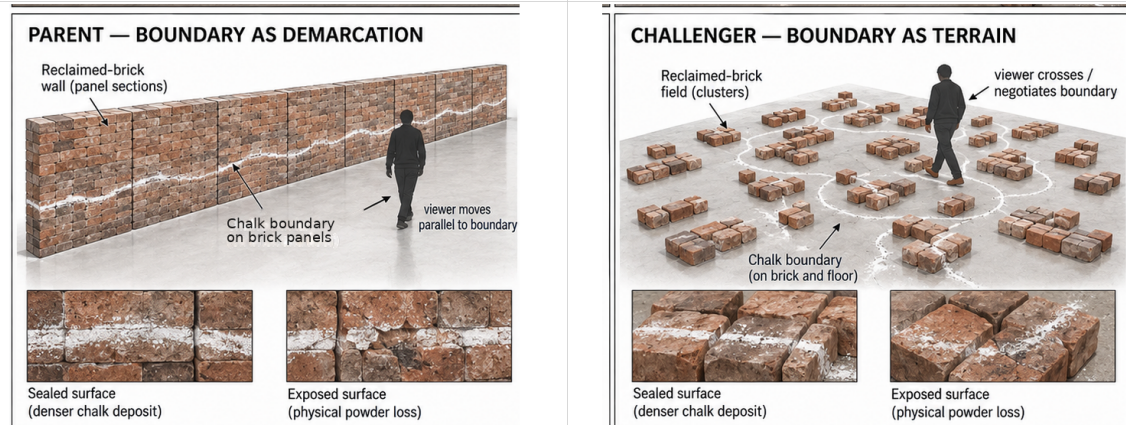
That spatial change and the gate outcome are separate facts. The recorded hard gate rejected the challenger before lane comparison, citing the dossier’s statement that “chalk may be reapplied mid-exhibition if legibility drops below threshold” against the brief requirement that the installation “remain readable after partial weathering.” In its explanation, the gate glossed the latter as a requirement to remain readable “without intervention,” although those words are not present in the benchmark brief; the dossier also does not define the legibility threshold at which reapplication would occur. The gate decision therefore depends on a contestable semantic reading of a maintenance detail rather than establishing an unambiguous hard contradiction. It remains a recorded outcome of the evaluated pipeline, not a comparative aesthetic judgment about wall versus floor.

This case motivates a design hypothesis rather than a result of the present experiment. When a potential violation is interpretation-sensitive or plausibly remediable, a future system might carry the challenger forward with a penalty, pessimistic score, or explicit risk flag rather than convert the concern into binary exclusion. The present experiment did not compare such a soft-gating policy. The consequence of a gate interpretation is also asymmetric by design: challengers

**Eroding Chalk Line Labyrinth — parent**



**Fragmented Borders: Erosion Field — challenger**



**Figure 7:** Brief 038, *Eroding Chalk Line Labyrinth* → *Fragmented Borders: Erosion Field*. Top: wall-based parent and floor-based Structural challenger. Bottom: simplified diagrams of the changed spatial relation and the proposed sealed/exposed weathering logic. The depicted slab arrangement and weathering/fabrication details are explanatory choices added after evaluation.

are screened at this stage, whereas the incumbent bypasses the challenger gate and remains eligible for comparison.

**Parent:** move alongside a chalk boundary on the perimeter wall

**Challenger:** move through a floor field in which the chalk boundary crosses the route

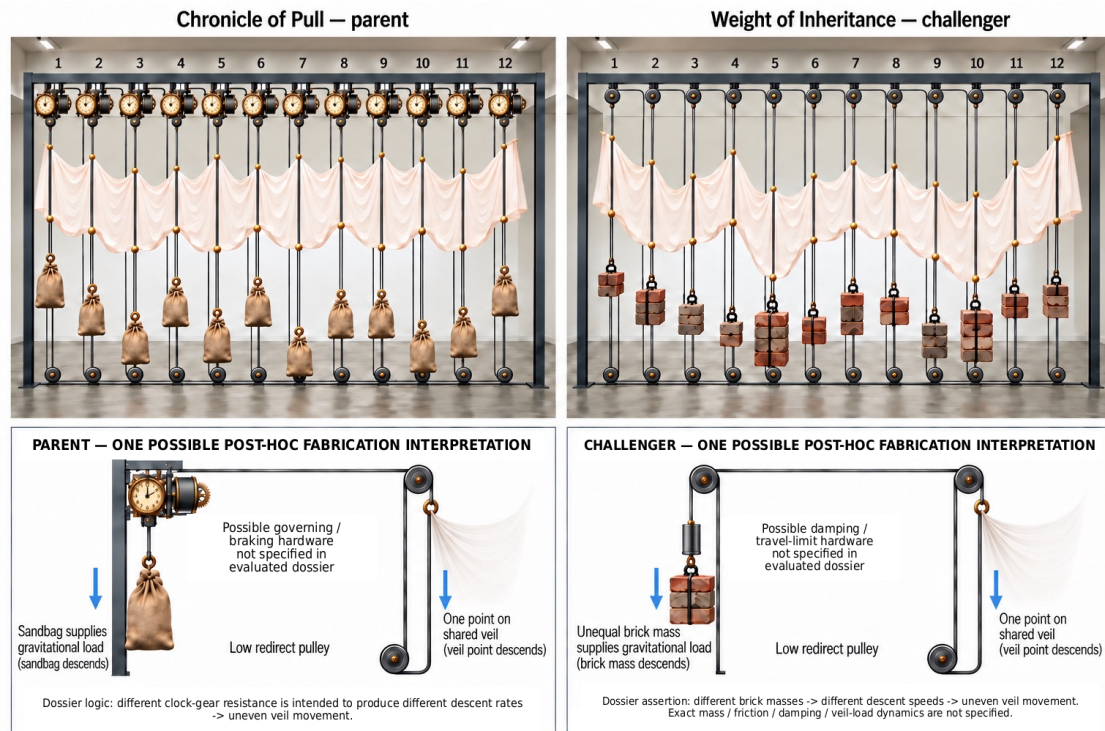
Because the recorded gate rejects the written challenger, it is not an adopted replacement and contributes no exact-parent RQ2 outcome. Figure 7 visualizes the wall-to-floor spatial change and one explanatory weathering implementation.

### 7.3 Judging Plausible Revisions: Uncertainty and Competing Readings

Section 7.2 examined what particular interventions materially or spatially change and how a gate decision can determine whether a challenger reaches comparison. We now turn to a different problem: what do the saved comparison traces reveal when plausible alternatives admit competing readings, when repeated comparisons do not yield a stable order, or when internal and held-out evaluators prioritize different properties? The positive mean RQ2 effect does not imply that every adopted intervention is preferred to its exact parent. Of the 228 adopted replacements, 179 have swap-consistent exact-parent judgments; in 53 of those comparisons, the external evaluator prefers the exact parent, while another 49 comparisons are orientation-sensitive. Those counts establish that exact-parent losses under the external evaluator and orientation-sensitive judgments occur, but not why. The cases below examine several concrete forms that this evaluative difficulty can take.

#### 7.3.1 Brief 030: Why Plausible Revisions Can Be Hard to Compare

Brief 030 asks for inherited obligation to become concrete and uneven while avoiding family photographs, heirlooms, handwritten letters, and sentimental domestic staging. Its `must_address` condition is “inherited obligation as uneven material relation,” while `must_avoid` excludes “family archive, domestic set, nostalgic soundtrack, confession.” In *Chronicle of Pull*, twelve



**Figure 8:** Brief 030, *Chronicle of Pull* → *Weight of Inheritance*. Top: matched views of the parent and selected Focused challenger. Bottom: one possible post-hoc fabrication interpretation built around the motion logic stated in the dossiers. For the parent, the dossier specifies differing clock-gear resistance intended to produce different descent rates but not the governing hardware; the illustrated brake/drum is one possible implementation. For the challenger, the dossier states that bricks of different masses descend at different rates but does not specify the dynamics linking mass, guide friction, damping, veil load, and speed; the illustrated routing and damping are hypothetical fabrication choices rather than evaluated mechanism details.

sandbag weights descend while cables and pulleys transmit their movement to different points of one shared voile curtain. Each sandbag is coupled to a separate frozen analog clock whose gear train is assigned a different resistance. The dossier states that these resistance differences are intended to produce different descent rates, so the veil moves downward in irregular pauses and drops over a 30-minute cycle. The sandbags are not specified as equal in mass, and the dossier does not provide engineering detail beyond this intended clock-resistance-to-descent relation.

The opportunity memo already proposes the main replacement before challenger generation: remove the clock-driven sandbags and use reclaimed bricks of unequal mass pulling the same veil through comparable cable-and-pulley paths. The selected Focused challenger, *Weight of Inheritance*, follows that hypothesis. It removes both the clocks and the sandbags and places twelve reclaimed bricks, specified as ranging from 2 to 15 kg, on individual guides. The dossier attributes their different descent speeds to the mass differences and again describes the shared veil descending unevenly, now over a 12-minute cycle.

The source-level change is therefore narrower than saying that the challenger simply makes the work “more material.” Both proposals are kinetic and both retain an uneven time-based descent of the veil. What changes is the stated source of that unevenness:

**Parent:** differing clock-gear resistance → differing load descent → uneven veil movement  
**Challenger:** differing brick masses → dossier-asserted differing descent speeds → uneven veil movement

The second chain is the challenger’s stated mechanism, not an engineering result demonstrated by the experiment. The dossier does not specify enough about guide angle, friction, veil load, damping, or other forces to establish how mass differences alone would produce the claimed speed differences. Figure 8 therefore schematizes the intended relation rather than validating it mechanically.

The external exact-parent comparison is orientation-sensitive (logical-pair score 0.5). One orientation prefers the challenger because the unequal brick masses make inherited obligation

more directly material; the reversed orientation prefers the parent, with lower confidence, for retaining the clock-mediated temporal structure and what the judge calls a stronger sense of bodily implication. The latter phrase is not physically unpacked in the rationale. The supported conclusion is therefore limited: the revision replaces an explicit clock-mediated source of kinetic unevenness with a mass-based one, and the two external orientations disagree about whether that substitution makes the proposal stronger.

This case points to an evaluation problem broader than installation art. The question is not merely how often evaluators disagree, but what such disagreement represents. In artistic tasks, competing judgments may reflect different defensible priorities—here, for example, direct material burden versus clock-mediated temporal structure—rather than only measurement noise around a single latent correct ordering. A broader Creative-AI question is therefore when disagreement should be collapsed into a consensus ranking and when its structure should instead be preserved and examined as part of the evaluative result. Whether artistic tasks require different treatment from domains with clearer task ground truth remains an empirical question.

### 7.3.2 Brief 036: Similar Local Changes Can Play Different Roles in the Artistic Operation

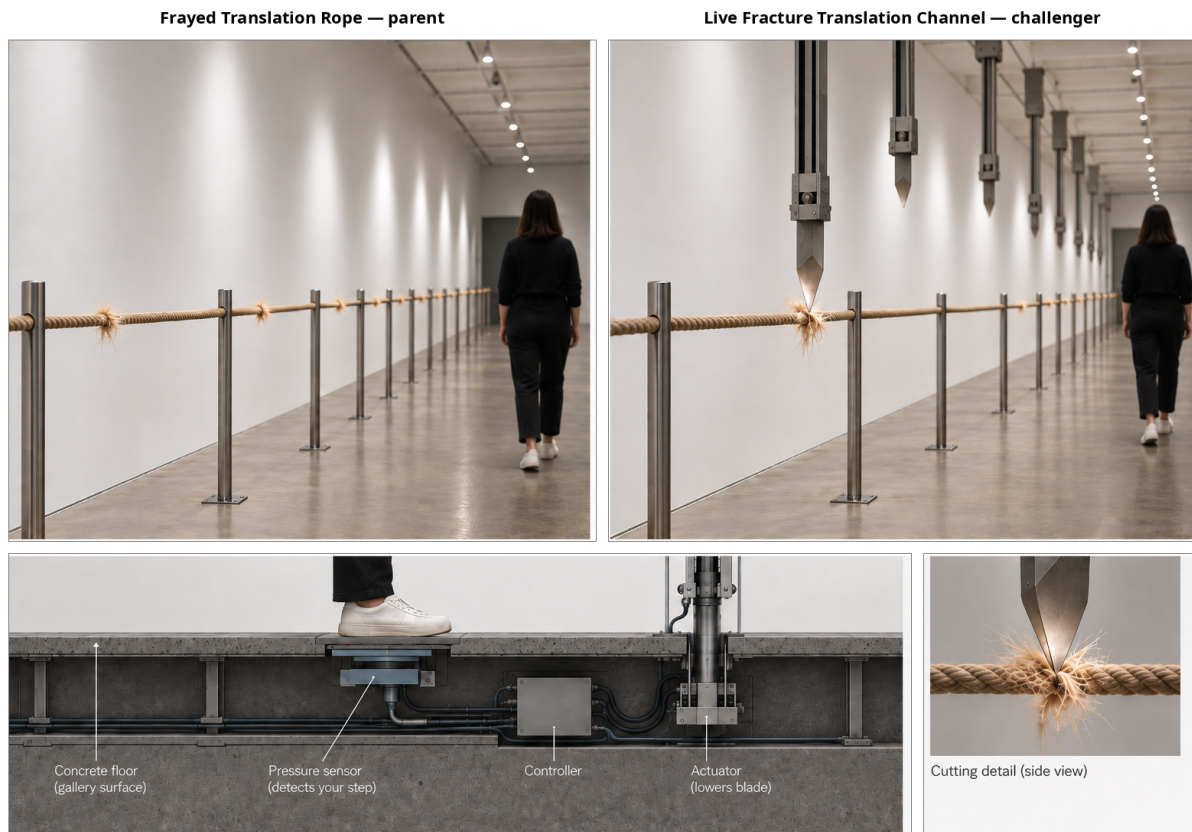
Brief 036 asks: “Design an installation about translation as a chain of minor losses, where each transfer changes a shared physical condition without representing languages as national symbols.” Its `must_address` condition is “translation through successive material transfer and loss,” and `must_avoid` excludes “flags, multilingual wall text, audio lessons, maps.” Two selected Focused reconstructions from this brief are useful because both make spectator action directly cause the next material event, while the role that action plays in the governing artistic relation is different.

**Rope lineage.** In *Frayed Translation Rope*, progressive fiber damage is already present along a tensioned hemp rope when the spectator encounters it. The selected Focused reconstruction, *Live Fracture Translation Channel*, adds pressure-sensitive floor locations connected to actuated blades. Stepping on one of those locations triggers the corresponding blade, so the spectator’s passage produces the next local cut rather than merely encountering an existing record of weakening. The external evaluator nevertheless prefers the parent in both swapped orientations. Its rationales cite the sensor-driven cutting apparatus as overengineered and safety-intensive relative to the simpler material weakening already present in the parent. The evidence supports a narrow statement: the revision makes spectator action more causally immediate, but the held-out holistic evaluator does not prefer that revision to the parent. Figure 9 provides the corresponding post-hoc explanatory visualization.

The author’s own artistic reading differs sharply from this model verdict. From that perspective, the challenger is substantially stronger because the spectator’s ordinary passage becomes part of the work’s operative causal structure, whereas the parent is comparatively inert. This is not additional evidence that the held-out evaluator is wrong. The disagreement instead raises the broader question introduced above: when artistic judgments diverge because they prioritize different properties of a work, should evaluation force a single consensus ordering, or preserve and analyze the competing value structures themselves?

**Sand-transfer lineage.** The parent shown in Figure 1, *Grain-by-Grain Funnel Cascade*, lets the same quartz-sand body descend automatically through successive funnels and calibrated meshes; each stage retains part of the material, so the amount continuing downward decreases. The selected Focused reconstruction, *Hand-Off Sand Transfer Loop*, preserves that successive loss but changes how the material advances: the next transfer occurs only when a visitor operates the next stage, releasing the diminishing sand body onward while residue remains behind. The spectator’s action is therefore not merely an additional interaction around the process; it is the action that performs the brief-required successive transfer. The external evaluator prefers this challenger in both swapped orientations, citing a clearer and more embodied chain of transfer and residue than in the automatic cascade. Figure 1, introduced in Section 1, visualizes this transformation. The evaluated challenger specifies ten sealed chambers with hand-operated valves and calibrated meshes; the reduced five-stage cascade, open receiving geometry, visible visitor, and specific valve details in the rendering are post-hoc explanatory choices.

Taken together, these two lineages show why a local descriptor such as “more causal spectator action” is too coarse to stand in for artistic improvement. In both revisions the spectator directly causes the next material event, yet that action occupies a different place in the artistic operation: in the sand lineage it performs the brief-required successive transfer itself; in the rope lineage it triggers a cutting apparatus that produces damage but does not itself transfer material from



**Figure 9:** Post-hoc explanatory visualization of the selected Focused reconstruction in Brief 036, *Frayed Translation Rope* → *Live Fracture Translation Channel*. Top: the parent presents progressively frayed sections as already-produced material states; in the challenger, stepping on a pressure-sensitive area triggers a corresponding blade and produces further damage live. Bottom: one plausible schematic of the trigger-to-cut relation. The specific sensor, controller, actuator, and blade geometry shown here are explanatory simplifications added after evaluation.

one stage to the next. Their external outcomes also differ—the sand challenger is preferred in both orientations, whereas the rope challenger is not—but this two-case contrast does not establish that the difference in operational role caused those verdicts. The more general lesson is representational: superficially similar local changes can play different roles in the governing artistic relation and therefore should not be treated as equivalent interventions.

### 7.3.3 Brief 050: Unstable Rankings and Cross-Evaluator Disagreement

Brief 050 asks: “Develop a work about interrupted ritual that stays secular and socially legible without candles, altars, incense, prayer language, or spectacle of collective healing.” Its *must\_address* condition is “interrupted ritual through a secular social operation,” while *must\_avoid* excludes “candles, altar imagery, incense, prayer, healing-event aesthetics.” Two lineages from this brief are useful not because one keeps the original and another replaces it, but because their saved within-lineage comparison traces expose two different ways in which evaluating plausible revisions can become difficult.

**Threaded Pause: unstable ranking of three interruption strategies.** The parent, *Threaded Pause*, places a half-finished brightly colored tapestry on a 3 m × 2 m motorized industrial loom where volunteers continue weaving the same work. After five minutes, a safety-rated rotary cutter severs the warp at its center and stops the loom, leaving a visible rupture. The Focused challenger, *Threaded Pause (Reconstructed)*, removes the cut: after the same five-minute period the motor stops and a brief high-frequency tone sounds, while the warp remains intact. The Structural challenger, *Resonant Pause*, keeps the cut but adds a sub-second white-noise burst and synchronized red LED flash.

All three candidates enter the three position-balanced listwise comparisons, and each call gives a different substantive answer. One ranks *Resonant Pause* first because the cut, sound burst, and flash produce the strongest sensory interruption. A second ranks the original *Threaded*

*Pause* first because the physical cut already provides a necessary rupture, while the added audio/visual apparatus in *Resonant Pause* dilutes that necessity and adds complexity. The third ranks *Threaded Pause (Reconstructed)* first because the motor halt and tone are read as a cleaner auditory-kinetic interruption that avoids a more visually gimmicky rupture. These three rankings produce the 3/3/3 Borda tie; the subsequent pointwise tie-break retains the parent. The informative part of the trace is therefore not retention itself, but that the three position-balanced calls resolve three distinct interruption strategies differently: (1) the original physical cut, (2) a non-cut motor-stop/tone interruption, and (3) the physical cut plus added sound/flash. The 3/3/3 tie therefore reflects instability over which form of interruption is preferable, not repeated judgments of one common modification.

**Procession Grid Freeze: a stable internal preference that the held-out judge reads differently.** The parent, *Procession Grid Freeze*, uses a 4 m × 4 m pressure-sensitive LED floor grid in which a blue light wave advances like a procession; after three complete traversals, an automatic timed command cuts downstream power and freezes the line mid-route. Both challengers replace that timed halt with nearly the same basic intervention: a participant-operated red STOP button triggers an official-style announcement and freezes the downstream LEDs. The important difference between them is spatial and social centrality. The Focused challenger, *Procession Grid Interruption*, places its console near the grid's terminus, away from the procession itself. The selected Structural challenger, *Bureaucratic Halt Grid*, places the console on a raised platform at the center of the grid, visible from all sides, so the act of stopping occurs inside the procession and becomes a conspicuous public exercise of authority. The Structural dossier also makes the frozen line a spatial dead-end around which spectators linger and negotiate the halted route.

Here the internal listwise judgments are unusually consistent. In all three position-balanced calls, *Bureaucratic Halt Grid* ranks first, *Procession Grid Interruption* second, and *Procession Grid Freeze* third, yielding Borda totals of 6/3/0. The rationales repeatedly reward the participant-controlled bureaucratic act and, specifically when distinguishing the two challengers, the Structural version's central placement and stronger integration of the stop authority with the procession; the automatic timed halt is criticized for lacking participant agency and a sufficiently explicit social operation. The held-out exact-parent evaluator then prefers *Procession Grid Freeze* to *Bureaucratic Halt Grid* in both orientations, although both judgments are low-confidence. Its rationales read the STOP console and prerecorded command in the opposite way: as a theatrical or game-like overlay, while the parent's interruption remains intrinsic to the moving procession.

The two Brief-050 traces therefore expose different comparison problems that the aggregate RQ2 frequencies cannot show. In *Threaded Pause*, the three position-balanced calls produce mutually incompatible rankings of three distinct interruption strategies. In *Procession Grid Freeze*, the internal comparator is stable and prefers the version that makes participant authority most spatially central and explicit, whereas the held-out evaluator treats that same explicitness as an added theatrical layer. The latter disagreement does not establish that the parent is objectively better; the external judgments are low-confidence. It shows more narrowly that a comparator can be internally consistent while still favoring a different artistic property from the held-out holistic evaluator.

#### 7.4 Terminal Selection Errors Arise at Different Stages

The RQ3 aggregate identifies recognition headroom but does not reveal where ranking errors enter. We therefore inspect a post-hoc stage-specific contrast within the Hybrid policy itself. For each brief, we compare the external Utility@1 of the final Hybrid top choice with the external utility of the proposal that Hybrid's own pointwise stage ranked first before the finalist tournament. Thus the diagnostic is  $\text{Utility@1}_{\text{Hybrid final}} - \text{Utility@1}_{\text{Hybrid pointwise}}$ , not the primary Hybrid-minus-Base contrast.

The tournament changes top-1 on 9 of 50 briefs. The mean final-minus-pointwise Utility@1 change is +0.0027, with 95% bootstrap interval [-0.0160, +0.0220]; brief W/T/L is 4/43/3 and the paired sign-flip test is not significant ( $p = .844$ ). The experiment therefore does not establish that the tournament stage is beneficial or harmful on average. The cases below are post-hoc explanatory traces selected from conspicuous Hybrid-versus-Base regressions. Base is used only to identify those regressions; the mechanism analysis itself follows Hybrid's pointwise screen and finalist tournament.

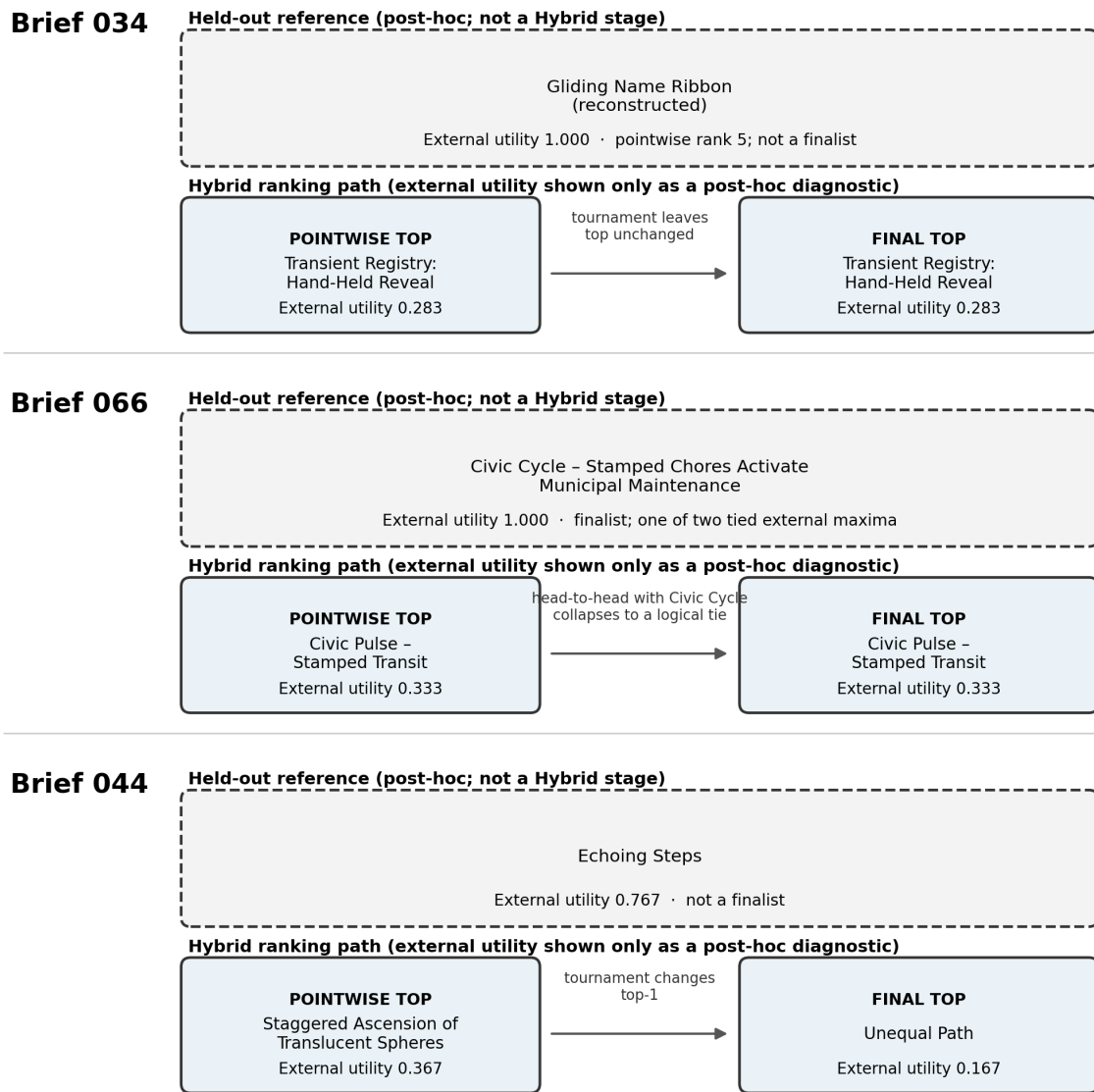
**Brief 034 — pointwise screening ranks the external maximum outside the finalist set.** The operative task concerns names that do not fit administrative forms and requires a reversible wall-based intervention that leaves existing seating untouched and can be removed without

trace. Its `must_address` condition is “misfit names and administrative forms through reversible wall-based operation,” while `must_avoid` excludes “personal data collection, adhesive residue, explanatory survey, mock application forms.” The externally highest-utility A5 proposal, *Gliding Name Ribbon (reconstructed)*, has utility 1.000, but Hybrid’s pointwise stage gives it an internal mean of 0.675 and ranks it fifth, outside the three-candidate finalist set. Its pointwise assessment identifies one explicit liability: the proposed magnetic attachment assumes a steel substrate behind plaster; without that substrate, the stated installation method would require damaging the wall. Production/institutional viability is consequently scored 0.45. That low viability score is one contributor, not a sufficient explanation of the exclusion. As a simple diagnostic, replacing only the 0.45 viability score with the theoretical maximum 1.00 while holding the other nine pointwise scores fixed would raise the mean only to 0.730, still below the fourth-place score of 0.775 and the finalist cutoff of 0.793. Hybrid instead ranks *Transient Registry: Hand-Held Reveal* first pointwise and leaves it first after the tournament; its external utility is 0.283. The case therefore exposes a broader mismatch between Hybrid’s multidimensional pointwise profile and the held-out external utility: the externally strongest candidate is scored low enough across the internal profile to miss pairwise comparison despite its external-utility maximum. The steel-substrate concern is one visible liability within that profile, not a demonstrated single cause of exclusion.

**Brief 066 — the decisive finalist comparison is itself unresolved.** The brief asks for domestic routines to enter public administration, shifting repetition from private care to collective procedure without recreating a home. Its `must_address` condition is “domestic routine transformed into collective procedure,” while `must_avoid` excludes “domestic set, kitchen imagery, family anecdote, cozy nostalgia.” Hybrid’s pointwise stage ranks *Civic Pulse – Stamped Transit* first; its held-out external utility is 0.333. One of the three finalists is *Civic Cycle – Stamped Chores Activate Municipal Maintenance*, one of two A5 proposals tied for the external maximum at 1.000. The tournament therefore compares these two candidates directly. In one orientation it prefers *Civic Pulse* with confidence 5; under exact reversal it prefers *Civic Cycle* with confidence 5. After both judgments are mapped back to proposal identity, their head-to-head result is a logical tie. Each candidate also beats the third finalist, leaving them tied on Copeland score and on the secondary orientation-level pairwise points; the next tie-break is pointwise mean, so *Civic Pulse* remains first. The two orientation rationales emphasize different properties—the municipal-vehicle and mist-ledger mechanism of *Civic Pulse* in one presentation, and the more direct chore-to-public-maintenance stamping relation of *Civic Cycle* in the other—but the saved pair does not establish a stable comparative preference between them. The case therefore shows how an externally high-utility candidate can reach the finalist set and be compared directly with the pointwise leader without producing a decisive tournament result.

**Brief 044 — position-stable finalist wins can still lower held-out utility.** The brief asks for a museum-foyer work about unequal waiting that operates during school visits, tolerates touching, and requires no attendant beyond normal front-of-house staff. Its `must_address` condition is “unequal waiting through touch-tolerant public operation,” while `must_avoid` excludes “queues created by the work, fragile parts, long instruction text, gamification.” Hybrid’s pointwise stage ranks *Staggered Ascension of Translucent Spheres* first, with external utility 0.367. The tournament then promotes *Unequal Path*, whose external utility is 0.167. Unlike Brief 066, the relevant pairwise preference is position-stable: *Unequal Path* beats *Staggered Ascension* in both A/B orientations, with confidence 5 in both, and also beats *Delayed Reveal Wall* in both orientations, with confidence 4. Across those judgments, the comparator repeatedly favors the visible sand flow and textured floor because they make unequal waiting bodily, tactile, and spatially consequential; it describes the ceiling-sphere proposal as more abstract or passive and the delayed wall mechanism as more generic or trigger-like. The tournament therefore does not fail through inconsistent pairwise judgments: it follows a coherent comparative preference that nevertheless moves external Utility@1 downward by 0.200. The external maximum in this pool, *Echoing Steps* at 0.767, had already been excluded before the tournament, so the tournament explains only the additional 0.367-to-0.167 drop, not the full distance from the best externally measured candidate.

Figure 10 summarizes these three traces while keeping the post-hoc held-out reference separate from the actual Hybrid ranking path.



**Figure 10:** Three terminal-selection traces with the held-out reference separated from the actual Hybrid ranking path. Dashed boxes show an externally highest-utility candidate from the same fixed A5 pool; they are post-hoc references, not stages in Hybrid. Solid boxes show Hybrid’s pointwise top and final top; the external utilities printed for those candidates are likewise post-hoc diagnostics. In Brief 034 the external maximum is not a finalist and Hybrid’s top choice remains unchanged. In Brief 066, *Civic Cycle*, one of two candidates tied for the external maximum, is a finalist, but its exact-reversal comparison with *Civic Pulse* collapses to a logical tie, so the pointwise leader remains first. In Brief 044 the external maximum is not a finalist, and the tournament changes the top choice from *Staggered Ascension of Translucent Spheres* to *Unequal Path*, reducing held-out external utility from 0.367 to 0.167.

Taken together, these traces distinguish three mechanisms hidden by the aggregate ranking metrics. Pointwise screening can place an externally high-utility proposal outside the finalist set when its internal multidimensional profile diverges sharply from held-out external utility; an externally high-utility candidate can reach the finalist set yet fail to displace the pointwise leader when their exact-reversal head-to-head collapses to a logical tie; and position-stable finalist wins can still promote a candidate with lower held-out external utility. These are explanatory cases rather than frequency estimates, and none changes the unresolved aggregate result for the tournament or the primary Base-versus-Hybrid comparison.

## 8 Discussion

### 8.1 What the Decomposition Reveals About Concept Development

For contemporary-art concept development, the central methodological implication of the experiment is that three consequential stages can diverge under evaluation: candidate-set construction, lineage transition, and terminal recognition. A weak surfaced proposal may reflect a weak field of alternatives, a lineage transition in which the selected replacement is preferred less than its parent under the external criterion, or a failure to recognize which surviving proposal deserved attention. Final-output evaluation can easily collapse these distinct failures into the same observed outcome.

The empirical evidence is correspondingly asymmetric. A5 candidate sets score higher than all three planned controls under the external measure, and adopted replacements are externally preferred to their exact parents on average, whereas terminal selection remains unresolved. The secondary selected-output check makes the separation especially visible for the central matched comparison: A5 preference over M5 is 0.6176 across the complete candidate pools but 0.520 when each system is reduced to the proposal it surfaces first. This selected-output contrast is descriptive rather than a separate inferential endpoint, but it shows that improving the candidate pool does not guarantee that terminal ranking will correctly identify and surface its strongest available proposal.

For creative-agent evaluation, the practical consequence is that candidate-set construction, lineage transition, and terminal ranking should be inspected separately. A compelling top-ranked output can conceal a weak candidate population, while a poor surfaced proposal can hide stronger material that has already been generated; exact-parent comparison separately reveals whether particular adopted replacements are externally preferred to the states they displace. Evaluating only the final recommendation collapses these distinct questions.

The decomposition is operational rather than ontological. We do not claim that creativity consists of three independent modules, nor that these stages correspond literally to how artists develop work. Their value is experimental: they expose failure modes that an end-to-end winner-only evaluation would otherwise merge.

### 8.2 The Full Structured-Search Package Yields Higher Measured Candidate-Set Quality

The matched M5 comparison provides the strongest evidence for the full A1rtist development procedure relative to substantially matched generic iteration. M5 begins from the same self-refined parents and preserves the same lineage count, successor count, incumbent retention, challenger eligibility, and lane-comparison scaffold, while A5 additionally uses field-context construction, specialized multi-role review, opportunity synthesis, preservation constraints, and differentiated Focused and Structural roles. The +0.2352 complete-pool effect therefore supports the full structured-development package under the external evaluation. It does not isolate a single causal mechanism: A5 changes both how inference is organized and how much contextual and analytical inference is supplied.

Separating those contributions would require further controlled comparisons rather than one idealized ablation. Useful tests would include inference-budget-matched structured versus generic reasoning and, where feasible, component ablations that remove field context, specialized review, opportunity synthesis, or differentiated challenger roles while keeping the remaining pipeline as comparable as possible.

The broader implication is therefore a hypothesis about how open-ended search should be studied, not a claim that the present experiment has isolated its cause. Iterative critique and revision are already established in systems such as Self-Refine and ResearchAgent [55, 8]. A1rtist makes the local move more explicit by specifying what a revision should preserve, what it is allowed to replace, and what kind of transformation it is attempting. Search is often characterized by depth, branching factor, sample count, critique rounds, or voting procedures; in weakly specified creative domains, the semantics and permissions assigned to a move may deserve separate study alongside search width and inference budget. The present experiment motivates that comparison but does not resolve it, and transfer beyond installation ideation remains an empirical question.

### 8.3 Contestable Refinement Treats Revision as a Proposal, Not an Obligation

Competitive refinement keeps non-replacement available as a real outcome rather than making every generated challenger compulsory. Across 250 lineages, 22 final states (8.8%) remain

content-identical to their incumbents, while 228 adopt a challenger. This is descriptive, not a count of explicit “keep” judgments—retention can follow challenger ineligibility or lane comparison—and in this evaluation it is an available but uncommon outcome.

Among the 228 adopted replacements, exact-parent comparison favors the challengers on average: candidate-weighted preference is 0.6601 and the brief-balanced centered replacement effect is +0.3347 (Section 6.2). Together, these observations support a bounded interpretation: non-replacement remains possible while the replacements selected by the evaluated policy are preferred to their exact parents on average.

Procedurally, the important distinction is that AIrtist treats a revision as a challenger to an existing state rather than as an obligatory overwrite. Section 7.3 shows why this distinction matters in an open-ended domain: revisions can alter the role that materials, spectator action, or other elements play in the governing artistic operation, and a coarse description of the local change does not by itself imply improvement. The experiment nevertheless does not isolate the causal value of keeping the incumbent eligible; that would require a matched mandatory-replacement policy.

#### **8.4 Disagreement in Artistic Evaluation Is Not a Single Phenomenon**

The saved comparison traces in Section 7 expose several forms of disagreement that a single agreement rate would collapse together. Some exact-parent judgments change under A/B reversal; repeated comparisons of the same candidate set can produce incompatible rankings; and an internally stable preference can still conflict with the held-out external evaluator. Section 7.3.2 additionally records an explicitly marked authorial counter-reading of one model verdict, outside the measured evaluation. These observations do not establish that all such disagreements have the same source, that they are equally meaningful, or that any one side is correct.

They motivate a broader question for Creative-AI evaluation: not only how often judgments disagree, but what properties the disagreement tracks and whether the difference persists across repeated comparisons and evaluators. In artistic tasks, some persistent disagreement may reflect different value priorities rather than only noise around a single latent ordering. Future work should therefore distinguish unstable judgments from persistent cross-evaluator differences and examine the reasons behind them before collapsing both into a single consensus ranking. Whether artistic tasks require different treatment from domains with clearer task ground truth remains an empirical question.

#### **8.5 Evaluator Rationales Show Limited Between-Group Asymmetries**

The post-hoc rationale coding does not identify a dominant explanation for the aggregate A5 advantage. The two most prevalent primary themes overall—material / causal consequence and brief / constraint fit—occur at similar rates in A5 and control wins. Three less prevalent themes show the clearest between-group asymmetries: site / process integration is more frequent among A5 wins, whereas realization feasibility / safety / maintenance and economy of means / over-elaboration are more frequent among control wins. Each still occurs in only a minority of wins, although the directions are the same across all three control campaigns. Because the rationales are generated by the same evaluator that issues the verdicts, these differences describe reported evaluator attention rather than independent validation of the artistic properties involved.

These asymmetries are best treated as leads for follow-up tests, not explanations of the A5 effect. The A5-favoring site / process pattern raises one question: does structured development more often integrate existing site or institutional processes into the artistic operation? The two control-favoring patterns raise two others: does structured revision sometimes make proposals harder to realize, and does it sometimes add more apparatus than the artistic gain warrants? The latter is especially notable because the challenger prompts explicitly discourage additive fixes; substantial revision may nevertheless be easier to achieve by adding mechanisms or layers than by subtracting, substituting, or reorganizing. The rationale coding cannot establish that these properties are more prevalent in A5 proposals, identify the responsible pipeline component, or show that feasibility and over-elaboration occur in the same cases.

#### **8.6 Recognition Failure Is Not One Problem**

Stronger externally measured material often already exists in the five-proposal pool, but neither terminal policy establishes superiority. Base has higher mean Utility@1 (0.6633 versus 0.6353) and lower regret, while Hybrid more often places an externally maximal candidate first (0.32 versus 0.26). Proposal construction and recognition are therefore empirically separable. Adja-

cent research-ideation systems report related gaps between generation and self-evaluation [50, 9], although their tasks differ from ours.

Section 7.4 localizes three distinct reasons why terminal ranking can fall short under held-out external utility. An external-maximum candidate can be screened out before finalist comparison by the pointwise surrogate; an external-maximum finalist can reach direct comparison yet fail to displace the pointwise leader when exact A/B reversal collapses the pair to a tie; and a position-stable tournament preference can still select a proposal with lower held-out utility. These cases distinguish pre-finalist surrogate mismatch, reversal-sensitive comparison instability, and stable criterion disagreement.

These are not the same ranking problem. Surrogate mismatch concerns shortlist construction: if held-out utility is the target diagnostic, pointwise scoring may need recalibration or a wider finalist set. Reversal-sensitive comparisons instead call for uncertainty handling—additional comparison, abstention, or preserving the tie. Stable internal-external disagreement is different again: because held-out utility is model-based rather than ground truth, the first question is which evaluative criteria differ, not how to reproduce the external evaluator. The cases are diagnostic, not frequency estimates; the post-hoc final-minus-pointwise analysis also does not show whether the tournament helps on average. Recognition should therefore be evaluated stage by stage, while artist-facing rank should allocate attention rather than present a definitive artistic verdict; Section 9.1 develops that consequence.

### 8.7 Domain Grounding Makes the Search Position Inspectable

AIArtist’s domain grounding encodes a recognizable *working position*: enacted artistic operation over topical illustration, necessary material and spatial relations over arbitrary styling, specific spectator relations over generic participation, field differentiation over first-obvious formats, and revisions that address the diagnosed relation rather than merely add explanation or spectacle. The A5–M5 advantage supports the full structured-development package, not any one of these commitments. Their methodological value here is that they are explicit and inspectable: the system records what it intends to preserve, what it treats as the dominant liability, which nearby format it seeks to escape, and what transformation it authorizes. This makes the search position more legible than a generic instruction to “make this more creative.”

The mechanism analysis also shows the limits of treating such explicit commitments as settled expertise. A gate can apply a contestable reading of a brief constraint; similar-sounding revisions can alter different relations within the artistic operation; and later comparison stages can remain unstable or disagree with the held-out evaluator. The point is not to attribute these outcomes to domain grounding. Rather, explicit grounding helps separate the assumptions that steer search from the later decisions that determine which revisions remain eligible and which candidates are ranked first.

A domain-specific prior is therefore better treated as a *contestable search position* than as expert truth injected into a generic model. Its commitments can organize search without being assumed to exhaust artistic quality or to be correct for every artist or project. The present experiment supports the structured package under its stated evaluation procedure, but it neither isolates the contribution of individual commitments nor tests whether artists would accept, reject, or renegotiate them. Section 9 takes that interaction question forward.

## 9 Intended Artist-Facing Use: Idea Search as a Substrate for Negotiated Co-Creativity

The evaluated system is an autonomous concept-development backend rather than an interactive co-creative interface. Its intended artist-facing use begins from an early-stage creative situation in which an artist has a direction but not yet a specification: they may know what matters, which conditions or materials constrain the problem, and which obvious solutions they want to exclude while remaining uncertain about what the work should become. AIArtist takes one such rough brief and autonomously constructs, develops, and ranks five complete project hypotheses. The experiment evaluates that computational layer. A future interaction layer would make some of the assumptions and decisions exposed by the backend negotiable to the artist rather than treating its search process as fixed.

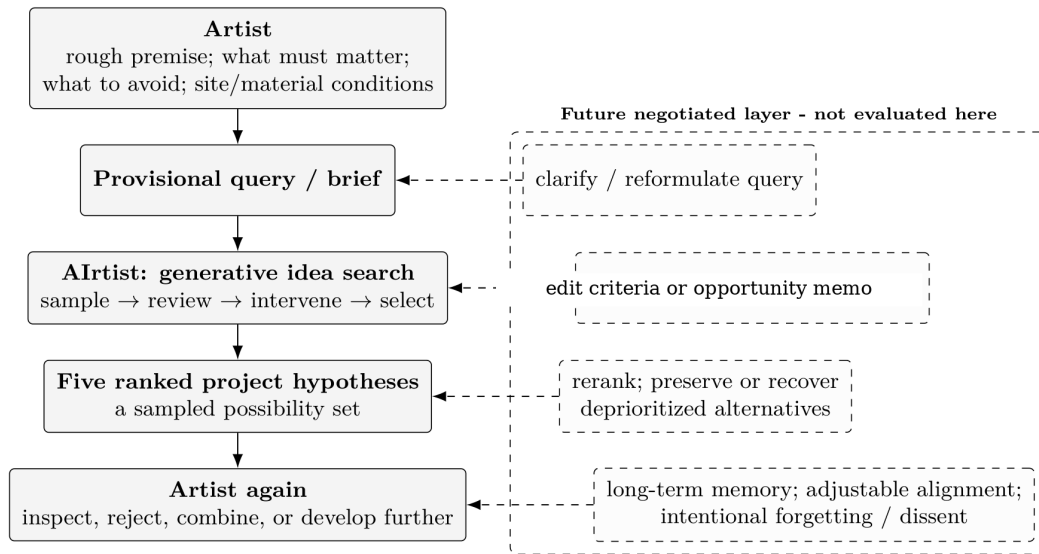
## 9.1 From a Provisional Query to a Sampled Possibility Set

A conventional generative interface often treats the prompt as a specification: the user states what is wanted, and the model attempts to produce an answer. Early creative work is different because part of the problem is precisely that the desired answer is not yet known, and the formulation of the problem itself may change as possible responses become more concrete. Human–AI design-space research has made this co-evolution of problem formulation and candidate solutions explicit in adjacent creative work [49].

We therefore use an information-retrieval analogy in a restricted sense. The free-text premise identifies what is at stake and functions like a provisional query; `must_address` conditions protect pressures the artist does not want development to lose; `must_avoid` conditions remove solution families already known to be unwanted; and site, material, institutional, or production requirements further constrain the region to be explored. The resulting input specifies an under-determined development problem rather than a target artifact.

The analogy concerns the relation between request and answer, not the retrieval mechanism. AIrtist does not search a database of pre-authored artworks. It performs generative search over proposal states in an open-ended space of possible artistic directions: candidate directions are constructed, reviewed, transformed, preserved or displaced, and finally ranked. The resulting five dossiers are comparatively developed project hypotheses whose artistic operation, spatial and material logic, spectator relation, and production implications are made explicit as described in Section 4. They are starting directions for further artistic work, not five finished artworks from which the system has identified a correct one.

Figure 11 summarizes this transition from an artist-authored provisional query through autonomous idea search to a returned sampled possibility set, and sketches the future negotiation layer discussed below.



**Figure 11:** From autonomous idea search to a possible future negotiation layer. The evaluated system expands an artist-authored provisional brief into a ranked sampled possibility set of five complete project hypotheses without artist feedback inside the run. The dashed right-hand layer is a conceptual design sketch, not an implemented or evaluated interface: query reformulation, editable opportunity memos and criteria, branch recovery, reranking, and longitudinal personalization remain future interaction mechanisms.

For an artist-facing system, the resulting design hypothesis is deliberately modest: candidate quality and breadth may need to remain separately visible and separately controllable. A future interface could, for example, cluster closely related proposals, preserve one or more minority directions, or expose operation-, material-, or spectator-level differences rather than assuming that five branches are meaningfully distinct. It could also expose alternative orderings of the same sampled set according to different criteria—for example conceptual necessity, feasibility, public or ethical risk, or distance from a creator’s habitual solutions—instead of collapsing all visibility into one fixed ordering. The present experiment does not evaluate such policies or criteria-contestable ranking; these remain artist-facing hypotheses for future study.

## 9.2 Artist-in-the-Loop: Negotiating the Search Process

The evaluated system implements autonomous search rather than artist negotiation over that search. We use *artist-in-the-loop*, by analogy with human-in-the-loop, for a future configuration in which the artist can intervene in selected search commitments without turning every internal model call into a conversational step. The dashed interaction layer in Figure 11 marks this proposed extension beyond the evaluated backend. This boundary matters because empirical work with artists indicates both that domain expertise remains consequential when using generative AI and that artistic co-creative practice is shaped by ongoing control, intuition, and adaptation to system affordances [46, 47].

Competitive refinement already makes intervention contestable in a limited computational sense: the incumbent remains eligible, and a proposed change must earn replacement. An artist-in-the-loop system would expose a different level of contestability. The artist could contest not only a generated candidate but also the system’s interpretation of the problem and the search commitments used to generate alternatives. The opportunity memo is a particularly natural negotiation surface because it makes several of those commitments explicit: which incumbent property should be preserved, what is treated as the dominant liability, what transformation is proposed for testing, and which additive repairs are treated as insufficient.

Three intervention points follow directly from the current architecture. The first is *query negotiation*. If the initial brief is provisional rather than a complete specification, the system could detect unresolved constraints, tensions between `must_address` conditions, or ambiguity about what is intended to remain open, then ask a small number of targeted questions before committing search resources. The relevant research question is not simply whether clarification improves proposal scores, but when clarification helps articulate an artist’s intention and when it instead prematurely determines the problem for them.

The second is *opportunity-memo negotiation*. Before challenger generation, an artist could inspect, edit, or lock memo fields; request an alternative diagnosis; or ask different agents to formulate competing readings of the same incumbent. An artist could accept the diagnosis but reject the proposed transformation, protect a mechanism the model wants to remove, declare that an inefficient material process is essential, or deliberately pursue a direction the default quality model regards as too didactic, decorative, spectacular, or familiar. A disagreement over what deserves protection, which liability should organize the next move, or which transformation is worth testing could then produce genuinely different searches rather than another rewrite under one hidden interpretation.

The third is *recoverable selection*. Ranking need not permanently erase losing candidates. A future interface could preserve minority alternatives, permit criterion-specific reranking, display proposals whose pairwise ordering is unstable, and allow the artist to recover a branch that the model deprioritized. Because the present study finds that recognition can fail even when stronger measured alternatives are already available, recoverability would serve not only as an interface convenience but as a safeguard against premature collapse of the sampled set.

This changes what AI initiative would mean. A proactive system could ask questions, point out contradictions, propose transformations, criticize a direction, or refuse an unsafe implementation, but each such action can also shift the formulation of the artistic problem. Artist-in-the-loop search therefore requires more than adding conversational turns: the interface must make sufficiently visible what the system is proposing to preserve or change and where the artist can accept, reject, or redirect that steering. The aim is selective involvement rather than continuous micromanagement: an artist might delegate broad initial search, intervene when the system surfaces a consequential conflict, and reclaim control when a diagnosis, transformation, or ranking decision would otherwise narrow the directions available for further development.

## 9.3 Longitudinal Collaboration, Memory, and Aesthetic Enclosure

The current AArtist has no persistent model of an individual artist. Every user receives the same fixed search machinery and bounded working criteria. Long-term creative collaboration, however, creates an obvious role for memory. Retrieval and personalization could make recurring concerns, materials, rejected clichés, philosophical commitments, past projects, and unresolved ideas available to later searches. Such memory also creates a possible failure mode that we call *aesthetic enclosure*. A system that becomes increasingly good at predicting what an artist has previously accepted may progressively reduce the chance of encountering what the artist has not yet learned to want. Personalization can therefore become another route to fixation rather than simply better alignment. Adjacent studies already show that generative assistance can

contribute to design fixation and homogenization of human ideation [51, 52], but they do not test longitudinal personalization or establish aesthetic enclosure as an observed effect.

The present experiment does not test longitudinal memory or personalization. A future personalized system should therefore test, rather than assume, that increasing fit to an artist’s prior preferences preserves productive difference over time. Possible countermeasures are design hypotheses rather than requirements established by the present study. They include different temporal layers of memory, intentional forgetting, adjustable degrees of alignment, or separate agents in which one remains closely aligned while others deliberately retrieve or generate dissenting and unfamiliar directions. Their purpose would be to let personalization accumulate useful continuity without silently turning an artist’s past choices into a boundary on future search.

Longitudinal collaboration would also require visible creative-data governance. An artist should be able to inspect what personal material is being treated as precedent, decide what may be retained or forgotten, and distinguish persistent artistic commitments from temporary experiments that should not harden into a permanent profile. In this sense, memory policy becomes part of the creative interface rather than only a background data-management choice.

## 10 Limitations and Threats to Validity

### 10.1 Validity of the External Model-Based Evaluation

**A strong but model-based external evaluator.** The principal outcomes in this study are derived from blind holistic pairwise comparisons performed by GPT-5.6 Sol using the extended-reasoning configuration specified in Section 5.3. At the time of the experiment, GPT-5.6 Sol was OpenAI’s flagship model and was described by OpenAI as its strongest model yet [60]. General benchmark strength does not by itself establish validity for contemporary-art judgment, but using a high-capability reasoning model matters for a task that requires comparison of long, structurally complex proposals rather than application of a simple surface metric. Every proposal pair is evaluated under the same model configuration, instructions, and blind A/B decision format, so comparisons across experimental conditions do not depend on changing evaluators or evaluation procedures. All principal outcomes nevertheless come from one model and configuration, and the degree to which its judgments correspond to those of professional artists, curators, producers, and other practitioners remains to be established directly.

**There is positive empirical evidence for human correspondence in adjacent evaluation tasks.** LLM-based judging is now an established methodology for open-ended model evaluation: Zheng et al. report that strong LLM judges such as GPT-4 agree with human preferences at rates above 80% on MT-Bench and Chatbot Arena, comparable in that setting to human–human agreement [61]. More relevant to the present task, recent studies have tested model judges against human evaluation of creative and early-stage conceptual work. Edwards et al. find that reasoning-enabled vision-language judges evaluating early engineering concepts approach expert-equivalent performance on measures including uniqueness, creativity, and drawing quality, and in some comparisons reach or exceed expert–expert agreement [62]. Addis et al. compare ChatGPT evaluations with Consensual Assessment Technique judgments of creative outputs produced by workers in cultural and creative industries; AI evaluations show internal consistency comparable to expert judges, and replacing one human judge with the AI evaluator does not reduce panel reliability [63]. A Nature Communications study of open-ended scientific ideation likewise reports encouraging alignment between expert and LLM judgments in a dedicated human-validation study, particularly for originality [64]. These settings are not contemporary-art evaluation, but together they provide concrete empirical grounds for expecting a high degree of correspondence between a strong reasoning-model evaluator and professional human judgment in at least some adjacent open-ended creative and conceptual tasks.

**That correspondence is task-dependent rather than guaranteed.** Creative evaluation is especially challenging because open-ended outputs lack a single ground truth. LitBench, for example, finds 73% agreement with human preferences for the strongest off-the-shelf judge on creative writing, while specialized reward models reach 78% [65]. The broader evidence therefore supports high human correspondence as a plausible expectation, not as a property that can be transferred automatically from one creative domain to another. Contemporary-art installation proposals combine conceptual, historical, material, spatial, and institutional considerations in a configuration not directly tested by any existing benchmark we identified.

**The present protocol probes several threats to measurement robustness.** Rather than assuming the external judge to be infallible, the experiment probes its behavior under exact

A/B order reversal, confidence-restricted consensus analysis, proposal-length diagnostics, and leave-one-brief and leave-one-category perturbations. The matched-control direction survives these tests. Exact reversal is particularly relevant because presentation-order effects are a documented concern in LLM-based pairwise evaluation; the length analysis likewise addresses the possibility that a judge simply rewards more elaborate responses [61]. These checks cannot establish agreement with professional human judgment, but they make it less plausible that the principal result is an artifact of one simple presentation, verbosity, or benchmark-composition effect.

**Internal and external evaluation are deliberately separated.** AIRTIST's internal ten-dimensional working model shapes review and ranking through explicit concerns such as prompt fidelity, enacted tension, artistic-operation necessity, field distinctiveness, spatial and visual encounter, material necessity, spectator specificity, reception resilience, and production/institutional viability. The external evaluator instead receives complete anonymized proposals and makes a holistic A/B judgment without access to system identity, lineage history, internal scores, or the external candidate utilities later derived from its judgments. The two evaluations necessarily attend to some of the same substantive properties: an external measure that was indifferent to whether a proposal was coherent, materially consequential, spatially specific, or artistically necessary would not be measuring the construct the system was designed to improve. The important separation is therefore not absence of conceptual overlap, but absence of access to AIRTIST's internal scores and decisions and the use of a distinct holistic judgment procedure.

**The underlying conception of proposal quality remains normative.** AIRTIST favors operational necessity, material and spatial specificity, explicit spectator relations, field differentiation, resistance to generic additive spectacle, reception resilience, and plausible institutional realization. These are explicit working commitments for the constrained installation-ideation problem studied here, not neutral or exhaustive criteria of artistic quality. Other artistic and curatorial positions may value decorative excess, explicit pedagogy, deliberate impracticality, institutional antagonism, or other properties differently. Making the commitments explicit makes the search procedure inspectable and contestable; it does not convert them into universal criteria for contemporary art.

**Evaluator rationales are diagnostic rather than independent validation.** The pair-specific free-text rationales used in the Mechanism and Failure Analysis reveal what the evaluator reports attending to. The coded analysis identifies limited between-group asymmetries in themes such as site / process integration, realization feasibility / safety / maintenance, and economy of means / over-elaboration; these themes do not dominate evaluator rationales overall. They are useful for interpreting the observed preference patterns but are not a separate source of professional artistic judgment or causal evidence about why one proposal is artistically superior. Verdict and rationale originate from the same evaluator. The post-hoc thematic coding is additionally LLM-assisted rather than independently human-coded, and its preserved artifact does not retain the exact coding-model configuration; it should therefore remain an exploratory mechanism diagnostic rather than an independent validation layer.

**Professional human validation is the next test, not a prerequisite for interpreting the present comparison.** The strongest claim supported here remains comparative: under a disclosed, blind, high-capability model-based evaluation instrument, AIRTIST produces the candidate-set and exact-parent effects reported in Section 6. Existing human-validation studies in adjacent creative and concept-evaluation domains, together with the stability of the present measurements under multiple perturbations, give positive reason to expect a high degree of correspondence with professional human judgments. A dedicated validation study with practicing artists, curators, producers, and related professionals is nevertheless needed to measure that correspondence directly and, more importantly, to determine whether the direction and magnitude of the AIRTIST effects persist under professional contemporary-art evaluation. High correspondence is therefore an evidence-based hypothesis for the next experiment rather than an assumption built into the present one.

## 10.2 Benchmark Generalizability and Text-to-Work Validity

The study evaluates 50 held-out commission-style briefs, IDs 026–075, disjoint from the 25 briefs available during system development. The main matched-control direction survives removal of every individual brief and every complete coverage category. This supports the aggregate comparison on the held-out benchmark, but it should not be confused with evidence from a natural sample of contemporary-art commissions.

The benchmark briefs are LLM-generated commission-style tasks rather than briefs written by practicing artists during actual project development. They were constructed to cover five deliberately balanced forms of pressure: thematic/conceptual, site-specific, material-constrained, institutional/production-constrained, and anti-cliche/difficult. These categories are useful for coverage and sensitivity analysis, but they are not an empirical taxonomy of contemporary-art practice, and their equal frequency is artificial. The experiment therefore tests controlled generalization across unseen synthetic tasks, not generalization to the idiosyncratic language, incompleteness, contradictions, private references, stylistic commitments, and evolving intentions that may appear in real artist-authored briefs.

The common commissioning context narrows the domain further. Proposals are interpreted within a temporary or medium-duration contemporary-art museum commission with a museum-controlled footprint and a shared production envelope. The results therefore do not establish performance under other commissioning regimes, such as public art, artist-run or commercial-gallery contexts, substantially lower-budget practice, or performance-centered production with different institutional constraints.

The benchmark size also places a specific boundary on subgroup interpretation. Each coverage category contains ten briefs. Leave-one-category-out analysis is useful as a robustness check of the overall matched-control direction; it is not evidence that AIArtist performs differently across artistic categories. Likewise, individual success and failure cases should not be interpreted as prevalence estimates for particular artistic mechanisms.

Beyond benchmark generalization, two representational gaps remain. First, the evaluated proposal is a structured textual dossier. Even before fabrication, a text can fail to carry qualities that would be evident through drawing, sound, scale model, embodied rehearsal, material sample, or spatial demonstration. The text-only external evaluation therefore measures the proposal as linguistically represented, not every artistic property the proposed work might possess. Second, the evaluated objects are unrealized project hypotheses rather than fabricated installations. A dossier can make a coherent case for material behavior, scale, circulation, duration, maintenance, public encounter, staffing, and budget and still fail when produced. No engineering, conservation, accessibility, fabrication, installation, maintenance, commissioning, or audience study validates the proposed works.

This text-to-work gap is unusually consequential in installation practice. Sound behaves differently in an actual room; materials wear unpredictably; visitors ignore intended routes; access and fire requirements alter spatial arrangements; natural light changes; repeated interaction modifies the work; and institutional labor that appears minor in prose may become constitutive in practice. Textual proposal evaluation can reason about such relations only counterfactually.

The post-hoc renderings do not close this gap. Their purpose is to make transformations in source dossiers visually legible—for example, a boundary moving from a frontal condition into a walkable field. Image generation can also supply convincing architecture, fabrication detail, lighting, scale, and finish that the text did not earn. Renderings are therefore explanatory visual hypotheses, were never shown to the evaluator, and provide no independent evidence that the represented work would function as depicted.

### 10.3 Generalization Across Generation Models

All core generation conditions use the same base-model family. This strengthens the within-study comparison because differences in base-model capability do not confound the search-policy contrasts: A5 and its controls are evaluated under the same underlying generation capability. The unresolved question is how the value of the architecture changes as that capability changes. A stronger model would plausibly improve the initial incumbents, but the same increase in capability could also improve specialized review, opportunity synthesis, and challenger generation when applied throughout the pipeline. There is therefore no clear reason to expect stronger base models to eliminate the value of structured development rather than shift the level at which useful intervention becomes possible. At the same time, the present experiment cannot establish that the observed gain will remain constant, increase, or decrease across model families or capability levels.

The task itself is open-ended and under-specified rather than one with a known optimal completion, so greater base capability does not by itself imply that diagnosis, preservation, alternative generation, and contestable comparison become redundant. Whether structured development continues to yield additional value as models improve is therefore an empirical scaling question rather than a result of the present study. A direct test would hold the architecture and evaluation procedure fixed while varying the generation model or reasoning capability,

and measure how both incumbent quality and the marginal A5–control effect change. Such experiments are required before treating the architecture as model-independent.

#### 10.4 Field Context as Search Material, Not Art-Historical Verification

The field-context stage is not intended to build an authoritative art-historical repository, establish global novelty, or generate remakes of specific artworks. Its role is narrower and operational: it supplies downstream review and opportunity synthesis with associative material about nearby artistic lines, recurring formats, possible clichés, productive differences, and contextual tensions. Recalled works, artists, texts, and neighboring practices can therefore function as inspirations, anti-inspirations, or cues for what a proposal might escape without the experiment treating them as verified art-historical evidence. Under this interpretation, the field map is search material rather than a claim about the true structure of contemporary art.

Model-mediated recall nevertheless creates an information-quality boundary. It can be incomplete, inaccurate, culturally narrow, or anachronistic, and an erroneous neighborhood can steer later reasoning toward an unhelpful comparison class. Provenance-bearing retrieval could improve factual grounding, broaden the evidence available to the system, and make specific precedent claims inspectable. It would not be an unambiguous upgrade, however. Retrieved material can be noisy, redundant, unevenly relevant to the artistic pressure in the brief, or overly anchored to literal documented precedents; using it effectively would require retrieval ranking and synthesis rather than simply inserting more sources into downstream prompts. Model-generated and retrieval-based field contexts therefore make different trade-offs among associative usefulness, provenance, coverage, noise, and search focus.

A useful next experiment would compare recall-only, retrieval-only, and hybrid field-context conditions while holding the downstream architecture fixed. The relevant question is not merely whether retrieval increases factual accuracy, but whether it improves proposal quality and field-aware differentiation without narrowing search around the most easily retrieved precedents or increasing derivative reuse. Such a comparison would test whether provenance-bearing field knowledge strengthens the role this stage already plays, rather than assuming that retrieval is automatically superior to model-mediated recall.

### 11 Conclusion

A consequential part of contemporary-art practice occurs before an artwork has stabilized: an artist may know what problem, pressure, site, material condition, or institutional constraint matters without yet knowing what the work itself should become. This paper treats that interval as an explicit computational task. Airtist develops an under-specified installation brief into several project hypotheses and models development as contestable transformation: proposed changes compete with the artistic state they are intended to replace rather than being accepted simply because additional inference has produced them.

Airtist produces a stronger measured five-proposal set than all three planned controls. Against M5, the substantially matched generic-iteration condition that begins from the same five self-refined parents, the centered complete-pool effect is +0.2352 (95% CI [+0.1472, +0.3212], Holm  $p < .0001$ ). Because A5 and M5 start from the same parents, the difference cannot be attributed to a luckier starting sample; it supports the full Airtist development package without isolating which component or residual compute difference is responsible. Across 250 lineages, 22 final states remain incumbent-identical. Among 228 adopted replacements, challengers receive candidate-weighted exact-parent preference 0.6601 and a brief-balanced centered effect of +0.3347 (95% CI [+0.2143, +0.4520]). Thus structured development can improve the measured candidate set while preserving non-intervention as an available final lineage outcome, without establishing the causal value of retention or any individual component.

Terminal selection remains less resolved. Base has higher mean Utility@1 than Hybrid (0.6633 versus 0.6353), while Hybrid more often places an externally highest-utility candidate first (0.32 versus 0.26); neither policy establishes superiority. Stronger measured candidates often already exist in the pool without being surfaced first. The experiment therefore establishes the recognition problem more clearly than the present solution. Candidate-set construction, adopted intervention quality, and terminal recognition should be measured separately because failure at any one can produce a weak surfaced result.

Contemporary-art installation is central to this argument rather than merely a difficult benchmark. Changing material causality, spatial organization, spectator position, or institutional procedure can change a proposal's identity rather than simply improve its presentation. The

mechanism traces show that superficially similar local changes can play different roles in the artistic operation, and that plausible alternatives can receive unstable or conflicting comparative judgments. Preservation, transformation, and selection should therefore remain separately inspectable rather than treating a local descriptor or a single comparative verdict as authoritative.

The claims remain bounded by the evaluation setting. The benchmark briefs are synthetic commission-style tasks rather than project notes written by practicing artists, and the evaluated outputs are textual proposals rather than realized works. The evaluation remains model-based: adjacent validation evidence gives positive reason to expect substantial correspondence with professional human judgment, but the degree of correspondence for contemporary-art installation proposals remains to be measured directly. No artist negotiates with the system inside the evaluated pipeline. Two especially important next steps are to test the measured preference signal against professional artistic and curatorial judgment while studying the structure of disagreement—which artistic priorities produce it, when consensus aggregation is appropriate, and when aggregation would erase meaningful evaluative plurality—and to study human-negotiated search around the same development backend so that artists can contest diagnoses, opportunity memos, candidate priorities, and recoverable branches.

The aim is not to automate the decision of what the artwork should be. It is to make the space in which that decision is formed richer, more inspectable, and less prematurely collapsed.

## A Exploratory Neural-Embedding Diagnostic of Candidate-Pool Semantic Dispersion

This appendix records an exploratory representation-level diagnostic that is not used as evidence for RQ1–RQ3 or for the main-text mechanism claims. It measures semantic dispersion in textual proposal representations, not artistic, visual, art-historical, political, or realized-installation diversity.

The analysis uses the same 50 held-out briefs and the five A5 and five M5 proposals available for each brief (500 proposals total). From each dossier it uses only `artistic_operation`, `spatial_logic`, `material_logic`, and `spectator_encounter`. Texts are embedded with the frozen Qwen/Qwen3-Embedding-0.6B encoder without a task prompt, and embeddings are L2-normalized. For a pair of proposals, semantic distance is  $1 - \cos(\mathbf{e}_i, \mathbf{e}_j)$ ; pool breadth is the mean distance across the ten unordered proposal pairs.

We compute two representations. In the concatenated-document representation, the four raw field values are joined in a fixed order with blank-line separators and embedded as one document. In the field-wise representation, each field is embedded separately; the four cosine distances for a proposal pair are averaged with equal weight before pool-level averaging. For each representation, A5 and M5 are compared within brief. Reported intervals are paired bootstrap 95% CIs over the 50 briefs (50,000 resamples), and  $p$  values are two-sided paired sign-flip Monte Carlo tests (200,000 random sign assignments). Table 16 reports the resulting aggregate and field-level comparisons.

Both aggregate representations point toward lower textual semantic dispersion for A5, but the effect is much smaller under equal-weight field-wise representation. The brief-level differences from the two representations are strongly associated (Pearson  $r = .821$ ; Spearman  $\rho = .806$ ), with the same sign on 80% of briefs. The field decomposition is heterogeneous: the negative difference is concentrated in `artistic_operation` and `material_logic`, while no directional difference is detected for `spatial_logic` or `spectator_encounter`.

A5 is also more verbose within this four-field representation: the mean concatenated length is 198.1 whitespace-delimited words per proposal for A5 and 146.1 for M5. However, the within-brief A5-minus-M5 length difference is not meaningfully associated with the corresponding breadth difference in this sample. For concatenated-document breadth, Pearson  $r = .010$  and Spearman  $\rho = -.045$ ; for field-wise breadth, Pearson  $r = .144$  and Spearman  $\rho = .083$ .

The result should therefore be read narrowly. It suggests modestly greater semantic concentration of A5 proposal texts under this frozen neural representation, especially in operation and material logic. It does not establish that A5 produces less diverse artworks, less visual or spatial variety, or less art-historical novelty. No independent encoder or human validation of this representation-level measure is included here, so the analysis remains hypothesis-generating and is intentionally separated from the primary results.

**Table 16:** Exploratory neural-embedding diagnostic of within-pool semantic dispersion. Breadth is the mean of the ten pairwise cosine distances among the five proposals in each pool. Differences are paired by brief; intervals are paired bootstrap 95% CIs and  $p$  values are two-sided Monte Carlo sign-flip tests. Field-level tests are exploratory.

| Representation                   | A5     | M5     | A5–M5   | 95% CI             | $p$      |
|----------------------------------|--------|--------|---------|--------------------|----------|
| Concatenated four-field document | 0.3510 | 0.3964 | −0.0454 | [−0.0575, −0.0335] | < .00001 |
| Equal-weight field-wise mean     | 0.3779 | 0.3889 | −0.0109 | [−0.0179, −0.0040] | .0033    |
| <i>artistic_operation</i>        | 0.3988 | 0.4190 | −0.0202 | [−0.0321, −0.0083] | .0017    |
| <i>spatial_logic</i>             | 0.3841 | 0.3791 | +0.0050 | [−0.0069, +0.0172] | .4237    |
| <i>material_logic</i>            | 0.3912 | 0.4196 | −0.0284 | [−0.0415, −0.0154] | .00013   |
| <i>spectator_encounter</i>       | 0.3375 | 0.3377 | −0.0002 | [−0.0107, +0.0102] | .9755    |

## B Post-hoc Coding of External-Evaluator Rationales

This appendix documents the post-hoc analysis of pair-specific free-text rationales produced by the same GPT-5.6 Sol evaluator that issued the pairwise preference judgments. The analysis therefore describes what the evaluator says it attended to; it is not an independent causal explanation of artistic superiority.

### B.1 Sampling, blinding, and coding

The three primary RQ1 control campaigns contain 3,750 logical cross-pool comparisons, each presented in two exact A/B-reversed orientations. For compactness in this appendix, C1 denotes A5 versus B5-self, C2 denotes A5 versus M5, and C3 denotes A5 versus B5-raw. For rationale coding, 811 logical pairs whose mapped winner changed with orientation were excluded. Forty-seven further pairs were ties in both orientations and were excluded because the analysis asks about reasons for directional preference. This left 2,892 eligible swap-consistent directional logical pairs.

A stratified sample of 600 logical pairs was drawn with 100 pairs from each campaign  $\times$  winner stratum (C1/C2/C3  $\times$  A5/control), spread across all 50 briefs. The first 25 pairs per stratum (150 total) formed a blinded codebook-development sample. A 12-category codebook was developed from those records and then frozen. The remaining 75 pairs per stratum (450 total) formed a separate frozen-codebook sample. During coding, the coder saw the two orientation-specific rationale texts for the logical pair but not system identity, canonical winner, candidate role, or experimental condition beyond the information already contained in the rationale text. The key was joined only after coding was complete.

Coding was LLM-assisted rather than independently human-coded. Each logical pair received one primary coded theme, up to two secondary coded themes, a locus tag (winner strength, loser weakness, both, or unclear), and optional specific failure-mode tags. The preserved coding artifacts contain the complete blinded inputs, frozen codebook, output fields, and final labels, but they do not retain the exact coder model/version or inference configuration used for this post-hoc pass; we therefore do not attribute the coding to a specific model. The operational coding protocol is nevertheless recoverable from the frozen codebook: both orientation-specific rationales are read as one record; generic boilerplate is ignored; exactly one primary category (or UNCLEAR) and up to two secondary categories are assigned only from considerations explicitly stated in the rationale; the primary code reflects the most concrete substantive contrast supporting the verdict; and substantively irreconcilable rationales are coded UNCLEAR. Table 17 gives the frozen categories and the operational distinction each one was intended to capture.

**Table 17:** Frozen rationale-codebook categories used for the 450-pair blind coding. Categories describe themes coded from the evaluator’s free-text rationales; coders were instructed not to infer properties that were not stated in those rationales.

| Category               | Operational meaning                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brief / constraint fit | Explicit compliance with or violation of a stated brief requirement or prohibition, such as required materials, wall-only installation, no surveillance, passive airflow, or a budget ceiling. |

| Category                                       | Operational meaning                                                                                                                                                                                       |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Realization feasibility / safety / maintenance | Physical or technical plausibility, safety, reliability, fabrication practicality, maintenance or reset burden, and operational viability when the issue is not primarily a formal brief violation.       |
| Conceptual specificity / distinctiveness       | Whether the artistic operation is specific, non-generic, and non-arbitrary rather than a familiar metaphor, decorative move, generic sensory effect, or interchangeable artistic format.                  |
| Conceptual legibility / directness             | How clearly the proposed mechanism communicates the intended relation or proposition. This concerns intelligibility of the relation, not whether the idea is distinctive.                                 |
| Material / causal consequence                  | Whether the cited strength or weakness concerns an actual material transformation or cause-and-consequence relation, rather than a merely illustrative, symbolic, atmospheric, or digital representation. |
| Spatial / bodily consequence                   | Room-scale or architectural consequence, bodily relation, circulation, passage, threshold, obstruction, choreography, or another spatial condition that is itself part of the artistic reason.            |
| Temporal / process logic                       | Duration, accumulation, persistence, recurrence, sequence, reset, rhythm, irreversibility, or another process that unfolds over time.                                                                     |
| Spectator encounter / access                   | The visitor's role, participation or non-participation, accessibility, forced disclosure, interaction burden, or another feature of how spectators encounter the work.                                    |
| Site / process integration                     | Integration of the artistic operation with real site or institutional processes such as staff workflow, service movement, irrigation, handling, maintenance, conservation, or other existing routines.    |
| Economy of means / over-elaboration            | Economy of means versus unnecessary apparatus, media, mechanisms, theatricality, gadget-like proliferation, polishing, or complexity that adds burden without comparable artistic gain.                   |
| Visual / perceptual force                      | Visual, sensory, atmospheric, or perceptual vividness when that quality itself is cited as the reason for preference, rather than merely described as part of the proposal.                               |
| Ethical / tonal framing                        | A cited problem of moralizing, didactic, therapeutic, confessional, spectacular, romanticized, or otherwise tonally inappropriate framing.                                                                |

## B.2 All-category results and stable contrasts in the frozen sample

The main-text Table 15 reports all twelve primary-theme categories, not only those with non-zero contrasts. Because campaign and winner strata were deliberately sampled in equal numbers, rates are first estimated within campaign  $\times$  winner strata and then reweighted to the C1/C2/C3 mixture of the 2,892 eligible pairs separately within A5 wins and control wins. They therefore retain the intended “more often in A5 wins versus control wins” interpretation while avoiding an artificial equal-campaign mixture imposed by the qualitative sampling design. They remain sample-based estimates rather than direct coding of every eligible rationale.

Three primary-theme contrasts have 95% brief-cluster bootstrap intervals separated from zero. All three contrasts have the same direction separately in C1, C2, and C3 (Table 18). These category-level intervals are exploratory and unadjusted for multiplicity. When a theme is counted if it appears as either the primary or a secondary coded theme, the corresponding A5-minus-control differences become +14.1 percentage points for site/process integration, –12.2 points for realization feasibility/safety/maintenance, and –18.0 points for economy of means/over-elaboration; all three intervals remain separated from zero.

The other nine categories remain visible in Table 15. Their primary-theme intervals overlap zero, so they are not promoted as stable directional differences. This includes material/causal consequence, spatial/bodily consequence, and conceptual specificity/distinctiveness: all are

**Table 18:** Campaign-specific A5-minus-control differences for the three stable primary-theme contrasts in the frozen 450-pair sample. The common sign across C1–C3 is a descriptive sensitivity check rather than a separately corrected confirmatory family.

| Primary coded theme                            | C1       | C2      | C3       |
|------------------------------------------------|----------|---------|----------|
| Site / process integration                     | +6.7 pp  | +9.3 pp | +14.7 pp |
| Realization feasibility / safety / maintenance | –13.3 pp | –5.3 pp | –5.3 pp  |
| Economy of means / over-elaboration            | –5.3 pp  | –9.3 pp | –8.0 pp  |

substantively common in individual rationales, but the frozen sample does not support treating any of them as a distinctive explanation of the A5 advantage.

### B.3 Illustrative rationale language

One A5-win rationale emphasizes operational integration: “The overhead cart tracking and wall ribbon avoid floor cuts and mist entirely while connecting service movement to real water flow.” By contrast, one control-win rationale emphasizes economy of means: “B is a clearer and simpler timed custody arrangement ... A adds sensor-triggered sound, tactile notes, and multiple lighting systems that dilute the core.” These examples illustrate the aggregate coding pattern; they are not additional observations beyond the coded sample.

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